OpenBilliards: a validated, bit-exact event-driven simulator for cue-sports mechanics, with a systematic assessment of the empirical laws of pool

OpenBilliards contributors

Abstract

To our knowledge, this paper presents the first systematic domains-of-validity assessment of the empirical laws of pool—the 30° and 90° rules, the throw, squirt, and swerve laws, gearing english, the ten-times-fuller and trisect systems, and the bank/kick diamond systems. The instrument is OpenBilliards, a validated event-driven analytic rigid-body simulator for cue-sports mechanics whose shipped WebAssembly binary is bit-exact across tested platforms at sub-millisecond typical per-shot cost and serves as the paper's normative physics definition. A battery of 24 experiments, internally registered as frozen designs before any sweep ran, executed 16,933 recorded simulations, distilled into 25 byte-stable pinned outputs that include a cross-simulator oracle run against the open simulator pooltool. Results are reported in ASME V&V 20 vocabulary—comparison error $E = S - D$ with validation uncertainty u_{val}—with every anchor classified as measured datum, derived reference form, or practitioner lore ("validated" claims rest on the measured class), and every registered-band miss is recorded in a machine-readable waiver ledger (38 rows, seven of them registered-band misses) rather than silently re-tuned. Two headline findings illustrate the method. The trisect "3× cut"

rule is exact only at its half-ball calibration point: the measured ratio declines from 3.99 to 2.59 over 5–45° cuts and stays within 10% of 3 only for 22.5–40° cuts. The battery also uncovered two inherited defects in our own kernel—a ball–ball slip-impulse law leaving slip-regime throw about 2.6% low, and a discontinuous cushion rebound-branch selection shifting bank arrivals by up to 0.85 diamonds—each fixed and confirmed by predicted-shift verification. Every figure regenerates from pinned outputs whose physics is proven identical to the shipped canonical binary: all 16,933 recorded simulations replay through that binary with stamp-normalized trajectory-digest equality (16,933/16,933), and successful reproduction is defined by hash equality. The kernel source is public under the MIT license, with the portions derived from pooltool under the Apache License 2.0; the canonical kernel revision of this paper, the experiment harness, and the pinned outputs are not part of that release.

Keywords: cue sports, billiards, event-driven simulation, verification and validation, reproducibility, multibody impact, WebAssembly determinism

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1. Introduction

Instruction in pool and billiards is transmitted through a canon of empirical rules: the 30° and 90° rules for post-impact cue-ball direction, the throw, squirt, and swerve laws, gearing english, the ten-times-fuller rule for frozen-ball kisses, the trisect system for draw shots, and the diamond systems for banks and kicks. The canon's analytic ancestry runs to Coriolis (Coriolis, 1835; Nadler, 2005), and its modern statements are maintained in Alciatore's technical-proof corpus and the companion high-speed-video and FAQ collections (Alciatore, 2008, n.d.c,n). Every rule in the canon is an idealization—a closed form or a geometric shortcut calibrated at one operating point—yet none is taught with a quantified statement of *where it holds*. A player learns that a well-struck draw shot departs at roughly three times the cut angle "below about 40°"; no source states the lower boundary of that claim, its speed window, or the error incurred at the edges.

The literature does not close this gap. Experimental work has validated single rules at single operating points: Bayes and Scott (1963) measured ball–ball collision losses, Wallace and Schroeder (1988) analyzed two-dimensional collisions (with the frictional correction of Onoda, 1989), and Cohen (2002) treated the corner-5 diamond system as geometry alone. Simulation authors have validated their own sub-models against their own footage (Mathavan et al., 2010, 2014), and the most complete open simulator, pooltool, ships a single self-consistency notebook (Kiefl, 2024). To our knowledge, no systematic, whole-canon, domains-of-validity study exists.

This paper conducts that study. The instrument is an event-driven analytic rigid-body simulator for cue-sports mechanics (Section 3): ball

motion between events is closed form, event times are polynomial roots, and the shipped WebAssembly binary—canonical kernel `0.1.0+fb3425ad`—reproduces every trajectory bit-exactly across the operating systems, architectures, and runtimes tested (Section 7), making the shipped binary itself the paper's physics definition of record. We use the term simulator deliberately: billiards is the vehicle, and the verification-and-validation (V&V) methodology together with the rules study are the contributions. One definition governs the title's word *validated* and every use of it below: validation means comparison against anchors classified as *measured data* (camera points, instrumented tables, high-speed video, a robot protocol), with comparison error and validation uncertainty reported and every miss ledgered — never pass/fail certification; comparisons against sources' own idealized forms are labeled model benchmarking or verification of the port, and rule-of-thumb lore is characterized, not validated (Section 4). The simulator's validity is established by a registered battery of 24 experiments spanning ball–ball impact and throw, stick–ball interaction, cushion rebound and the diamond systems, curved (Coriolis) trajectories, frozen contacts and the break, and closed-form kinematics with jump shots, plus a cross-simulator oracle run—25 pinned outputs in all, byte-stable across repeated runs.

The battery is designed to be hard on its own instrument (Section 4). We adopt ASME V&V 20 vocabulary throughout (ASME, 2009): each anchor names its validation variable and validation points, reports the comparison error $E = S - D$ between simulation value S and source datum D, and carries a validation uncertainty u_{val} whose numerical, input, and source-data com-

ponents are estimated per source family (Section 4), including digitization uncertainty for points lifted from published figures. Verification—agreement of the code with its own closed forms and with a cross-simulator oracle whose event times match to within 7.4×10^{-14} s where the model registries coincide—is kept strictly separate from validation, agreement with physical measurement. Evidence spans four source tiers (published curves and tables — classified per anchor as measured data or derived reference forms — high-speed video, a published specification window, and the cross-simulator oracle), every experiment design was frozen and registered in-repository before any sweep ran, and misses of registered bands are never re-tuned: they are recorded in a machine-readable waiver ledger (38 rows) kept with the study's artifact bundle.

Figure 1 is the battery's one-look summary — one verdict-segmented domain bar per rule family, annotated with the headline comparison errors the following sections quantify. Most rows benchmark derived forms or lore; "validation" is reserved for measured-datum rows (Section 4). Throughout the figures, headline errors E and markers in orange are the waiver-marker convention (Section 5). The map's 239/240 cushion count is verification-of-port; its measured-datum content is the 12/12 camera points (Section 5.3); its miscue-limit row states the registered input-envelope convention, revised post-registration under the scoped compatibility note of Section 8 (simulated results unchanged). Three numbers preview what the battery finds. The trisect "3× cut" rule is exact only at its half-ball calibration point: the measured ratio declines monotonically from 3.99 to 2.59 over 5–45° cuts and stays within 10% of 3 only for 22.5–40° cuts—even friction-free perfect balls,

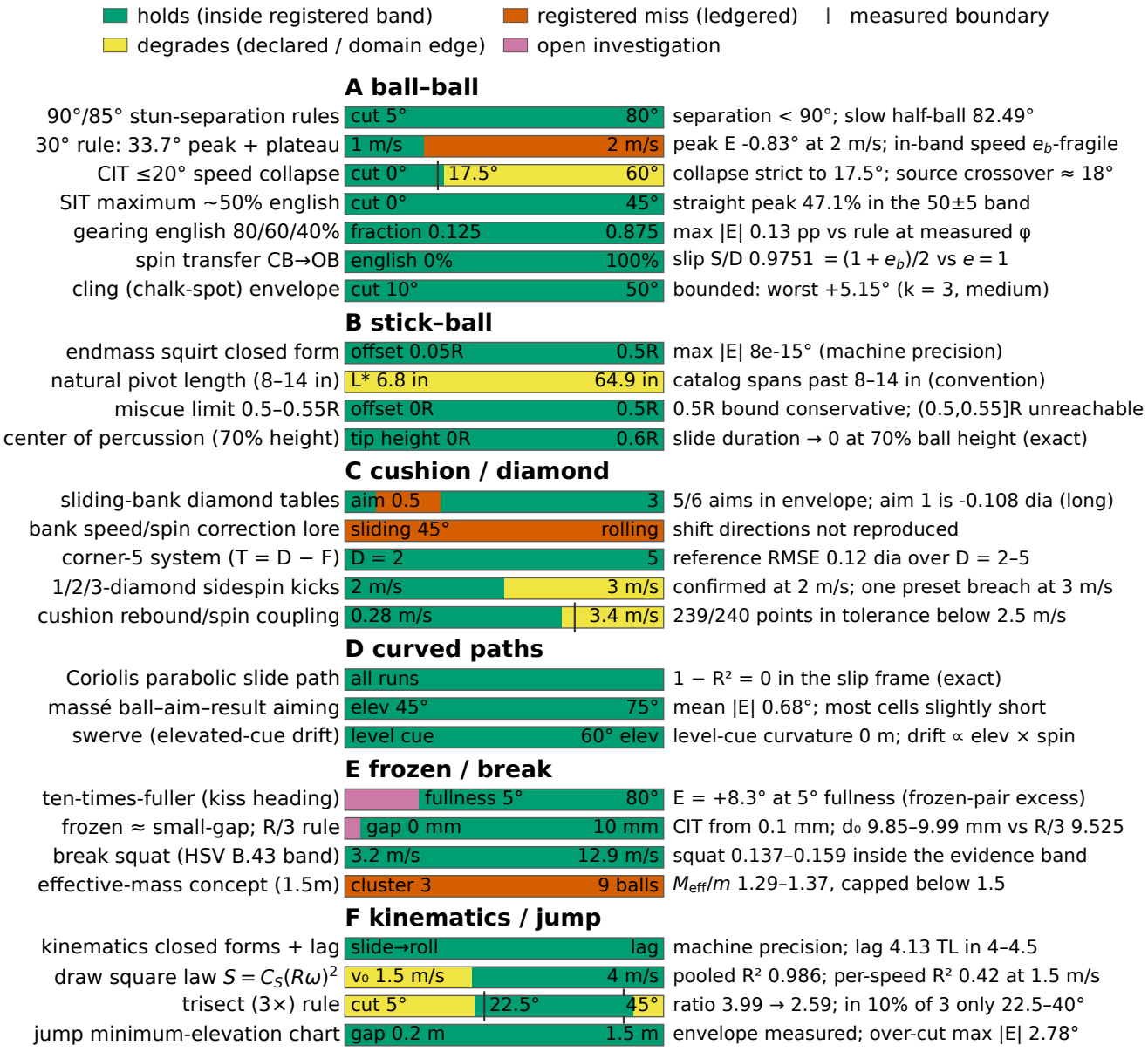


Figure 1: **Verdict map.** One row per rule family of Table 10, spanning all three anchor classes. Bars span the registered sweep domains, verdict-segmented (legend at top; in-bar ticks: measured boundaries), with one headline note per row (values: pinned outputs, Supplementary Material S1).

calibrated at the same half-ball anchor, give a small-cut limit of 4.0, and the measured small-cut ratio (5° cut) stays at 3.99–4.01 across the restitution and friction sensitivity arms because each arm re-anchors its own calibration; the breakdown is geometric in that calibration-anchored sense, with e_b and friction setting only the operating point (the decomposition is in Section 5.6) (experiment F2; Fig. 14). The 30°-rule peak height sits inside the registered $33.7 \pm 0.3°$ band only at the slowest registered speed ($E = -0.57°$ at 1.5 m/s and $-0.83°$ at 2.0 m/s, against a frictionless-elastic reference form), while the peak location and the 27–34° plateau hold at every tested speed (experiment A2; Fig. 4). And the standard claim that cut-induced throw collapses across speeds for cuts at or below 20° holds strictly only to 17.5° on the registered grid—consistent with the source's own crossover near 18° (experiment A3; Fig. 3).

The methodology also proved to have teeth against the instrument itself: the battery uncovered two inherited defects in our own kernel—a ball–ball slip-impulse law that left slip-regime throw about 2.6% low (found by experiment A1; hereafter the *slip-law defect*) and a discontinuous settled-rebound branch selection in one compliant cushion model that jumped registered bank arrivals by 0.49–0.85 diamonds (found by experiment C1; hereafter the *branch-selection defect*)—both fixed and confirmed by predicted-shift verification, and Section 6 presents the episode as a case study.

1.1. Contribution claims

This paper makes five contributions, each sized by verified novelty against the closest prior art we could locate. The novelty sizing rests on two structured prior-art research sweeps (run 2026-07-21, coverage through July 2026)

across scholar-indexed journals and preprint servers in the simulation, sports-engineering, and computational-billiards literature together with the cue-sports gray literature (the Dr. Dave technical-proof and high-speed-video catalogs, *Billiards Digest*, the pooltool documentation), with every load-bearing claim verified against its primary source; the sweep summaries are retained internally but per-query transcripts were not logged, so we report the search as scoped rather than as an exhaustively documented systematic-review protocol, and we claim "first" only as first-we-could-locate. The closest prior art found is listed with each claim.

C1. Systematic domains-of-validity quantification of the empirical laws of pool (30°/90° rules, throw/squirt/swerve laws, gearing english, 10-times-fuller, bank/kick diamond systems, Coriolis results) with registered, hash-pinned experiment scripts — the paper's headline contribution, addressing a confirmed literature gap. Closest prior art: pooltool's single self-consistency notebook (Kiefl, 2024); Cohen (2002) (geometric corner-5 only); Bayes and Scott (1963); Wallace and Schroeder (1988) (single-rule experiments).

C2. Whole-engine multi-source validation battery (published curves and tables + HSV footage + published specification windows + cross-simulator oracle) under an explicit anti-overfit out-of-sample doctrine, with a machine-readable waiver ledger — the paper's headline methodology contribution. Closest prior art: Mathavan et al. (2010, 2014) (single sub-models, own footage); no engine-wide precedent.

C3. Compliant frozen-cluster break resolution (Hertz + Lankarani–

Nikravesh, Jacobi concurrent advance) embedded in an otherwise closed-form event-driven kernel — a major section of this paper and the seed of a planned companion paper. Closest prior art: Gismeros Moreno et al. (2022) (full-game penalty time-stepping); Cross (2023) (small-N analytics); pooltool's 10% momentum-theft hack (Kiefl, 2024).

C4. The shipped canonical `.wasm` as the paper's normative physics definition: bit-exact cross-platform reproduction of every figure from the shipped binary — a major section of this paper and, to our knowledge, without peer-reviewed simulator precedent. Closest prior art: Rapier's enhanced-determinism mode (engineering documentation); deterministic-WebAssembly runtimes (WebAssembly Community Group, 2015); the nearest scientific neighbors — versioned WebAssembly tool distributions in bioinformatics and the climate community's bit-for-bit reproducibility and ensemble-consistency practice — pin binaries for reuse or test statistical consistency of rebuilt source, but neither makes a shipped versioned binary a paper's normative physics definition.

C5. Terminal rest projection ("round-trip law": simulate output is valid simulate input) and quartic event-detection robustness — deliberately sized as an engineering subsection, not a headline. Closest prior art: Leckie and Greenspan (2006) flagged the fragility; Algorithm 1010 (Orellana and De Michele, 2020) + two ACM Remarks (De Michele, 2022, 2024).

The remainder of the paper is organized as follows. Section 2 situates the work in the measurement, simulation, and computational-billiards literature. Section 3 describes the instrument, and Section 4 the registered V&V design, the anchor tiers, and the waiver-ledger doctrine. Section 5 reports the rules study across the six experiment groups, and Section 6 the cross-simulator agreement study together with the defect-discovery case study. Section 7 defines the reproduction protocol built on the canonical binary; Section 8 states limitations, and Section 9 outlines future work.

2. Related work

The literature behind this paper spans nearly two centuries, and it organizes naturally into five threads: the analytic foundations of billiard mechanics; the single-model validation era whose best practice we generalize; the event-driven simulators and computational players whose formulation we inherit; the learning-era systems that consume billiards physics as ground truth; and the verification, validation, and reproducibility standards that this domain has so far left unapplied.

2.1. Foundations, 1835–1995

The analytic mechanics of billiards begins with Coriolis's 1835 treatise (Coriolis, 1835), which derived, among other results, the parabolic path of a sliding cue ball after impact, the aiming construction for the massé shot, and the invariance of the friction direction during impact. The treatise remained difficult to obtain for most of its history; Nadler's annotated English translation (Nadler, 2005) restored it to working accessibility. These results are not merely ancestral here: experiments D1–D3 re-derive the parabola,

the massé construction, and the swerve behavior event-exactly and measure where each idealization degrades (Section 5; Fig. 11).

Controlled measurement arrived more than a century later, one rule at a time. Bayes and Scott (1963) measured the speed loss in billiard-ball collisions; Wallace and Schroeder (1988) analyzed billiard-ball collisions in two dimensions, and Onoda (1989) corrected their treatment of post-collision rolling. These short papers established the genre this study generalizes — and they also exhibit its archetypal limitation: each supplies a validation point without a validity domain. Marlow (1995) closed the era with the first book-length synthesis to attempt parameterized realism across the whole game; several of its catalog constants were later superseded by measured curves, a history reflected in the anchor-tier discipline of Section 4, which ranks published measured curves above catalog constants.

2.2. Single-model validation era

Between the late 1990s and the mid-2010s, individual sub-models of cue-sports mechanics acquired rigorous derivations and, in the best cases, instrumented validation. Shepard's monographs (Shepard, 1997, 2001) bridged playing lore to rigid-body mechanics, including a squirt formalization that precedes the endmass theory of Cross (2008); Alciatore's technical-proof corpus (Alciatore, 2008, n.d.c) maintains the modern statements of the empirical laws together with measured curves and is the largest single anchor source for the battery of Section 4. Cohen (2002) formalized the corner-5 diamond system, but geometrically, without dynamics. Han (2005) introduced the parameterized impulse model of cushion impact for carom billiards that our kernel ships, in corrected form, as `han2005_default` (Section 3); Doménech

(2008) developed non-smooth two-stage models of ball collisions; and Cross (2008) derived the endmass theory of cue–ball deflection that anchors the closed form of experiment B1 (Fig. 7). Analytic treatments of single-shot mechanics continue to appear (Kim, 2022, 2024).

The strongest validation practice of this era is that of Mathavan, Jackson, and Parkin: high-speed imaging to measure billiards dynamics (Mathavan et al., 2009), a cushion-impact theory validated against that footage (Mathavan et al., 2010), and numerical simulation of frictional ball–ball collisions on cloth (Mathavan et al., 2014). Instrumented measurement, quantified comparison error, published parameter sets — this is the discipline our methodology adopts per anchor. It remained, however, validation of single sub-models against the authors' own footage; to our knowledge, no whole-engine, multi-source battery — much less a systematic domains-of-validity study — followed from it. Experiment C4 overlays our rebound-coupling predictions directly on the measured curves of Mathavan et al. (2010, Figs. 7–10) (Fig. 10), and Section 3 records where our pool-specification cushion geometry deliberately departs from their snooker rail profile — a domain separation, not a disagreement.

A parallel thread in multibody impact mechanics feeds the break resolver of contribution C3. Continuous contact-force models with hysteresis damping originate with Lankarani and Nikravesh (Lankarani and Nikravesh, 1990, 1994), and Stronge's energetic coefficient of restitution (Stronge, 2018) supplies the consistency condition our resolver enforces. Applied to pool, Corral et al. (2020) modeled pool-ball interaction with these tools, and Gismeros Moreno et al. (2022) extended the approach to multiple simultaneous

impacts — the closest prior art to C3. The difference is architectural: they integrate the entire scene under a smooth penalty formulation, whereas we confine compliant integration to the frozen-cluster episode inside an otherwise closed-form event stream (Section 3). Cross (2023) treats small frozen clusters analytically, delimiting what conservation laws alone can determine — precisely the degeneracy that makes a compliant resolver necessary at break scale.

2.3. Event-driven simulators and computational players

Leckie and Greenspan (2006) formulated pool physics as event-driven simulation: closed-form ball states between events, with event times obtained as real roots of polynomials of degree at most four. Our kernel descends from this formulation. Leckie and Greenspan also flagged its central numerical hazard — the fragility of polynomial root finding near grazing configurations; we address it with the quartic solver of Orellana and De Michele (2020), incorporating both published corrections (De Michele, 2022, 2024), together with the first-crossing branch selection discussed in Section 6 (contribution C5).

That formulation was built to serve computational players, and a productive lineage followed: the PickPocket search-based player (Smith, 2007), the Deep Green pool-playing robot (Greenspan et al., 2008), and the computational-pool tournaments with their winning entries (Archibald et al., 2009, 2010, 2016); robotics-assistive systems continue the line (Park and Park, 2022). Throughout this lineage the physics substrate is instrumental — a planning model — and was never independently validated against physical measurement; reported evaluation concerns the agent's tournament

performance rather than the simulator's comparison error.

The closest contemporary system is pooltool (Kiefl, 2024), an open-source event-driven billiards simulator distributed as a Python package. Our relationship to pooltool is cooperative: it serves as the cross-simulator oracle of our battery, and where the two model registries coincide we quantify agreement event-by-event (Section 6). Where the simulators deliberately differ — break-cluster resolution, cushion geometry, and the velocity-dependent cushion restitution $e_c(v)$ — each departure is stated with its evidence (Sections 3 and 6). The two codebases also share a reference-implementation lineage, and the case study of Section 6 shows the battery detecting a slip-impulse convention inherited through exactly that shared ancestry (the slip-law defect) — an argument for independent cross-validation even between closely related simulators.

Table 1 condenses this landscape along the axes this paper contributes to: simulation paradigm, published validation posture, determinism guarantees, treatment of the break's frozen-cluster degeneracy, cushion modeling, and reproducibility artifacts. Every cell describing another system transcribes that system's published paper or code as cited; where the cited sources make no statement on an axis, the cell records that rather than a guess. Commercial simulators are omitted as a class — closed systems with unpublished physics and no research API (Kiefl, 2024) — and per-shot runtime comparisons are deferred to the measured setting of Section 3.6.

2.4. Learning-era consumers

Billiards has meanwhile become a standard vehicle for machine-learning research on physical reasoning. Fragkiadaki et al. (2016) trained visual

Table 1: Capability comparison across the open simulators discussed in this section. Cells for other systems transcribe their published papers and code (Kiefl, 2024; Archibald et al., 2009, 2010; Gismeros Moreno et al., 2022; Leckie and Greenspan, 2006); "not claimed" and "not stated" mean the cited sources make no statement on that axis. HLN = Hertz–Lankarani–Nikravesh.

	This work	pooltool	FastFiz	Gismeros Moreno et al.
Paradigm	event-driven analytic; canonical WebAssembly kernel	event-driven analytic; JIT-compiled Python	event-driven analytic; Leckie–Greenspan (2006) lineage	smooth continuous-contact multibody, MATLAB
Published validation	registered multi-source battery with domains of validity (Sections 4–6)	none claimed in the software paper; documented examples reproduce known rules	none published; evaluation targets agent play	qualitative high-speed-video comparison; single-formulation study
Bit-exact determinism	hash-identified canonical binary (Section 3.4)	not claimed	not claimed	not claimed
Frozen-cluster break	bounded compliant HLN episode (Section 3.3)	documented 10% momentum-reduction workaround	not stated	simultaneous impacts by construction; four-ball scenes reported
Cushion model	corrected Han impulse at the WPA nose height; fitted $e_c(v)$ presets	Han 2005 impulse family; constant restitution	not stated	continuous-contact force model; constant restitution
Reproducibility artifact	open-source kernel (MIT; pooltool-derived parts Apache-2.0); bundle with expected output hashes and identity checker, not yet published (Section 7)	open-source package and documentation	unmaintained source archive	none stated

predictive models to play simulated billiards; DeepPHY (Xu et al., 2025) benchmarks agentic vision–language models on physical-reasoning tasks that include billiards; and CueTip (Memery et al., 2025) builds an interactive, physics-aware pool assistant on a simulator backend. The common pattern across all three is that the simulator is trusted as ground truth: whatever model error it carries transfers, unquantified, into trained policies and benchmark conclusions. Contributions C1 and C2 make that transferred error measurable for the cue-sports domain, and the reinforcement-learning program this motivates is deferred to Section 9.

2.5. The reproducibility gap

The standards needed to close the remaining gap exist; they have simply not been applied in this domain. ASME V&V 20 (ASME, 2009) defines the vocabulary — validation variables and points, comparison error $E = S - D$, validation uncertainty u_{val} — that Section 4 adopts while keeping verification and validation strictly distinct; the FAIR principles (Wilkinson et al., 2016) and their research-software extension FAIR4RS (Chue Hong et al., 2022) define the artifact posture. To our knowledge, however, no prior cue-sports simulator reports bit-exact expected outputs, a cross-platform determinism audit, or a machine-readable deviation ledger. The substrate that makes bit-exactness practical is WebAssembly, whose execution semantics are deterministic up to an enumerated set of escapes (WebAssembly Community Group, 2015), each excluded by the kernel by construction (Section 7). This paper accordingly imports the V&V 20/FAIR toolkit into cue-sports simulation and adds the artifact class the toolkit lacks: the hash-identified canonical binary as the physics definition of record (Section 7).

3. The simulator

The instrument under study is an event-driven analytic rigid-body simulator for cue-sports mechanics. Between discrete events, every ball state evolves in closed form; events are detected by root-finding on low-degree polynomials and resolved by a catalog of impact models, each carrying an explicit published provenance and a frozen parameter set. The kernel ships as a single WebAssembly binary that this paper treats as the normative definition of the physics (Section 3.4). Consistent with that posture, this section describes the simulator through its model provenance and its verifiable outputs rather than through source code (public at a later revision; Section 7.1); Fig. 2 summarizes the architecture.

3.1. Event-driven analytic core

The formulation descends from the event-based pool physics of Leckie and Greenspan (2006): rather than time-stepping, the kernel advances each ball along a closed-form trajectory determined by its current motion regime and solves for the earliest time at which any regime transition or contact occurs. Cloth interaction is modeled with three independent friction regimes—sliding, rolling, and spin about the vertical axis—each with its own coefficient, so that a ball's planar trajectory within a regime is exactly quadratic in time and airborne flight is exactly ballistic. The design deliberately excludes air drag and Magnus lift: admitting either would destroy the polynomial character of the trajectories, and with it the guarantee that every event time is a root of a polynomial of degree at most four (the degree-four case arising from ball–ball proximity between two quadratic trajectories).

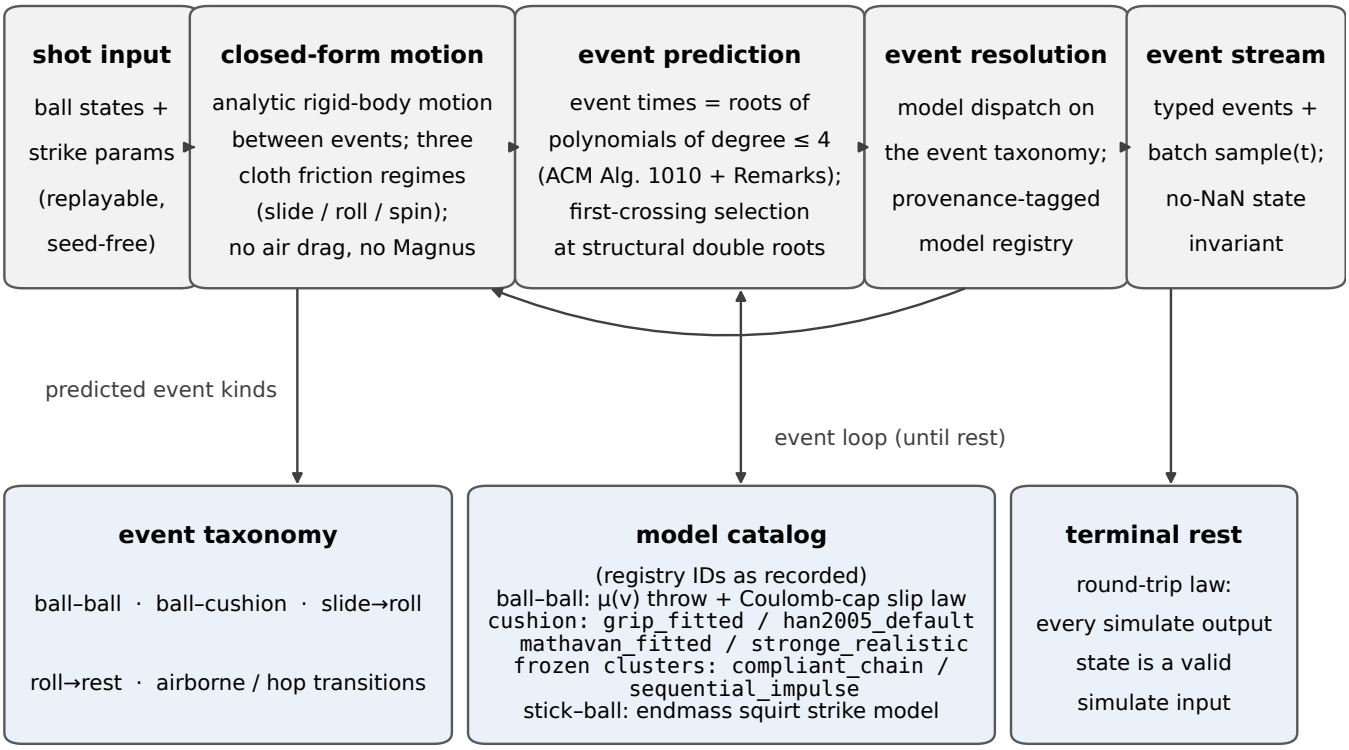


Figure 2: **Overview.** Architecture of the event-driven kernel. Between events, each ball state evolves in closed form under one of three independent cloth-friction regimes (slide, roll, spin) or under ballistic flight; candidate event times (ball–ball, ball–cushion, slide→roll, roll→rest, and airborne transitions) are roots of polynomials of degree ≤ 4, located with the Algorithm 1010 quartic solver (Orellana and De Michele, 2020). Each resolved event dispatches to a provenance-tagged impact model from Table 2; break-scale frozen clusters dispatch to the bounded compliant episode of Section 3.3. Terminal rest projection closes the round-trip law (Section 3.5), and the shipped WebAssembly binary is the physics definition of record (Section 3.4).

Quartic event detection uses the closed-form solver of Orellana and De Michele (2020), incorporating both published corrections (De Michele, 2022, 2024). Leckie and Greenspan (2006) already flagged root-finding fragility as the Achilles' heel of event-driven billiards simulation; the kernel's posture at the residual hard cases—structurally coincident roots at grazing and stick-onset configurations—is a deterministic first-crossing selection rule, discussed as a robustness property in Section 3.5 and, as a case study of a defect this posture replaced, in Section 6. Each resolved event then dispatches to an impact model from the catalog of Section 3.2, and the resulting event stream—not any sampled rendering of it—is the simulator's output of record.

3.2. Model catalog with provenance

Table 2 lists the physical channels, the model identifiers under which they are recorded in every battery output, their published sources, and their frozen parameters. The complete governing equations of every catalog row, with each source's deviations itemized and the frozen constants consolidated, are reproduced in Supplementary Material Section S3. Two doctrinal points govern the catalog. First, a *parameter freeze*: catalog values were frozen before any battery sweep ran, and every subsequent miss became a ledger row rather than a re-tune (Section 4). The worked example is the declared ball–ball restitution $e_b = 0.95$, which differs from the $e = 1$ idealization used by several reference derivations and produces a coherent, declared family of comparison errors across the battery, reported as declared model content, not error (quantified per experiment in Section 5). Second, *provenance honesty*: where the sources themselves drift, both readings are recorded—the kernel's

ball–ball friction curve $\mu(v_{\text{rel}}) = 0.009951 + 0.108\, e^{-1.088\, v_{\text{rel}}}$ follows the current revision of the velocity-dependent throw fit in Alciatore's technical-proof corpus (Alciatore, n.d.d), while an earlier document in the same corpus prints a different decay constant; the discrepancy is ledgered as a source revision drift rather than silently harmonized.

The ball–ball slip-friction impulse uses the Coulomb-cap law (slip impulse $\mu\,(1 + e_b)/2 \cdot v_n$, recorded in outputs as `CoulombCap`). This law is itself an outcome of the paper's methodology: the battery surfaced an inherited miscaled convention (the slip-law defect), the corrected law became the default, and the legacy law is retained behind an explicit option with byte-identical replay proofs (case study in Section 6).

Cushion impact offers one default and three alternative models: the corrected Han carom impulse model as validation default (Han, 2005), a compliant Stronge energetic-restitution alternative (Stronge, 2018), and two fitted equipment presets carrying velocity-dependent restitution responses—the $e_c(v)$ hook that couples rebound speed and spin to approach conditions. Identifiers, frozen parameters, and—importantly—each preset's *declared validity domain* are the corresponding rows of Table 2; the fitted presets are out-of-domain on several US-pool bank and lag observables (quantified in Section 5), and neither was re-tuned.

Cushion *geometry* deserves its own paragraph, because it is a documented trap. The kernel uses the US-pool cushion-nose height of 63.5% of ball diameter, per the WPA equipment specification; Mathavan et al. (2010) place the cushion contact at $7R/5$—70% of ball diameter—which is correct for the snooker equipment their model was fit on and systematically skews US-pool

bank angles if transplanted. We treat this as a domain separation between equipment classes, not a disagreement with the source; the fitted Mathavan preset accordingly pairs with the snooker table configuration, and the bank experiment sweeps nose height over a 0.60–0.67 sensitivity bracket around the specification value to bound the geometric channel. The remaining rows of Table 2—endmass squirt for stick–ball interaction (Cross, 2008; Alciatore, n.d.f; Shepard, 2001) and the three-coefficient cloth model verified against the closed-form kinematics of the technical-proof corpus (Alciatore, n.d.c)—are exercised by experiment groups B and F respectively (Section 5).

Table 2: Model catalog with provenance: physical channel, model identifier as recorded in every battery output, published source, frozen catalog parameters, and validity-domain note. All parameters were frozen before any battery sweep ran; deviations and domain boundaries are ledger rows (Section 4), never re-tunes.

Channel	Model ID	Source	Frozen parameters	Domain note
Ball–ball impact	`CoulombCap` slip law	Alciatore, n.d.d; TP A.5 (Alciatore, n.d.c)	$e_b = 0.95$; $\mu(v) = 0.009951 + 0.108\, e^{-1.088v}$	corrected default after the slip-law fix; legacy law retained as explicit option
Cloth friction	three-regime catalog	TP closed forms (Alciatore, n.d.c)	$\mu_s = 0.20$, $\mu_r = 0.01$; independent spin decay	one parameter set must satisfy lag and bank data jointly (Section 4)
Cushion (default)	`han2005_default`	Han, 2005, corrected	kinematic $e_0 = 0.85$; flat $\mu_w = 0.20$	validation default; aim 1.0 banks 0.11 diamonds beyond the TP B.28 envelope
Cushion (compliant)	`stronge_realistic`	Stronge, 2018	spin ratio 1.8; calibrated $\mu(v_{\text{slip}})$	sliding-bank calibration gaps ledgered
Cushion (fitted)	`mathavan_fitted`	Mathavan et al., 2010 + video joint fit	tangential compliance; $e_c(v)$ response, $ee_0 = 0.80$	in-domain: kick/rebound coupling, snooker pairing; out-of-domain: shallow US-pool banks, lag
Cushion (fitted)	`grip_fitted`	first-party fit	velocity-sloped $e_c(v)$	out-of-domain on banks/kicks in three frames
Stick–ball	endmass squirt	Cross, 2008; Alciatore, n.d.f; Shepard, 2001	per-shaft endmass catalog, 1.7–12.7 g span	commanded tip-center offset convention

Continued on next page

Table 2 *(continued)*

Channel	Model ID	Source	Frozen parameters	Domain note
Break cluster	`CompliantChain` (default); `SequentialImpulse` (legacy)	Lankarani and Nikravesh, 1990, 1994; Stronge, 2018	frozen Hertz stiffness; 10^{-6} s internal step, ≤ 3000 steps per episode	Section 3.3; frozen-pair throw excess open

3.3. Compliant frozen-cluster break resolution (C3)

The break shot confronts an event-driven kernel with its degenerate case. A racked set of balls is a *frozen-contact topology*: in the battery's recorded rack, the apex-wedge pairs sit at a surface separation of $0.5\,\mu$m—far below the $10\,\mu$m cluster-contact threshold—while seeded micro-gaps appear only at back-row boundaries (the battery's seed-0 topology carries one open 0.16 mm boundary between rows three and four, truncating the connected cluster at the cue ball plus six balls). Instantaneous pairwise impulse resolution is ill-posed on such clusters: conservation laws alone underdetermine the outcome of a simultaneous multi-ball impact (Cross, 2023). The failure mode is not hypothetical. Decomposing the break into a serial chain of pairwise impacts starves the multi-ball momentum exchange, and the battery's break experiment explicitly gates on the absence of the resulting defect class, in which the cue ball arrives at the rack and stops nearly dead (a settled squat fraction below 0.05); no such row appears anywhere on the registered grid under the shipped resolver.

The kernel therefore embeds a bounded compliant episode inside the otherwise closed-form event stream. When an impact reaches a frozen cluster, the cluster is advanced with Hertz contact stiffness and Lankarani–Nikravesh hysteresis damping (Lankarani and Nikravesh, 1990, 1994), all contacts integrated concurrently in Jacobi fashion rather than sequentially, with friction applied as force-based Coulomb traction during the episode and restitution kept consistent with the energetic definition of Stronge (2018). The episode is strictly bounded—a 10^{-6} s internal step and at most 3000 steps, with the stiffness and damping constants frozen in the catalog and recorded in each

output's parameter block—after which control returns to the analytic event loop. This placement distinguishes the design from both neighboring approaches. Smooth multibody treatments of pool impacts (Corral et al., 2020; Gismeros Moreno et al., 2022) integrate the *entire* game with penalty time-stepping; we integrate only the degenerate cluster episode, preserving closed-form determinism everywhere else. At the other pole, pooltool documents a 10% momentum-reduction workaround for the break (Kiefl, 2024); the battery realizes that baseline class as the kernel's own `SequentialImpulse` legacy arm—a pairwise resolver retained as an explicit option—so the rejected alternative is runnable like-for-like rather than merely cited.

The model contrast is quantified by experiments `E1` and `E3` (Figures 12 and 13; measured values in Section 5.5): the compliant resolver reproduces the ten-times-fuller kissed-ball slope and the qualitative effective-mass scaling seen in high-speed break footage, both of which the sequential arm erases—and the compliant rows survived the kernel's mid-program re-baseline bit-identically, because force-based Coulomb traction consumes no impulse cap. One compliant-side residual — a speed-dependent frozen-pair combination-throw excess over the measured anchor, with no counterpart in the sequential arm — remains open and is quantified where it is measured (Sections 5.5 and 8).

3.4. Determinism doctrine: the canonical WebAssembly binary

The kernel's determinism contract attaches to a specific artifact: the shipped WebAssembly binary, version stamp `0.1.0+fb3425ad` (SHA-256 prefix `5e3b6ee4`; full hash and manifest in Supplementary Material Section S2), *is* the physics definition of record, and native and Python builds of

the same kernel are declared development conveniences, held to the binary by byte comparison where they are exercised but never themselves normative. The doctrine has one home, Section 7: why WebAssembly can carry the obligation, the stamp-lineage proof connecting the battery's native build to the canonical binary, the replay contract, the determinism audit matrix, and the measured platform sensitivity that makes the binary — not the source — the right canonical artifact.

3.5. Robustness: rest projection and event-detection continuity (C5)

Two engineering properties, sized deliberately below the headline claims, close the loop between the analytic core and the determinism doctrine. First, *terminal rest projection*: every simulation ends by projecting all balls to exact rest states, so that any `simulate` output is a valid `simulate` input—the round-trip law. Combined with the no-NaN invariant, this makes arbitrarily long shot chains (replays, rule adjudication, and the battery's own multi-shot protocols) closed under the API with no accumulating representational debt. Second, *continuity at structural double roots*: where an event polynomial's physical root is structurally coincident—the configuration class Leckie and Greenspan (2006) identified as fragile—the kernel selects the first crossing by a deterministic fixed-resolution scan rather than by bracketing, which is continuous in the incidence parameters by construction. The value of this posture is measurable: the legacy bracketing selection produced the discontinuous bank arrivals of the branch-selection defect, and under first-crossing selection the discontinuous cell count fell from four to zero with the divergence class measured empty on every pinned artifact (case study in Section 6).

3.6. Computational performance

Performance is an engineering property of the implementation, reported here for context; nothing in the paper's verification or validation claims depends on it. All timings in this subsection were measured on one stated host — an Apple M2 Ultra (macOS 26.5, 128 GB), single-threaded on both engines — and are indicative rather than normative: unlike the battery outputs, the performance artifact is hardware-pinned and carries no expected hash.

Table 3 summarizes the two measurements. The first block times the canonical WebAssembly binary of Section 3.4 (stamp `0.1.0+fb3425ad`, executed in Node.js through the same loader the application uses) on the kernel's continuously-gated budget workloads: a typical two-ball shot resolves in about 0.19 ms at the median, a full 15-ball break in about 7.9 ms, the heaviest seeded break workload — a 22-ball snooker break at its worst rack seed — in about 14.5 ms, and the gated any-shot ceiling workload — the same break at the slider-maximum $v_0 = 12\,\mathrm{m/s}$ — in about 16.6 ms, while whole-table state sampling for rendering costs about 5 μs per call. Per-shot cost is therefore three to four orders of magnitude below real time for interactive play, and a full re-simulation is cheap enough to be the replay and adjudication primitive of Section 3.4.

The second block is a matched-scenario comparison against the oracle of Section 6: the 30-shot ORACLE-X A3 grid plus the five committed oracle fixtures, constructed verbatim under the parity-convention model registry. On these shots the two engines' event sequences agree 38/38 *after* the declared time-zero normalization of Section 6 — on 15 rolling-strike cases pooltool

additionally emits a zero-duration sliding-to-rolling transition at the strike instant, which that bookkeeping rule drops from both streams — so the timed workloads are sequence-identical up to that documented rule. Each scenario was timed for 120 repeats after warm-up on both engines — the kernel's native build through its Python binding (binding overhead included), pooltool 0.6.0 at the pinned oracle commit (Python 3.12, numba 0.62) — and the entire measurement was run twice, with run-to-run median drift below 3% on both engines. The kernel resolves these shots at 0.025–0.118 ms per shot at the median (median of medians 0.064 ms) against pooltool's 0.75–2.97 ms (median 1.62 ms): a per-scenario ratio of 22–30× (median 26×). pooltool's steady state deliberately excludes just-in-time compilation, which is a one-time cost of about 13.7 s in a cold-cache process (0.15 s with its default warm on-disk numba cache); the kernel has no corresponding warm-up.

Three caveats keep this comparison honest. First, it is a comparison of implementations on one machine, not of formulations: pooltool is a JIT-compiled Python package whose stated design priorities include speed (Kiefl, 2024), and its absolute per-shot cost of 1–3 ms is itself far below real time; the ratio should be read as characterizing an ahead-of-time compiled kernel against a JIT-compiled interpreted-host package, on this host only. Second, the compared scenario set is the oracle registry's two- and three-ball verification grid, not a break-heavy workload; kernel break costs appear in the first block of Table 3 and no cross-engine break comparison is made, because the two engines deliberately differ in break-cluster resolution (Section 3.3). Third, for the closest prior art on the multibody side, Gismeros Moreno et al. (2022) report — on different hardware, as published — wall-clock costs of

2.4×10^4 to 1.0×10^5 s for 3–4 s of simulated four-ball play under their full MATLAB formulations (440–1,730 s for their reduced variants); we cite their numbers as reported and draw no ratio against them, since neither hardware nor workload is matched. The architectural point stands independently of any specific figure: closed forms between events make per-shot cost proportional to the event count rather than to simulated time, which is what confining numerical integration to the bounded break episode (Section 3.3) preserves.

4. Verification and validation methodology

The experiment battery reported in Section 5 was designed, registered, and executed under a single doctrine: comparisons are stated in ASME V&V 20 vocabulary (ASME, 2009), every acceptance criterion was frozen before any sweep ran, and every miss is recorded in a machine-readable ledger rather than absorbed by parameter adjustment. This section defines the vocabulary, the four evidence tiers the anchors draw on, the registration discipline, the waiver-ledger doctrine, and the cross-simulator oracle protocol.

4.1. Vocabulary: ASME V&V 20

Every validation anchor declares a validation variable, one or more validation points, and a comparison error $E = S - D$, where S is the simulation value and D the source datum (ASME, 2009). V&V 20 composes the validation uncertainty from numerical, input, and experimental-data components, $u_{\mathrm{val}}^2 = u_{\mathrm{num}}^2 + u_{\mathrm{input}}^2 + u_D^2$; Table 4 states, per source family, how each component is estimated in this battery, and the paragraphs below give

Table 3: Computational performance on the stated host (Apple M2 Ultra, single-threaded). Top: canonical-binary budget workloads (`0.1.0+fb3425ad`, Node.js). Bottom: matched-scenario cross-engine comparison on the 35 ORACLE-X shots whose event sequences match 38/38 under the declared time-zero normalization (Section 6); 120 timed repeats per scenario after warm-up, JIT compilation excluded (reported in the text). Values are medians; the full artifact records p95/min, both measurement passes (run-to-run p50 drift $< 3\%$), and host provenance.

Canonical WebAssembly binary, per-shot p50	
Typical two-ball shot	0.19 ms
Carom power shot (3 balls, $v_0 = 7\,\mathrm{m/s}$)	0.09 ms
15-ball pool break	7.9 ms
22-ball snooker break (worst of four rack seeds)	14.5 ms
22-ball snooker break, slider-max $v_0 = 12\,\mathrm{m/s}$	16.6 ms
(any-shot ceiling)	
Whole-table state sample (22 balls, per call)	5.0 μs
Matched-scenario comparison, 35 shots, per-shot p50 median (range)	
This work (native kernel, Python binding)	0.064 ms (0.025–0.118)
pooltool 0.6.0 (steady state, JIT excluded)	1.62 ms (0.75–2.97)
Per-scenario ratio	26× (22–30×)

the estimation procedures honestly, including where a component is zero by construction and where it cannot be estimated at all.

Numerical uncertainty. $u_{\text{num}} \approx 0$ at the scale of every registered band, and we say why rather than assert it: between events ball motion is closed form and event times are polynomial roots polished to machine precision, the kernel's solution-verification floors measure 10^{-16}–10^{-13} in the native observables of each channel (Sections 5.2, 5.6), and the canonical binary is bit-exact across runs and across every platform–runtime configuration the determinism matrix actually tests — V8-based runtimes (Node, Chrome) on macOS arm64 and Linux x86_64, native builds on both operating systems, and the Python-SDK legs the battery itself ran on (Section 7) — so run-to-run and cross-platform variance contribute exactly zero *on the tested configurations*. The claim is scoped to those legs: the matrix's untested cells (non-V8 engines, wasmtime, Windows) are enumerated in Section 8 and are not covered by it; every battery output was produced and replay-verified on tested legs. The one time-stepped component, the compliant break resolver, is step-converged on its battery observables at both energy scales: the E1 kiss-energy retention curve (spread $\leq 6.3 \times 10^{-4}$ under step halving/doubling) and the E3 break-scale squat and effective-mass observables (spreads $\leq 2.8 \times 10^{-4}$ on the squat fraction and $\leq 7.7 \times 10^{-4}$ on M_{eff}/m under the same bracket, with every evidence-band verdict invariant; both post-hoc sweeps recorded in the artifact bundle; Section 5.5). The $u_{\text{num}} \approx 0$ statement carries one further scoped exception: the Mathavan-port cushion family integrates its contact in ΔP impulse steps ($N = 10^3$, first-order bias control) rather than in closed form. Its discretization error is measured directly rather than proxied: halv-

ing and doubling the impulse step around the shipped defaults moves no scored overlay point by more than 0.0022 m/s on speed observables or 0.044° on angle observables (recorded maxima 0.0021 m/s and 0.0431° on the overlay's 4-dp values; the quoted bounds sit one rounding quantum above), with every per-figure within-tolerance count invariant across the three step arms (post-hoc convergence sweep recorded in the artifact bundle; the kernel's own anchor suite independently records ΔP convergence stable to 2×10^{-6} on its scenario family). The 0.005 m/s and 0.07° residuals of the figure-versus-equations source-discrepancy row are a different quantity — the protocol comparison of the kernel against the source's *printed equations*, whose pin tolerance is itself 0.005 m/s — and are not a discretization measurement.

Input uncertainty. The frozen catalog is a declared point choice, not a measured distribution, so no prior input uncertainty was propagated at registration. After registration, one-at-a-time brackets — $e_b \in [0.92, 0.98]$ around the declared 0.95, and the $\mu(v)$ source-revision spread the ledger itself records — measure the input-uncertainty response of every headline domain boundary post hoc; the measured responses are woven into Section 5 and Table 10, and the analyses are declared as post-registration characterization recorded in the artifact bundle (a pinned sensitivity report), never re-tunes. One scoping statement applies to the whole sensitivity pack: unlike the battery itself, whose 16,933 recorded simulations replay end-to-end through the shipped wasm binary (Section 7), the sensitivity scripts ran on the native Python surface only, under the same source-identity argument the pre-replay battery used (the Python environment's kernel source-identical to the canonical binary); they are not part of the wasm replay corpus, and extending the

tap/replay harness to the sensitivity pack is left as an artifact-maintenance item. Relatedly, the battery runs one simulation per grid cell by design: the kernel is deterministic and byte-stable across repeated runs (Section 7), so replicate statistics would measure exactly zero variance; the error bars available to a deterministic simulator are grid quantization and parameter sensitivity, and those are what the brackets report.

Provenance of e_b and its bracket. The declared ball–ball restitution $e_b = 0.95$ is a catalog point choice, and we state its provenance at the precision it actually has: the value sits at the center of the 0.92–0.98 phenolic ball–ball range printed, without attribution, in Alciatore's physical-properties compilation (Alciatore, n.d.a) — a community constant. We could locate no systematic first-hand measurement behind that range: the book-length treatment the community often associates with such constants (Marlow, 1995) is out of print and reached us only through secondary quotations, and the Marlow discussions we could verify (e.g. in Mathavan et al., 2010) concern *cushion* restitution, not ball–ball impacts. No source states an uncertainty for the range, and the battery's own reference forms variously idealize $e = 1$ (TP A-14) or fix $e = 0.94$ (TP A.5's own evaluation). The bracket endpoints $[0.92, 0.98]$ therefore coincide with the compilation's range but are a *declared sensitivity range*, not a measured interval. One modeling asymmetry is deliberate and should be visible: e_b is held speed-constant from the battery's slowest throw-sweep impacts (0.45 m/s ball–ball speed, experiment A3) to its fastest break impacts (12.9 m/s, experiment E3) while the cushion channel carries a velocity hook $e_c(v)$ — we know of no published pool measurement that either supports or contradicts an $e_b(v)$ slope at break speeds. What

the artifact can measure is constant-e_b sensitivity, and we scope it as exactly that: re-running E3 with every contact held at each point of a 62-point grid over $[0.92, 0.98]$ moves the squat fraction by up to 0.0201 and M_{eff}/m by up to 0.0551 — observed grid maxima, not interval bounds: the response is non-monotone and event-discrete in e_b, with an interior jump discontinuity near $e_b = 0.958$ at which one clip-target band membership flips (Section 5.5) — but a speed-dependent $e_b(v)$, which would mix effective values across the break's contact spectrum, is not expressible in the kernel and is not bounded by these constant-e_b runs, so the speed-constancy assumption itself remains a declared limitation (Section 5.5; analysis recorded in the artifact bundle).

Source-data uncertainty. Three distinct procedures produced the source comparisons, and "digitization" honestly applies to only one of them. (1) The cushion-impact figures of Mathavan et al. (2010) were pixel-digitized (2026-07-02) from a 600-dpi `pdftoppm` render of the publisher PDF by pixel-cluster and per-column color sampling, calibrated against each figure's own frame and tick marks, with overlapping markers merged to cluster centroids; the per-series spreads recorded in the shipped fixture are $u_D = 0.01$–$0.04\,\text{m/s}$ on speed observables and 1.0–2.0° on angle observables (the Fig. 7a camera points carry $u_x = u_y = 0.04\,\text{m/s}$). No repeat-digitization study was performed; the recorded values are cluster spreads from the single digitization pass. (2) The technical-proof comparisons are *transcriptions*, not pixel digitizations: each source's printed closed forms, fitted constants, and tabulated values are transcribed and self-checked against the numbers printed in the source PDF (every battery script embeds its transcription check), so their u_D is a transcription floor rather than a pixel spread. What survives for this

family is source-revision drift, which is ledgered as a source row rather than absorbed as uncertainty: where a source exists in mutually inconsistent revisions, both readings are transcription-checked against their own equations (the worked example is the TP A.17 revision row, where TP A.17's printed $\mu(v)$ decay constant of 0.77 conflicts with the 1.088 of the current TP A-14 revision and of the kernel). The five measured TP B.28 bank tables are printed numeric tables transcribed exactly; the source states its measurements were careful but gives no error bars, so no measurement u_D can be estimated for them. What the shipped fixture records instead is *equipment* context: the spread between the two on-system tables is at most 0.25 diamond per point — carried as a brand-agnostic equipment window, with the full five-table envelope as the brand-bias envelope — which characterizes table-to-table variability, not measurement uncertainty. (3) High-speed-video observables carry the capture chain's stated precision — ∼1–2 px centroiding (≈ 0.5–0.8 mm at ball scale) at 320 × 240, with speed *ratios* independent of the frame-rate assumption — and are consumed as evidence bands, not point data. For rule-of-thumb lore anchors (the FAQ charts and practitioner systems: the trisect multiplier, the ten-times-fuller flat retention, the corner-five corrections, the jump chart), the sources state no uncertainty and no measurement conditions, so u_{val} *cannot be estimated*; those rows are labeled lore in Table 10, and their comparisons are characterizations against the rule as printed, not validation with quantified uncertainty. Finally, one anchor family carries a composite *acceptance tolerance* that is deliberately not an uncertainty: experiment C4 accepts a point when $|E| \le u_y + 0.05\,|D|$, where u_y is the recorded digitization uncertainty and the 5%-of-$|D|$ term is an accep-

tance allowance inherited verbatim from the kernel's continuous-integration
655 anchor suite; the two terms are reported separately wherever quoted, and the composite is never called u_{val}.

Table 4: Per-source u_{val} decomposition. $u_{\mathrm{num}} \approx 0$ for every family (bit-exact binary; closed-form solution verification at 10^{-16}–10^{-13} floors) except where noted: the compliant-resolver families carry measured step brackets and the Mathavan ΔP port a measured step-convergence bound ($\leq$ 0.0022 m/s / 0.044° under step halving/doubling). u_{input} is measured post hoc by the one-at-a-time brackets ($e_b \in [0.92, 0.98]$; $\mu(v)$ revision spread); per-boundary responses in Section 5 and Table 10. Lettered marks resolve to the notes at the end of the table.

Source family (anchor class)	u_{num}	u_{input}	u_D (procedure)
Mathavan Fig. 7a camera points (measured)	ΔP stepping ($N = 10^3$): measured bound[a]	incident-speed digitization $u_x = 0.04$ m/s propagated through the local response slope ($\mathrm{d}S/\mathrm{d}V \approx 0.91$): 0.036 m/s (model constants: source's own fit)	$u_y = 0.04$ m/s per point (pixel digitization, cluster spread); combined $u_{\mathrm{val}} \approx 0.054$ m/s
Mathavan Figs. 8–10 numerics (derived; verification-of-port)	ΔP stepping: same measured bound[a]	none: source scheme at source constants	0.01–0.02 m/s speeds; 1.0–2.0° angles (pixel digitization)

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Table 4 *(continued)*

Source family (anchor class)	u_{num}	u_{input}	u_D (procedure)
TP master curves and constructions (derived)	≈ 0	e_b bracket: A2 peak family shifts ∓ 0.88–$0.96°$ per ± 0.03; $\mu(v)$ revision: boundaries $+1.6$–$2.9°$	transcription floor (self-checked against printed values); revision drift ledgered
TP B.28 bank tables (measured)	≈ 0	restitution-dominated: e_0 0.75–0.95 moves arrivals -0.34 to $+0.35$ dia (Section 5.3)	exact transcription; source states no error bars, so measurement u_D is not estimable — the $\leq$0.25-dia on-system inter-table spread is an equipment window, not u_D
High-speed-video catalog (measured, band form)	$\approx$ 0 (compliant-step brackets)[b]	bracketed where consumed (Section 5.5)	1–2 px centroiding ($\approx$ 0.5–0.8 mm); evidence bands, not point data

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Table 4 *(continued)*

Source family (anchor class)	u_{num}	u_{input}	u_D (procedure)
Platinum robot squirt band (measured; consumed as in-sample calibration consistency)	≈ 0	not independently estimable: the catalog's per-shaft end masses were derived from the same Platinum pivot data (Shepard inversion), so the comparison closes a calibration loop	published band (8–14 in), no per-shaft uncertainty stated; convention-dominated
FAQ / practitioner lore (lore)	≈ 0	bracketed where quoted	*not estimable*: no stated uncertainty or conditions; characterization only
Cross-simulator oracle (verification tier)	u_{val} not applicable (code against code); agreement floors 7.4×10^{-14} s, 1.12×10^{-4} ° (Section 6)		

Notes. [a] measured ≤ 0.0022 m/s and $\leq 0.044°$ under step halving/doubling; verdicts invariant. [b] compliant-step brackets: E1 retention spread $\leq 6.3 \times 10^{-4}$; E3 squat/M_{eff} spreads $\leq 2.8/7.7 \times 10^{-4}$.

Verification is kept strictly separate from validation, and the anchors themselves are classified so that the word "validation" cannot lean on sources that are not measurements. Every anchor carries one of three classes, printed per rule in Table 10: a *measured datum* (an instrumented measurement: the Mathavan Fig. 7a camera points, the five TP B.28 bank tables, the high-speed-video observations, the Platinum robot band); a *derived idealized ref-*

erence (a source's own closed form, construction, or numerical model output: the TP 3.3 frictionless-elastic peak, the gearing and trisect constructions, the Mathavan Figs. 8–10 numerics); or *rule-of-thumb lore* (practitioner systems and FAQ charts with no stated conditions). We reserve *validation* for measured-datum comparisons. Comparisons against derived references are *model benchmarking* against an idealized form; and where the kernel implements the reference's own model verbatim, the comparison is *verification of the port* — the central Group A case, since the kernel's catalog implements TP A-14's fitted $\mu(v)$ verbatim, making the A3 master-curve comparison code-against-its-own-source-model (Section 5.1); the same classification governs C4's headline count (Section 5.3). Verification rows in the classic sense compare the code against its own closed forms or against the cross-simulator oracle: the internal worksheet forms embedded in the outputs of experiments A1, A3, A4, A6, A7, and E2 (closed-form re-derivations of the kernel's own impact laws), the D1 tangent-frame criterion, the C3 mirror-symmetry check, and both ORACLE-X ledger rows. Under the classification, the battery's validation content proper anchors to the measured class — the TP B.28 envelope (C1), the Mathavan Fig. 7a camera points (C4), and the frame-tracked high-speed-video bands (E2, E3). The robot band (B1/B2) is measured but consumed as *in-sample calibration consistency*, not independent validation: the catalog's per-shaft end masses were derived from the same Platinum pivot measurements (Section 5.2). Applying the measured-only rule strictly also reclassifies two rows earlier drafts filed as measured: E1's numeric datum is the instructional retention constant 0.10 (the video documents the scenario, not a measured retention), and F3 carries no numeric video anchor

(its airborne-cut reference is analytic contact geometry) — both are benchmarking. They join the A2 peak band, the trisect multiplier, and their kin as benchmarking against derived forms and lore, labeled as such wherever quoted. Verification of the numerical core itself (quartic event detection, the terminal-rest round-trip law) is described with the simulator in Section 3.

One boundary of the standard deserves an explicit statement. V&V 20 quantifies validation uncertainty *at* the validation points and declares extrapolation beyond them out of scope: the standard declines to certify model behavior where no measurement exists. That excluded question—over what range of inputs an empirical law keeps holding—is precisely this paper's thesis. The battery is therefore designed to sweep across each rule's presumed domain and to report where E grows beyond the registered bands, turning the standard's scoping boundary into the object of study.

4.2. Four-tier anchor discipline

Each acceptance criterion anchors to one of four evidence tiers: (i) published sources—further classified per anchor, under the taxonomy above, as measured data (the Mathavan Fig. 7a camera points, the five measured through-diamond bank tables of TP B.28 (Alciatore, n.d.h)) or derived reference forms (the transcribed technical-proof master curves and constructions (Alciatore, n.d.d), the Mathavan Figs. 8–10 numerics (Mathavan et al., 2010)); (ii) high-speed-video catalog observations (Alciatore, n.d.b); (iii) a published specification window—the WPA lag criterion of 4.0–4.5 table lengths, which earlier drafts of this program filed as "first-party table-session data" and which we correct here: it is a specification anchor, not a measurement of ours, and the first-party table-session captures

remain staged under quality assurance (Section 8) and are consumed by *no* registered anchor in this battery; and (iv) a cross-simulator oracle, pooltool (Kiefl, 2024) at a pinned commit, described below. Tiers and classes are declared per anchor, so a reader can weight each verdict by the strength of its source.

The discipline is deliberately anti-overfit: the default constants the battery scores were frozen before any battery sweep ran — unchanged since their 2026-07-01 introduction, with the last calibration touching the family nineteen days before registration; the per-literal chronology, including the younger and not-static equipment-catalog registry, is stated in Section 4.3 — and one parameter set must satisfy independent datasets simultaneously. "Out-of-sample" is scoped precisely in Section 4.3: it means no re-tuning against any battery observable, not blindness, since most anchors pre-existed the registration as continuous-integration tests. The worked example is experiment F1 (slot F13, Fig. 14): the same cloth catalog must reproduce both the lag window (a pass) and the bank-table dataset (a registered miss of the frozen all-six criterion), with neither dataset re-tuned against the other — quantified once, in Section 5.6. Where the joint requirement fails for a fitted preset, the failure is ledgered as a validity domain rather than repaired by re-tuning. Source-data uncertainty is recorded per anchor under the per-source procedures and magnitudes of Table 4: pixel-digitization spreads for the Mathavan figures, transcription floors for the technical-proof forms and tables, capture-chain precision for the high-speed-video bands.

4.3. Frozen registered designs

Every experiment received a registered design—sweep grid, measured output, and acceptance or reporting criterion—committed on 2026-07-21, *before* any sweep ran. The registered battery comprises 24 experiments in six groups (A1–A7, ball–ball impact and throw; B1–B4, stick–ball; C1–C4, cushion and diamond systems; D1–D3, curved paths; E1–E3, frozen contacts and the break; F1–F3, kinematics closed forms and the jump) plus the ORACLE-X cross-run, i.e. 25 pinned outputs. Roughly 60% of the registered rules were already covered by standing continuous-integration anchor tests; those experiments promote the anchors to figure grade rather than re-derive them, and anchors whose calibration targets fail are retained as documented ignored gates—never deleted—with each failure carried as a ledger row (e.g. the steep rolling-bank and Han banking targets of Section 8).

Registration provenance is stated here from the version-control history, because several of its facts qualify the out-of-sample claim. (1) *Defaults with the first anchors; the registry after them; both long before registration.* The four default constants the battery scores ($\mu_s = 0.20$, $\mu_r = 0.01$, $e_b = 0.95$, cushion $e_0 = 0.85$) entered the kernel as literature-sourced defaults at the kernel scaffold commit (2026-07-01), the same day the first continuous-integration anchor suite — including the lag anchor — was created; the banking anchor suite followed two days later (2026-07-03). Checked per literal against the version history, none of the four default literals has changed since that introduction; later commits only added preset entries alongside them (fitted cushion presets, 2026-07-02; cloth presets, 2026-07-04). The `EquipmentCatalog` registry that packages these constants is younger and

was *not* static: it was introduced on 2026-07-04, three days *after* the first anchors, and its entries were revised through 2026-07-12 (catalog v8) — nine days before registration — so the freeze claim is scoped to the four battery-scored defaults, not to the registry as a whole (a post-registration catalog v9 added two visual-variant ball sets whose physics parameters byte-copy their existing class). One catalog value is explicitly anchor-informed: the flat rail-friction intercept $\mu_w = 0.20$ was pinned by a 2026-07-02 calibration whose hard constraint was the kick-anchor windows — the same anchor family that experiment C3 later promotes to figure grade — so C3's verdict is calibration-consistent rather than out-of-sample, and its results paragraph says so. None of the four defaults (nor μ_w) changed between that calibration and the 2026-07-21 registration — nineteen days — and none has changed since. (2) *Blind versus anchor-informed bands.* The registration index labels every design's prior coverage, and we restate the split here rather than leave it at "roughly 60%": six designs promote standing anchors (A3, B1, C1, C3, C4, F1), eight extend partial coverage (A2, B3, C2, D1, D3, E3, F2, F3) — fourteen of 24 — and ten were registered blind, with no prior kernel evidence on the observable (A1, A4–A7, B2, B4, D2, E1, E2). For the anchored and partial rows the registered bands were set by authors who already knew how the kernel scored against the standing anchors: "frozen before any sweep ran" is exact for the battery's sweeps, but it is not blindness for those fourteen rows, and both halves of the flagship one-parameter-set exhibit of Section 5.6 (lag, banks) pre-existed as anchors during kernel development. What the freeze certifies is narrower and checkable: no catalog parameter and no acceptance band moved after registration, including after misses were measured. (3) *The*

22-to-24 amendment. The registration index was drafted at 22 designs; A6 and A7 were added during the same registration pass, and the index as first committed (2026-07-21) already carries all 24 — the "22-experiment" count survives in that commit's message and in an internal planning document, and we disclose the amendment here. No design, A6/A7 included, was registered after any sweep of any experiment had run: the first battery scripts landed 43 minutes after the registration commit, and the A6/A7 scripts the following day. (4) *Attestation boundary.* All of these timestamps are commits in an author-controlled development repository whose history is not public (the public repository's history begins at a later snapshot); this is *internal* registration, not an external timestamping service, and a reader must weigh it as self-attested. We state the dates (registration, 2026-07-21; battery-script commits, two on 2026-07-21 and three on 2026-07-22), and we scope every out-of-sample claim in this paper to what this history supports.

Deviations from a registered design are themselves declared and ledgered rather than silently absorbed. Two worked examples: experiment A4 registered an english grid of $-100..100\%$ but executed $0..100\%$ in steps of 2.5%—sign-symmetric up to squirt direction, so the negative arm adds no information on the locus under study—declared in the output's parameter block; experiment E3 executed the elevation subset $\{0, 3\}^\circ$ of its registered 0–5° range and realized its registered "momentum-theft baseline" as the kernel's own sequential-impulse legacy arm, the external hack being a documented rejected alternative.

Every experiment script emits a self-describing output: the kernel version hash, the full shot parameters, fixture identifiers, and a canonical event-log

hash, following the FAIR and FAIR4RS principles (Wilkinson et al., 2016; Chue Hong et al., 2022). The complete mapping from figures and tables to pinned outputs is given in Supplementary Material Section S1. When the kernel itself changed mid-program (the two inherited-defect fixes discussed in Section 6), the entire battery was re-baselined under the new pin with per-row drift characterized on byte-verified endpoints, rather than selectively re-run (Section 7).

4.4. Waiver-ledger doctrine

V&V 20 is explicit that validation is not a pass/fail exercise (ASME, 2009): a missed band with a characterized mechanism is a result, not a failure to be engineered away. The battery operationalizes this with a machine-readable waiver ledger (38 rows at the time of writing). Each row records the anchor identifier, the observed delta against its tolerance, a physical rationale, a disposition, a tracking reference, and the kernel identity whose artifacts produced its numbers. Nothing is ever silently re-tuned: a registered band that is missed stays missed in the record, with the miss quantified. Table 5 summarizes the nine row categories.

Table 5: Waiver-ledger categories and row counts (38 rows total; the machine-readable ledger belongs to the artifact bundle of Section 7).

Category	Rows	Meaning
Resolved	9	Formerly ledgered deviation resolved by a landed fix (including ledger-evaluation corrections); retained with the resolving reference
Registered-band miss	7	A registered acceptance band or source claim was missed; measured and characterized, band not re-tuned
Declared model content	7	Deviation is declared catalog parameter or design content (or a declared registration deviation), not a defect
Convention sensitivity	4	Result depends on an input or measurement convention; both readings recorded, one adopted with rationale
Source discrepancy	3	The source material itself carries the inconsistency (revision drift, figure-vs-equations artifact, unspecified-condition lore)
Preset domain	3	A fitted equipment preset is outside its validity domain on this observable; domain documented, preset not re-tuned

Continued on next page

Table 5 *(continued)*

Category	Rows	Meaning
Open investigation	3	Characterized residual or queued maintenance with an open tracking reference
Frame artifact	1	The registered criterion was frame- or definition-sensitive; the physics verifies exactly in the correct frame
Oracle waiver	1	Cross-simulator tolerance exceedance attributed to a documented oracle-side mechanism

The ledger's maintenance law keeps it honest over time: row identifiers are stable and rows are never deleted; a fixed deviation moves to the resolved category with the resolving reference retained; and when a ledger *evaluation* is itself found wrong, the row is recategorized in place with the correction recorded inside it (two such corrections exist in the ledger, one where the original entry had scored the wrong observable and one where a one-sided evidence bound had been misread as a two-sided band).

One decision rule separates repair from disclosure, and it was applied uniformly. A deviation is *fixed* only when root-cause analysis identifies a mechanism error — the implementation demonstrably deviating from its own intended mathematical law — and every fix must land behind a preserved legacy option, carry written predicted shifts registered before the re-run, and trigger a full battery re-baseline (Section 6). A miss with no such diagnosed mechanism — the model disagreeing with external data or lore — is *ledgered*

and never re-tuned. A discrepancy traced to a defensible derivation or convention choice rather than an error is recorded as a convention row, not fixed; the permanent oracle parity pin is the standing example. Applied to the nine resolved rows: two are mechanism fixes under the rule (the slip-law defect, a mis-scaled Coulomb-cap slip impulse inherited from the reference lineage; the branch-selection defect, a numerically discontinuous branch selection at a structural double root); six are script-bookkeeping re-pins that changed no measured column; and one is a ledger-evaluation correction in which the original row had scored the wrong observable. The seven registered-band misses stayed misses because none has a diagnosed mechanism error: the banking direction reversals, for instance, are candidate model-form limits of the affected presets, not identified implementation defects, and they remain ledgered with exactly that framing (Section 8).

Two rows illustrate how the doctrine reads in practice, each quantified once in Section 5. First, the registered C1 criterion requires all six through-diamond bank aims inside the measured TP B.28 envelope; the default cushion model is inside on five of six, reported as a miss under the frozen criterion even though the shortfall is smaller than the source's own table-to-table equipment spread — context that is recorded, not invoked as u_{val} to excuse the criterion (Section 5.3). Second, the same mechanics turn misses into findings: the trisect-rule row, a registered-band miss in ledger terms, is simultaneously the paper's flagship domains-of-validity result (Section 5.6). This stance follows the non-pass/fail reading of V&V 20 (ASME, 2009), and it is consistent with the value that artifact-review practice places on documented non-reproduction (the ACM artifact review and badging framework)

and with CODECHECK's principle of recording, rather than fixing, discrepancies found during independent execution. The open rows are restated as limitations in Section 8.

4.5. Oracle protocol

Wherever the two simulators' model registries coincide, pooltool (Kiefl, 2024) at a pinned commit serves as a cross-simulator oracle (experiments A3, A6, D1, and F1/F2; the matched registry runs Han-model cushions only). The comparison is event-by-event: both simulators emit their event streams for identical shot parameters, and event times, event sequences, and post-event states are compared point by point. The registered tolerances are $\leq 1\,\mu$s on event times and ≤ 0.1 mm on rest positions where models coincide. Exceedances are governed by waiver bounds, not discretion: the oracle suite carries an event-level waiver fixture with per-observable ceilings, and every tolerance exceedance must name an attributed oracle-side or convention-side mechanism to be waived. The measured agreement, and the six waived exceedances with their single attributed oracle-side mechanism, are reported in Section 6 (slot F14, Fig. 15).

One convention rule is permanent: oracle parity is pinned to the *legacy* slip law (the convention pooltool shares by inheritance, corrected in our kernel by the slip-law fix), keeping the oracle channel a like-for-like verification instrument across kernel evolution (rationale and the measured convention gap in Section 6).

5. Results: the empirical laws of pool

This section reports the battery group by group. All quantities are post-cascade baselines under the canonical kernel `0.1.0+fb3425ad`; the battery itself executed on a native build whose kernel sources are git-identical to the canonical binary (the source-identity proof travels with the baseline manifest), and per-figure provenance is tabulated in Supplementary Material Section S1. Comparison errors follow the $E = S - D$ convention of Section 4, verification results are labeled as verification, and every empirical rule carries a domain-of-validity statement whose one-row summary is collected in Table 10; deviations are recorded in the study's waiver ledger (categories in Table 5). Figure slots F2–F13 are referred to as "slot Fn" to keep the slot namespace distinct from experiment IDs F1–F3.

Two figure conventions hold throughout the paper. First, every figure caption opens with a bold claim-type prefix so the epistemic weight of the artwork is visible before its content: **Validation** (comparison against measured data or observation), **Verification** (code against closed forms, its own source model, or another code), **Comparison** (benchmarking against derived reference constructions, conventions, or practitioner lore), and **Performance** (reserved for computational measurements); the architecture schematic carries the neutral prefix **Overview**. Second, registered ledger rows visible in artwork use a single waiver-marker language: an orange (Okabe–Ito vermillion) open ring on the affected data point — or, where no single point applies, an orange panel-anchored note naming the row's ledger category. Table 6 maps every figure slot to its claim type and experiments.

Table 6: Figure roadmap: every figure slot with its claim type and experiment IDs. Slot F0 is the battery-wide verdict map of Section 1; slots F2–F13 appear in this section; slot F14 in Section 6. Slot names F0–F14 are a distinct namespace from experiment IDs F1–F3.

Slot	Claim type	Experiments	Where
F0	Validation (summary map)	battery-wide	§1
F1	Overview (architecture)	—	§3
F2	Verification	A3	§5.1
F3	Comparison	A1, A2	§5.1
F4	Comparison	A4–A6	§5.1
F5	Comparison	A7	§5.1
F6	Verification	B1–B4	§5.2
F7	Validation	C1	§5.3
F8	Comparison	C2, C3	§5.3
F9	Validation / verification-of-port	C4	§5.3
F10	Verification	D1–D3	§5.4
F11	Validation / comparison	E1, E2	§5.5
F12	Validation	E3	§5.5
F13	Comparison	F1–F3	§5.6
F14	Verification	ORACLE-X	§6

5.1. Group A — ball–ball impact and throw

Seven experiments probe the ball–ball channel: stun separation (A1), the 30° rule (A2), cut-induced throw (A3), spin-induced throw (A4), gearing english (A5), spin transfer (A6), and the chalk-mark ("cling") envelope (A7). Each design records its strike convention and gates it per run (stun via the recorded pre-impact roll ratio, natural roll via a roll-in check), and E is evaluated at the first ball–ball event against the source forms (Alciatore, n.d.c) at kernel-measured impact conditions. Two threads run through the group. First, under the corrected Coulomb-cap slip-impulse law (the slip-law fix; Section 6), slip-regime throw now matches the source forms essentially exactly. Second, the declared $e_b = 0.95$ ball–ball restitution versus the sources' $e = 1$ idealizations produces a coherent ∼2.6% family of stick-regime differences, declared as catalog content rather than model error.

The 90° and 85° rules (A1).. The ideal stun-separation rule (TP 3.1: cue ball and object ball depart at exactly 90°) and its real-ball refinement (≈ 85°) were swept over cuts 5–80° at impact speeds 0.5–3 m/s with stun enforced at impact (64 runs; Fig. 4a). Measured separation never exceeds 90°, and the deficit is largest exactly where the taught rule is silent: for cuts of 15° and below (nearly full hits) the impact is friction-capped, and separation collapses 8.5° (at 15° cut) to 19.2° (at 5°) below the ideal — identically at all four speeds, because the stick regime is speed-free. At the half-ball reference the separation runs 82.49° (0.5 m/s) to 86.37° (3 m/s), a 3.88° speed spread the rule does not carry; the speed-averaged value 84.63° sits 0.37° below the informal 85° figure, and TP A.5's own fixed-parameter evaluation (83.259° at $e = 0.94$, $\mu = 0.06$) is met within −0.77° at slow speed, the residual being the

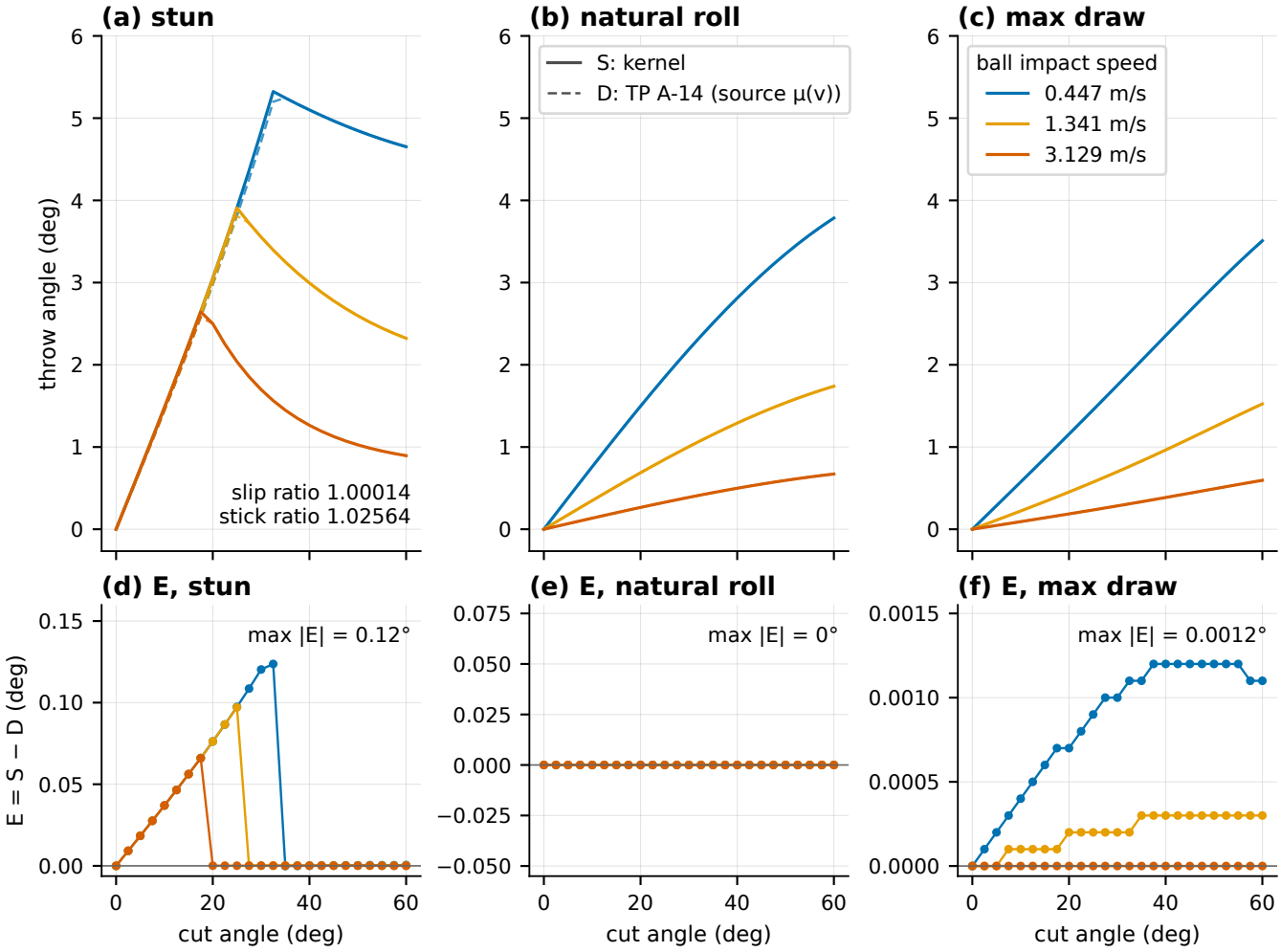


Figure 3: **Verification.** Cut-induced throw (experiment A3): kernel throw versus cut angle at ball-impact speeds $\{0.447, 1.341, 3.129\}$ m/s for the (a) stun, (b) natural-roll, and (c) maximum-draw states (225 runs), overlaid on the TP A-14 master forms (Alciatore, n.d.d) — the source's own fitted $\mu(v)$ model evaluated at kernel-measured impact conditions, so this comparison is code-against-source-model verification rather than validation against a measurement. Panels (d)–(f) make the comparison error visible: $E = S - D$ per state, with the per-family maxima annotated. The panel-(a) callout gives the slip- and stick-regime tan-argument ratios — the slip Coulomb-cap expectation is 1.0 and the stick ratio is $2/(1 + e_b)$, the declared e_b channel; values in the text.

declared parameter difference ($e_b = 0.95$, velocity-dependent $\mu(v)$). Straight-in rows verify the degenerate closed form — cue-ball residual speed fraction 0.02498–0.02500 against $(1 - e_b)/2 = 0.025$ — and are bit-identical across the kernel cascade. On the verification side, the mean slip-regime friction ratio over the 49 slip-classified impacts is 0.99988 against the Coulomb-cap law (expected 1.0) and 1.02612 against the retired legacy overlay (expected $(1 + e)/2e = 1.02632$) — direct evidence of which impulse law the kernel implements; the experiment's worksheet reproduces the kernel to 0.0017°, with the legacy form retained as a documented overlay at 0.1293°, the measured size of the slip-law shift on this grid. Domain statement: the 90° rule is an upper bound approached only for thin, fast cuts (deficit 0.87° at 80°/3 m/s); the 85° half-ball refinement is accurate on average but carries a −2.5° to +1.4° speed dependence over 0.5–3 m/s; below a 15° cut both rules fail by construction.

The 30° rule (A2).. Rolling-ball carom deflection θ_c was measured against TP 3.3's closed form (the friction-free, perfectly elastic ideal in the lineage of Wallace and Schroeder, 1988) over ball fractions 0.05–0.95 in steps of 0.025 at 1.0, 1.5, and 2.0 m/s with natural roll gated per run (111 runs, none censored; Fig. 4b). The source pins the peak exactly: 33.749° at ball fraction 0.529. Peak *location* is speed-robust — refined peak fractions 0.532, 0.526, and 0.523, inside the registered 0.53 ± 0.02 band at every speed. Peak *height* is not: 33.72° at 1.0 m/s ($E = -0.03°$) lies inside the registered $33.7 \pm 0.3°$ band, but 33.18° at 1.5 m/s and 32.92° at 2.0 m/s miss it ($E = -0.57°$ and $-0.83°$). The miss is characterized, not re-tuned: the reference form carries no restitution or friction loss, and the deficit grows with speed because the

absolute tangential impulse grows even as $\mu(v)$ decays. The 27–34° plateau claim holds at all speeds — measured quarter- to three-quarter-hit plateau widths 0.512, 0.496, and 0.488 in ball fraction versus the ideal 0.516 — and θ_c never exceeds 34° anywhere on the grid (grid maximum 33.7103°). The peak-height band is thus a slow-speed statement of the source form (Table 10). The post-registration one-at-a-time brackets of Section 4 restate this verdict at its honest precision. The robust content is the *speed-ordered departure family*: in every bracket arm the peak height declines monotonically with speed, by 0.75–1.00° from 1.0 to 2.0 m/s, and the peak-location band (0.53 ± 0.02) holds at every speed in every arm (refined fractions 0.517–0.539). The fragile content is band *membership*: a ±0.03 change in the declared e_b translates the whole peak-height family by $\mp$0.88–0.96°, and the $\mu(v)$-revision spread by +0.33–0.53°, so which single speed lands inside the ±0.3° band flips across arms (none at $e_b = 0.92$; 2.0 m/s only at $e_b = 0.98$; 1.5 m/s only under the 2005-era decay constant). We therefore quote the registered in-band-at-1.0-m/s verdict together with its e_b-fragility, and rest no conclusion on the band membership alone.

Cut-induced-throw master curves (A3).. The group's central comparison — *verification of the port*, not validation: the kernel's catalog implements TP A-14's fitted $\mu(v)$ verbatim, so this is code against its own source model (Section 4) — sweeps cuts 0–60° at ball-impact speeds $\{0.447, 1.341, 3.129\}$ m/s under stun, natural roll, and maximum draw (225 runs) against the TP A-14 master curves (Alciatore, n.d.d), both as the source's printed master forms (transcribed and self-checked, Table 4; no pixel digitization) evaluated at nominal conditions and as the closed form evaluated at kernel-measured im-

pact conditions (Fig. 3). Under the corrected slip law the slip-regime throw is the TP A-14 form itself: the mean tan-argument ratio over 132 slip impacts is 1.00014 (pre-fix legacy value 0.9745; the collapse of this ratio to unity was a registered prediction of the slip-law fix, Section 6), while the stick-regime ratio $1.02564 = 2/(1 + e_b)$ is exactly the declared restitution difference. Comparison errors: the roll and maximum-draw families close at $\max|E| \leq 0.0012°$ (maximum draw, slow) — roughly two orders of magnitude below the pre-cascade baseline of 0.0886° — and the stun family's $\max|E| = 0.1237°$ occurs at the two grid cells where kernel and source disagree on stick/slip classification (the $(1 + e_b)/2$-scaled cap boundary). The rule's features quantify as follows. (i) The slow-stun peak reaches 5.32° near the half-ball hit, confirming the taught "maximum ≈ 5°". (ii) The ≤ 20°-cut speed-collapse rule (the 1/7 friction cap) holds strictly to 17.5° on this grid: the spread across speeds is exactly zero at cuts ≤ 17.5° and 0.55° at 20°, the fast curve leaving the cap between 17.5° and 20° — consistent with the source's own ≈ 18° crossover. Stated with its brackets: at the 2.5° grid step the measured crossover is the interval $(17.5°, 20.0°]$; the closed-form cap-departure boundary at the fast speed evaluates to 18.0°, moving only to 17.9–18.2° over $e_b \in [0.92, 0.98]$ but to 20.9° under the $\mu(v)$-revision spread, under which the grid-measured collapse extends to 20.0° (the slow-speed half-ball throw spans 4.76–4.91° over the e_b bracket). (iii) The "∼4× throw reduction under roll or draw" holds at medium and fast speed (medium-speed roll ratios 3.5–4.6) but relaxes to ≈ 2.2–2.3× at slow speed, where $\mu(v)$ saturates — a validity boundary of the rule, not a kernel miss.

Spin-induced-throw maxima (A4).. The TP B.25 and FAQ speed-effects claims (Alciatore, n.d.g,n) were swept over outside english 0–100% in steps of 2.5 (percentage points of maximum sidespin, pp), cut angles 0–45°, and the three A3 speeds (492 runs); the registered −100..100% grid was executed as 0..100%, a declared deviation — straight-shot SIT is sign-symmetric up to squirt direction. All four registered claims hold against the source's derived constructions (Fig. 5). The peak locus sits at stun + slow + ≈ 50% english: the measured peak on the slow straight shot is at 47.10% of maximum sidespin — inside the registered 50 ± 5% band, 3.06 pp below the source's closed-form construction (50.16%) and 2.07 pp below the same construction at kernel constants — with peak SIT 5.15° against the source's 5.12°. Below 25% english, SIT is speed-independent to 0.021° (the stick branch makes the claim exact; the spread is 0.09° at 25% itself, where the fast curve reaches its branch point). Beyond the peak, SIT declines monotonically at every speed (SIT at 100% english over peak: 0.755 measured versus 0.768 for the source form, slow). The required-english locus rises monotonically with cut angle at every speed, tracking the source construction within 3.7 pp over the 0–45° locus grid while the SIT value at the peak matches within 0.04°. The SIT maximum is a branch-boundary feature — it sits at the measured stick-to-slip kink — so its location inherits the $(1 + e_b)/2$-scaled cap boundary, about 1 pp below the $e = 1$ source kink; gearing cross-checks agree with the $80\% \sin\phi$ rule within 2.01 pp (see A5).

Gearing english (A5).. TP A.26's zero-throw construction (Alciatore, n.d.e) prescribes the gearing english percentage $PE_g(\phi) = 80\% \sin\phi$ — 80/60/40% english at 0/25/50% ball fraction. Over the seven-point fraction ladder (1/8–

7/8, at 1 m/s), the measured zero-throw and zero-spin-pickup crossings coincide to ≤ 0.0075 pp of english: the two definitions of "gearing" are operationally one crossing. Against the rule evaluated at the measured cut angle at impact, $|E| \leq 0.13$ pp everywhere (largest at the thinnest hit); against the nominal aimed fraction, E runs -0.29 to -1.29 pp, the difference being default-cue squirt shifting the delivered cut — an input convention, not mechanics. The half-ball crossing is speed-invariant to 0.18 pp over 0.5–2.0 m/s, and the spin-rate-factor identity underlying the construction verifies directly to $|E| \leq 3.2 \times 10^{-4}$. These zero-crossing anchors are bit-identical across the slip-law kernel cascade — gearing zeroes the contact slip, so a slip-impulse rescale cannot move them (the predicted invariance) — while the off-gearing slip throw rescaled exactly as predicted (half-ball sanity throw $4.5222° \to 4.6407°$). The construction is thus reproduced essentially exactly once input conventions are stated; squirt compensation is the only correction the rule needs (Table 10).

Spin transfer (A6).. TP A.27's friction-capped transfer curves were measured from object-ball spin immediately post-collision over english × speed grids on straight and 30°-cut legs (Fig. 5). The source's only exact printed value is the stick-branch plateau $\mathrm{STP}_{\max} = 5/14 = 35.71\%$; the kernel transfers exactly 5/14 of the contact slip, and the measured plateaus of 36.79–36.81% sit a speed-independent +1.1 pp above the nominal form — an offset fully accounted by default-cue squirt tilting the contact kinematics (E against the form at measured kinematics: -0.003 to -0.012 pp). On the slip branch, the transfer ratio against the $e = 1$ worksheet is $0.97508 = (1 + e_b)/2$ exactly — the corrected slip law's predicted signature (the pre-fix value was

0.95007 = e_b). The measured stick-to-slip kink sits 1.2–3.6 pp of english below the source's $e = 1$ kink, consistent with the same scaled cap boundary seen in A3/A4, and within +0.2 to +0.5 pp of the kernel-law boundary at measured kinematics. The cut-30° transferred-spin zero crossing reproduces A5's gearing point (spin-rate-corrected $|E| \leq 0.01$ pp; within 0.50 pp of A5's crossing range) — two independent observables, one construction. One bookkeeping item remains open: this experiment's internal verification worksheet still transcribes the retired legacy slip law, leaving an ungated residual of 0.3146 rad/s that equals the slip-law shift by design; its re-pin is queued.

Cling envelope (A7).. Chalk-spot ("cling") contact friction was modeled by scaling the ball–ball friction curve by $k \in \{1, 2, 3\}$ (TP A.17's scale factor) over the A3 stun grid at cuts 10–50° and the three standard speeds (45 runs; the $k = 1$ arm is byte-identical to A3's stun rows). The extra throw $\delta(k) = S(k) - S(1)$ is published as the aiming-rule uncertainty band (Fig. 6): the worst cell is +5.15° at medium speed, 50° cut, $k = 3$, with the slow cell (+5.06°) within 0.1° and fast topping out at +3.14°. On the source's own total-throw observable there is no speed-ordering reversal anywhere on the grid — at $k = 3$/50°, slow 9.90° > medium 7.75° > fast 3.08° — and an earlier ledger reading to the contrary, evaluated on the δ observable, was retired on re-evaluation (a resolved ledger row; the δ ordering survives only as a subordinate envelope remark). The sweep adds a speed-boundary refinement to the source: friction independence below the cap-departure cut is exact ($\delta = 0$ to machine precision), but the boundary itself is speed-dependent — 32.5° at slow, 25.0° at medium, and 18.0° at fast speed on the kernel law (closed-form values 32.45/24.99/18.02°, moving by at most ±0.34/0.24/0.15°

over $e_b \in [0.92, 0.98]$ and to 34.1/27.7/20.9° under the $\mu(v)$-revision spread; the 10°-grid verification brackets are $[30, 40]$, $[20, 30]$, and $[10, 20]$°) — so TP A.14's "independent below about 30°" is a slow-speed statement; at fast speed the safe zone shrinks to $\approx 18°$. In practical terms, the first cut at which a $k = 3$ chalk spot adds $\geq 0.5°$ of throw moves from 20° (fast) to 40° (slow). One source discrepancy is recorded rather than resolved: TP A.17's printed $\mu(v)$ fit uses decay constant 0.77 (2005 era) while the current TP A.14 revision and the kernel use 1.088; both documents transcription-check against their own forms, and the difference is carried in u_{val}. The slip and stick branch ratios (1.00006 and 1.02564) are invariant under k, as the scaling multiplies μ identically in S and D.

Taken together, group A locates the throw family's first failure axis at shot speed — behavior traced to the velocity dependence of ball–ball friction (Marlow, 1995; Alciatore, n.d.d) rather than to any single-point miss; the boundaries seed the corresponding rows of Table 10.

5.2. Group B — stick–ball

The stick–ball channel is the battery's cleanest. All four outputs are physics-identical across the slip-law kernel cascade (the strike resolves before any ball–ball impact), and because the kernel's strike model implements the published closed forms directly (Alciatore, n.d.f; Cross, 2008; Shepard, 2001; Coriolis, 1835; Alciatore, 2008), this group is verification in the V&V 20 sense: machine-precision agreement is the *expected* result, framed as implementation evidence, not physics validation (Section 4). The group's measured content lives instead in its protocol conventions — and one scoping note governs the whole group: the Platinum comparisons are *in-sample calibra-*

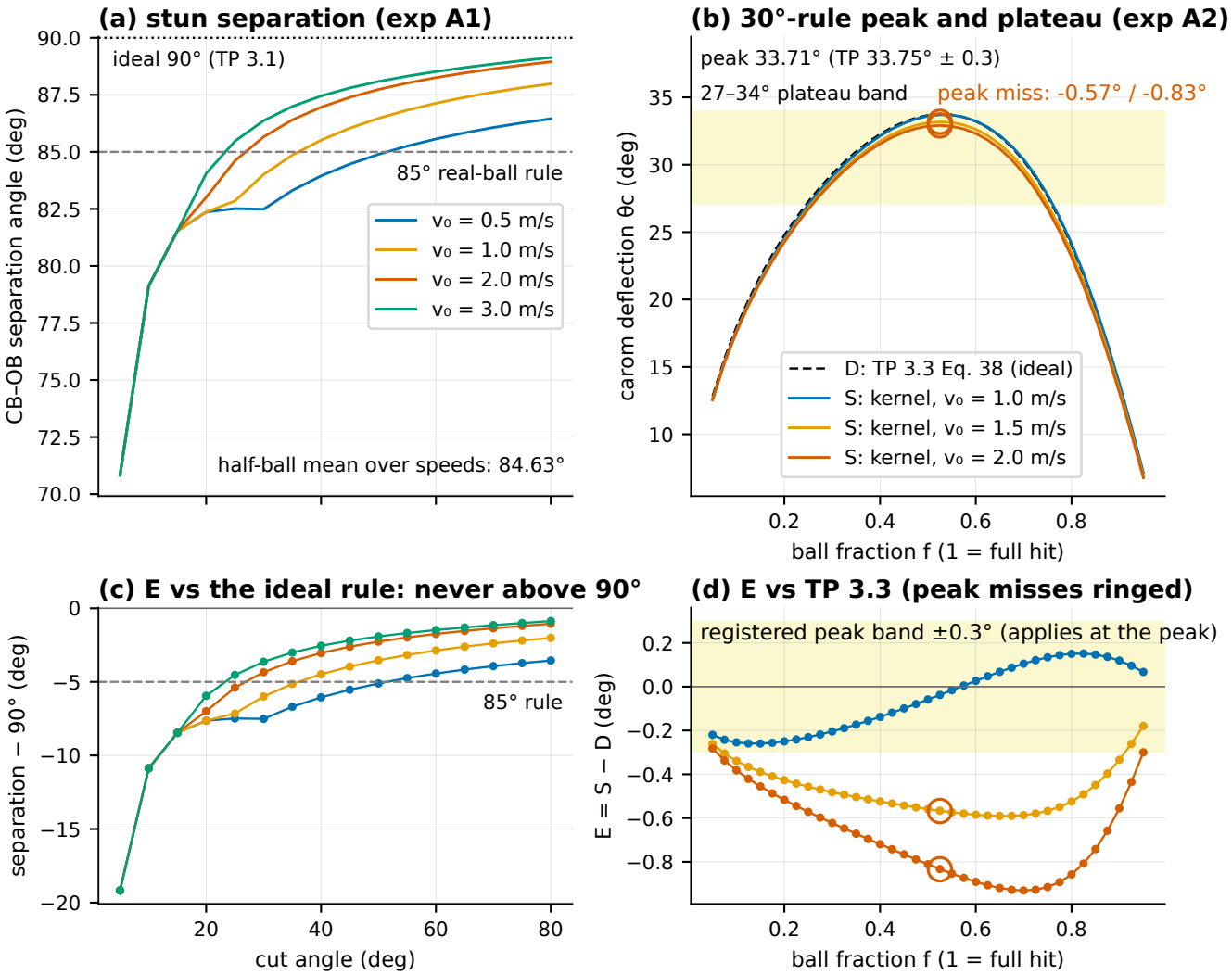


Figure 4: **Comparison.** Stun separation and the 30° rule (experiments A1, A2) against idealized reference forms. (a) Separation angle versus cut and speed against the ideal-90° construction (TP 3.1) and the 85° real-ball rule of thumb; in the friction-capped region (cuts $\lesssim 15°$) the deficit is speed-free. (b) Rolling-ball carom deflection θ_c versus ball fraction against the frictionless-elastic TP 3.3 closed form, with the 27–34° plateau shaded and the speed-dependent peak-height band departures ringed ($E = -0.57°$ at 1.5 m/s and $-0.83°$ at 2 m/s). (c) Residual view of (a): separation minus 90°, negative everywhere on the registered grid. (d) Residual view of (b): $E = S - D$ against the TP 3.3 form per speed, with the registered ±0.3° band (which applies at the peak) shaded and the peak misses ringed; values in the text.

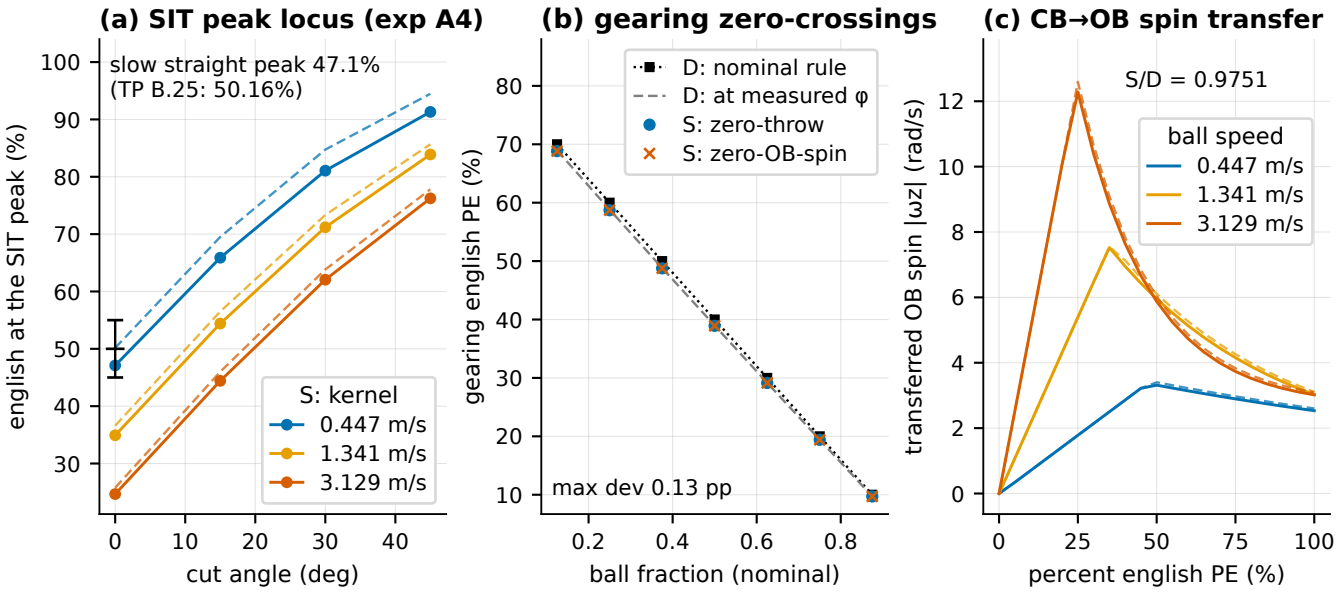


Figure 5: **Comparison.** Spin-induced throw, gearing english, and spin transfer (experiments A4, A5, A6). (a) SIT peak locus: the english fraction at the throw maximum over cut × speed (solid: kernel S; dashed: the TP B.25 construction D per speed), with the registered $50 \pm 5\%$ band drawn as a black error bar at its registered condition, the slow straight shot — the required english rises monotonically with cut angle at every speed. (b) Gearing zero-crossings against the $PE_g(\phi) = 80\% \sin\phi$ construction (Alciatore, n.d.e). (c) Cue-ball to object-ball spin transfer per ball speed (solid: kernel S; dashed: the TP A.27 form D at measured kinematics), with the slip-branch ratio $S/D = (1+e_b)/2$ against the $e = 1$ worksheet tagged on the panel (re-pin open); values in the text.

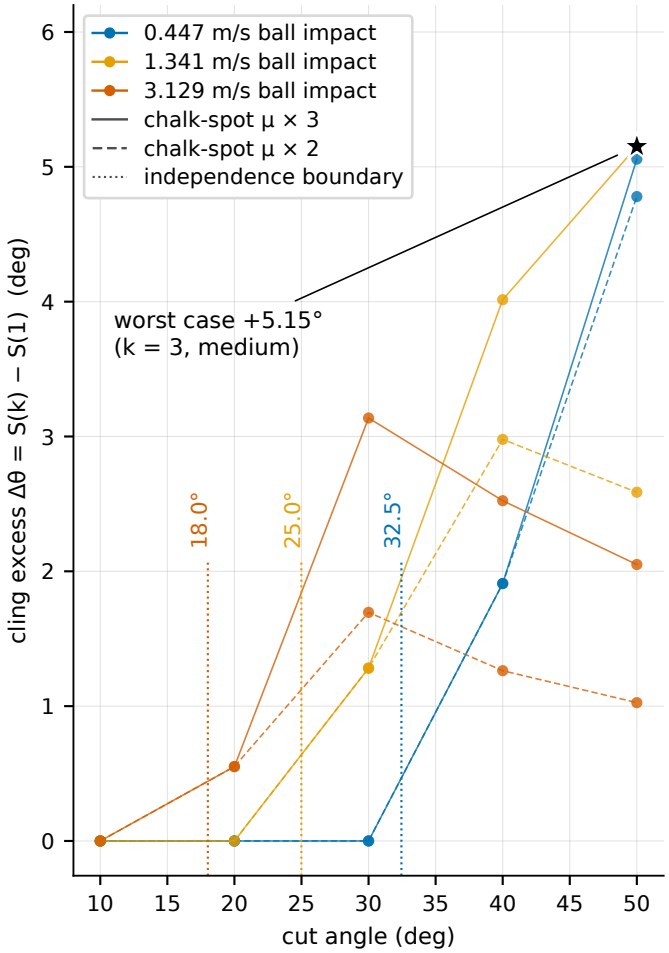


Figure 6: **Comparison.** Cling envelope (experiment A7): extra throw $\Delta\theta(k) = S(k) - S(1)$ under chalk-spot friction multipliers $k \in \{2, 3\}$ across speed × cut, published as the aiming-rule uncertainty band. Markers carry the registered 10°-grid samples (connecting lines only guide the eye); dotted verticals mark the speed-dependent friction-independence boundary (the μ-cap departure cut per speed); the worst case is starred, and the TP A.17 versus TP A-14 $\mu(v)$-revision discrepancy is recorded; values in the text.

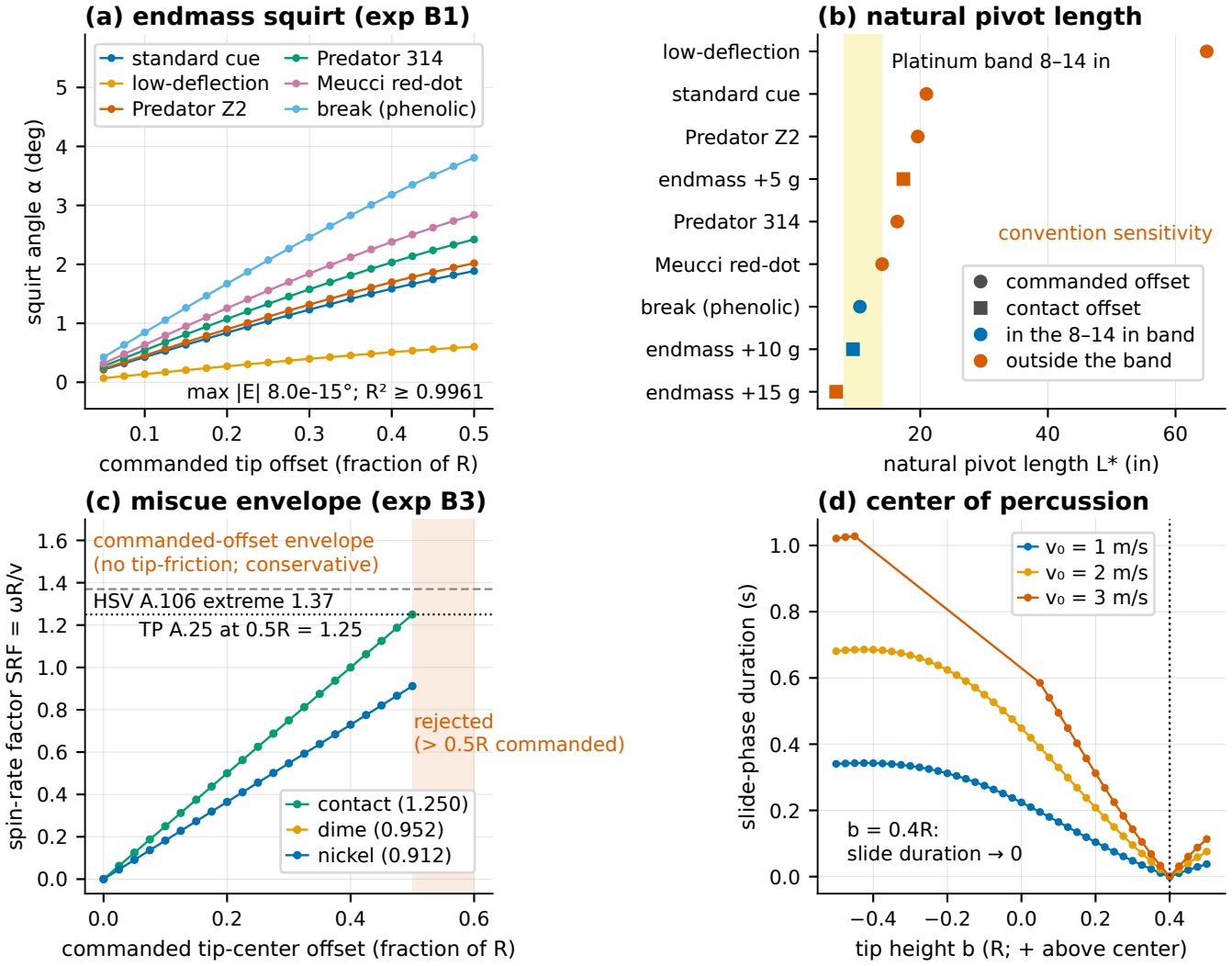


Figure 7: **Verification.** The stick–ball channel (experiments B1–B4). (a) Endmass squirt versus the TP A-31 closed form (Alciatore, n.d.f; Cross, 2008): machine-precision agreement, with the maximum error and per-cue linear-fit R^2 on the panel; the speed-invariance spread over 1–4 m/s is exactly zero. (b) Natural pivot length versus the Platinum protocol band; circles read the commanded offset, squares the contact offset, and blue versus orange encodes inside versus outside the 8–14 in band. (c) Miscue envelope: spin-rate factor versus commanded offset — exactly linear — with per-convention ceiling SRFs in the legend, the two literature reference levels, and the rejected tail (Table 8). (d) Center of percussion: slide-phase duration versus tip height, reaching zero at 70% of ball height (overspin above it); open markers are rows censored by the table runway; values in the text.

tion consistency, not independent validation, because the catalog's per-shaft end masses derive from the same Platinum pivot lengths via the Shepard inversion (Table 4; Section 4); holdout measurements do not exist in the published record.

Table 7 summarizes the four experiments; the per-experiment prose is Supplementary Material Section S4. Two findings do not read off the table. First, the Platinum convention sensitivity: commanding the robot protocol's published 6/12 mm as *tip-center* offsets (the protocol page's literal wording, and the frozen `B1` protocol reading) undershoots all four robot-measured shafts by 0.32–0.42°, while re-reading the same numbers as *contact* offsets closes the gap almost exactly (+0.02° on the lowest-squirt shaft); in the pivot-length observable the catalog spans 10.5–64.9 in against the published 8–14 in band, and a contact re-reference (factor ≈ 0.729) would bring two further shafts inside. Both readings are recorded and neither is adopted as a recalibration — the convention gap is an order larger than any plausible protocol u_D. Second, the miscue envelope is a conventions result (Table 8): the kernel's miscue rule is input validation on *commanded tip-center* offsets with no tip-friction model anywhere, so the envelope is one-sidedly conservative — the tip friction implied by its ceilings (0.39–0.58 across conventions) sits at or below the 0.577–0.659 the literature attributes to chalked tips, contact offsets in $(0.5, 0.55]R$ are unreachable under every convention, and the registered 0.525–0.6R sweep tail was realized as 16 designed rejection probes. A post-registration kernel amendment re-binds the miscue limit to the contact offset; every previously accepted input reproduces bit-exactly, while the nickel-tip rejection probes among these 16 would now be accepted — the one

disclosed behavioral change class (amendment note in Section 8).

Table 7: Group B (stick–ball) summary: headline result, measured domain boundary, and figure panel per experiment. Per-experiment prose in Supplementary Material Section S4. Lettered marks resolve to the notes at the end of the table.

Experiment	Headline result	Domain boundary	Fig.
B1 endmass squirt	$\max \lvert E \rvert = 8 \times 10^{-15}$ ° vs the TP A-31 closed form (333 runs); near-linear in offset ($R^2 \geq 0.9961$); speed invariance exact	protocol conventions only[a]	7a
B2 natural pivot length	closed form to 4.8×10^{-10} m; one per-cue L^* leaves $\leq 0.077°$ worst-case residual across the legal offset ladder	tip-center vs contact measurement convention; catalog 10.5–64.9 in vs the 8–14 in robot band	7b
B3 miscue limit	SRF $= (5/2)(x/R)$ exactly linear ($\lvert E \rvert \leq 2.2 \times 10^{-16}$; 252 runs); per-convention ceilings in Table 8	one-sidedly conservative commanded-offset envelope[c]; $(0.5, 0.55]R$ unreachable	7c
B4 center of percussion	slide duration exactly zero at $b = 0.4R$ (70% of ball height) at all speeds; SRF slope 2.5, $R^2 = 1$	none mechanical; 19 long-slide cells censored by the table runway[b]	7d

Notes. [a] verification; in-sample for the Platinum catalog. [b] a registered exclusion, not a waiver. [c] registered convention; revised post-registration to bind the contact offset (Section 8); accepted inputs reproduce bit-exactly, the nickel-tip rejection probes would now be accepted (artifact unchanged), $(0.5, 0.55]R$ still unreachable.

In summary, every boundary surfaced in this group is a protocol or equipment convention rather than a mechanics discrepancy, and all four experiments' physics is bit-identical across the kernel cascade; the corresponding rows of Table 10 therefore state *conventions* as the first failure mechanism.

5.3. Group C — cushion and diamond systems

Group C moves from single impacts to the cushion channel and the diamond aiming systems built on it. Throughout, four cushion presets are compared: `han2005_default` (the corrected Han model, Han, 2005),

Table 8: Miscue-envelope conventions (experiment B3): the kernel's commanded tip-center miscue rule expressed per tip convention, against the source's tip-friction envelope (TP 2.1 (Alciatore, n.d.c); observed extreme per the high-speed video catalog (Alciatore, n.d.b)). "Contact mode" drives the contact offset directly (tip radius $\to 0$); implied μ_{tip} expresses each ceiling through the source's friction model. Kernel rows state the registered envelope (kernel `0.1.0+fb3425ad`); the post-registration amendment re-binds the limit to the contact offset, raising the nickel- and dime-tip reachable contact offsets to exactly $0.500R$ (SRF ceiling 1.250) while $(0.5, 0.55]R$ stays unreachable; registered values unchanged, with the one behavioral change class (B3's nickel-tip rejection probes) disclosed in Section 8.

Convention	Max contact offset (R)	SRF ceiling	Implied μ_{tip}
Contact mode (tip $r \to 0$)	0.500	1.250	0.577
Nickel tip (10.6 mm)	0.365	0.912	0.392
Dime tip (8.95 mm)	0.381	0.952	0.412
Source, recommended maximum	0.500	1.25	0.577
Source, measured extreme	0.550	1.37 (observed)	0.659

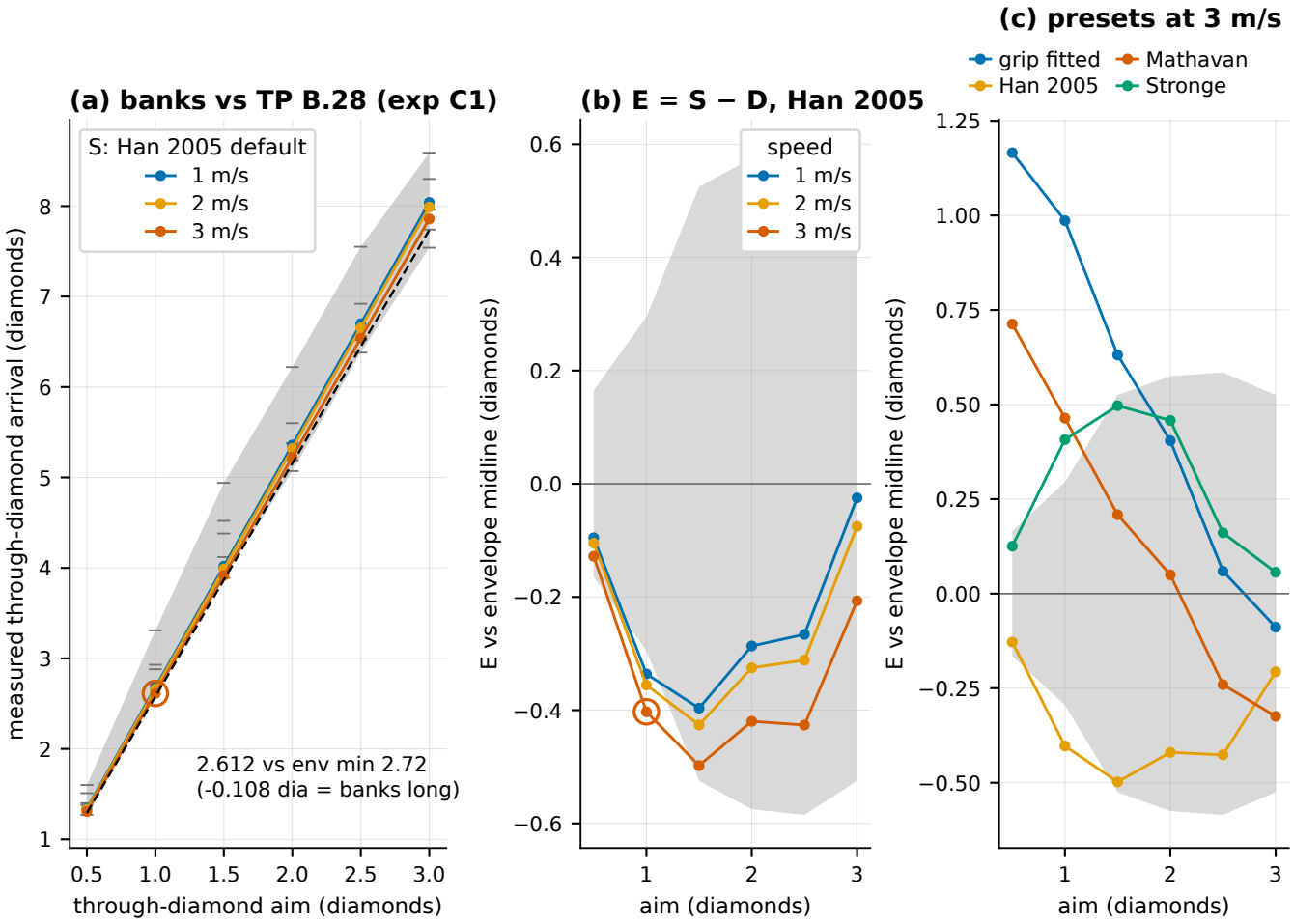


Figure 8: **Validation.** Sliding banks (experiment C1). (a) Through-diamond bank arrivals for six aims at 1–3 m/s against the TP B.28 five-table measured envelope (Alciatore, n.d.h) (grey band; short horizontal dashes mark the individual measured tables), with the twice-plus-tenths rule line overlaid (dashed); `han2005_default` sits inside on five of six aims at the 3 m/s reference speed, and the aim-1.0 miss is ringed (a through-diamond arrival *below* the envelope means the bank runs *longer*). (b) Residual view: $E = S - D$ against the envelope midline for `han2005_default` at all three speeds, where the grey band is the envelope's half-width about its midline. (c) All four cushion presets at the reference speed on the same residual axes (`han2005_default` and `stronge_realistic` sit inside the envelope on 5/6 aims, `mathavan_fitted` on 4/6, `grip_fitted` on 3/6); the post-fix settled-rebound continuity record is 0/156 discontinuous cells with one multi-crossing cell (`grip_fitted` at 1 m/s); one-at-a-time parameter-sensitivity values in the text.

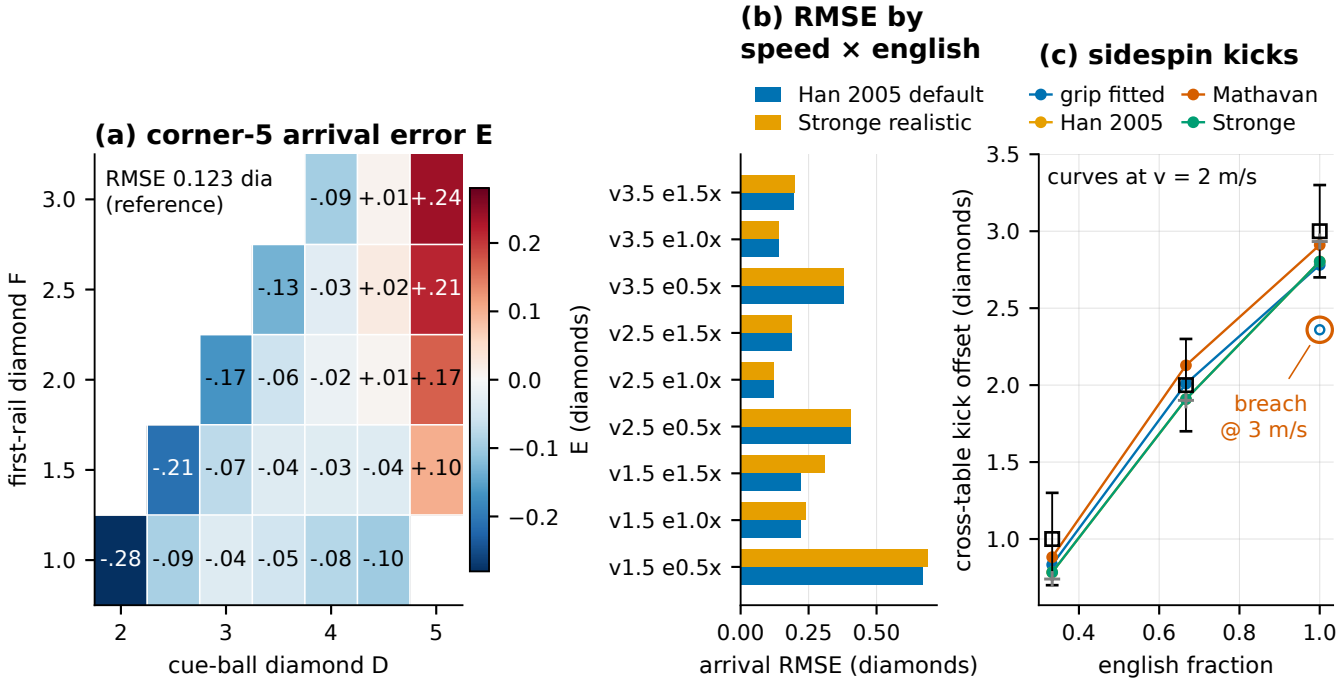


Figure 9: **Comparison.** Diamond systems (experiments C2, C3) against practitioner system lore. (a) Corner-5 third-rail arrival error $E = S - T$ over the 24 (D, F) cells at the reference condition (`han2005_default`, 2.5 m/s, nominal english) under the primary $T = D - F$ reading (in-cell values are E in diamonds). (b) Arrival RMSE across the speed × english conditions for `han2005_default` and `stronge_realistic` — english deviation dominates speed; the y-tick encoding `v<speed> e<mult>x` reads cue speed in m/s and english as a multiple of the nominal setting. (c) 1/2/3-diamond sidespin kick offsets versus english fraction for all four cushion presets at 2 m/s, with the ±0.3-diamond rule window on the rule targets (open squares) and the CI fixture pin (grey +; `mathavan_2010` preset); the single registered-window breach — `grip_fitted` at 3 m/s full english — is ringed; values in the text.

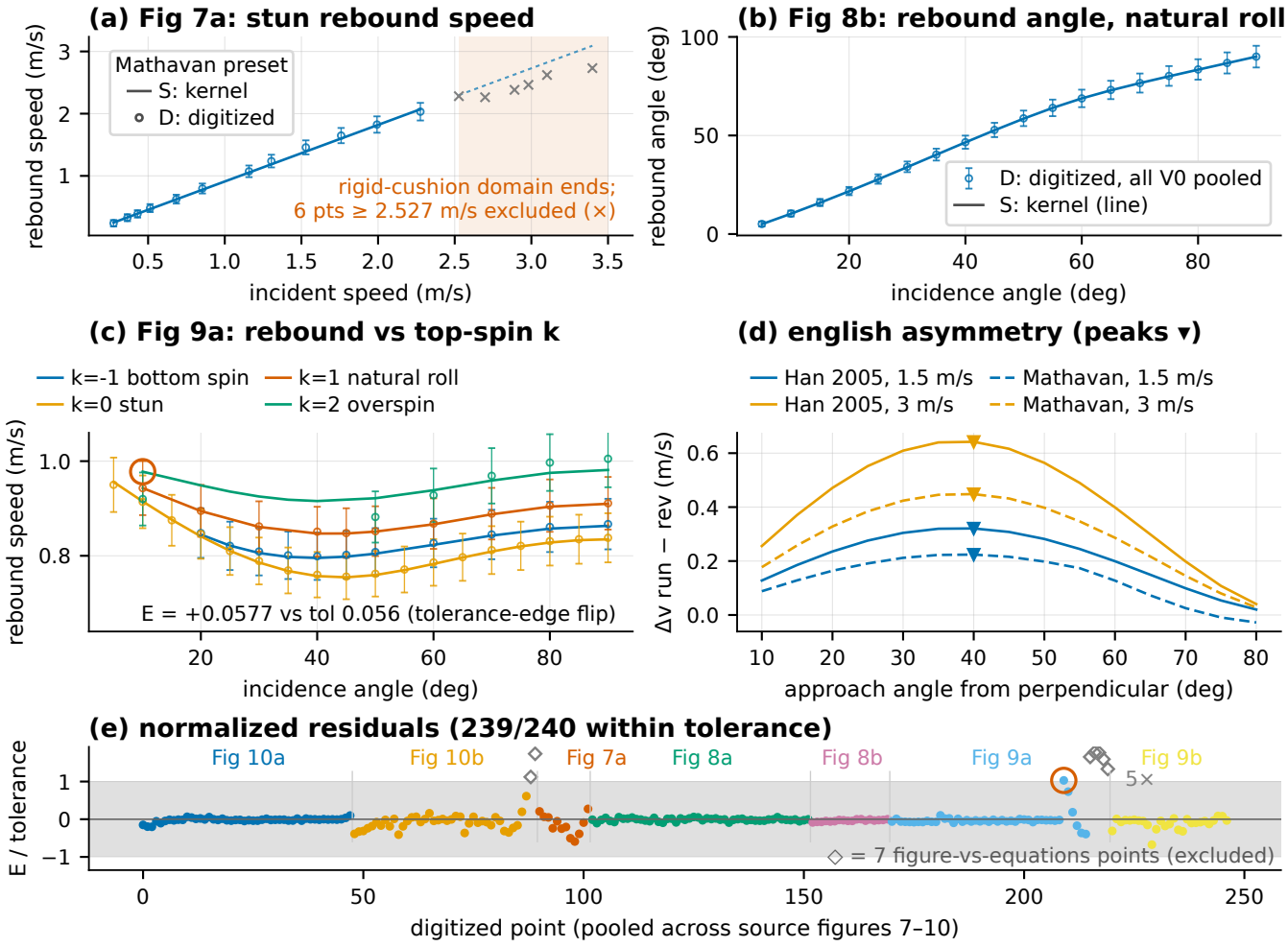


Figure 10: **Validation.** Cushion rebound coupling (experiment C4) against the digitized Figs. 7–10 of Mathavan et al. (2010). (a)–(c) Kernel curves (S; the `part_a_mathavan_2010` preset, the kernel's Mathavan dual-contact resolver at the source's own snooker fit) over the digitized points (D; bars are the per-point composite acceptance tolerance $u_y + 0.05\,|D|$ — digitization uncertainty u_y plus the 5%-of-$|D|$ acceptance term inherited from the CI anchor suite — not experimental error bars); (a) marks the points beyond the source's rigid-cushion domain (×), (c) rings the single tolerance-edge miss. (d) Running-minus-reverse rebound-speed asymmetry, peaks marked. (e) Normalized residuals E/tol, all 240 anchored points, with the seven figure-versus-equations points marked ◇. Anchor classes per Section 4: the Figs. 8–10 comparisons are verification of the ported scheme; the Fig. 7a camera points are the measured-datum validation content (12/12, $\max|E| = 0.067$ m/s); values in the text.

`stronge_realistic` (Stronge, 2018), and the fitted `mathavan_fitted` (Mathavan et al., 2010) and `grip_fitted` presets.

Bank tables (`C1`). Experiment `C1` aims stun banks through six diamond aims (0.5–3.0) at 1–3 m/s and reads the through-diamond arrival of the solved corner-bank reference line against the five measured tables of TP B.28 (Alciatore, n.d.h). At the documented 3 m/s reference speed the `han2005_default` arrivals are 1.31, 2.61, 3.92, 5.23, 6.54, and 7.86 diamonds (Fig. 8). Five of the six aims fall inside the five-table measured envelope; the frozen registered criterion requires all six, and aim 1.0 misses it: the arrival measures 2.6121 against an envelope minimum of 2.72, i.e. 0.11 diamond below the envelope — the simulator banks slightly longer at that aim than every measured table. TP B.28 prints no error bars, so no $|E| < u_{\text{val}}$ conclusion is available for this datum (Table 4); the 0.108-diamond shortfall is smaller than the $\leq$0.25-diamond spread between the source's own on-system tables — equipment variability, recorded as context but not invoked against the frozen criterion (Section 4). The TP B.27 "twice-plus-tenths" reference-line replication passes separately on its own gates — the measured reference line (tangent ratio 0.786 at 4 m/s) sits inside both the 15° and 30° windows (Alciatore, n.d.c) — and that verdict is not extended to the envelope. Among the other presets, `stronge_realistic` is likewise inside on five of six rows, while the fitted presets depart: `mathavan_fitted` on four of six (outside at the shallow aims 0.5–1.0) and `grip_fitted` on three of six, with a sign-reversed speed dependence at shallow aims.

The one-at-a-time sensitivity study on the registered "diamond-short" axis identifies restitution, not friction, as the lever on bank length. Sweeping

the kinematic restitution `e0` from 0.75 to 0.95 around the 0.85 default moves the mean arrival by −0.34 to +0.35 diamond, and the energetic `ee0` of the Mathavan family from 0.70 to 0.98 moves it by −0.38 to +0.70 diamond, whereas cushion nose height contributes at most 0.11 diamond. The flat rail-friction axis instead saturates: lowering μ_w from the 0.2 default to 0.1 shifts arrivals by −0.30 diamond, but raising it to 0.25 or 0.30 produces exactly zero change in every cell — the stun-bank contact already resolves in the stick branch, so additional friction cannot be consumed.

The adjacent banking calibration targets that fail are ledgered rather than re-tuned: the Stronge-pairing 45° sliding bank and the Han/Stronge draw rebounds *reverse* the source's claimed speed and vertical-spin directions (the fitted Mathavan presets get the spin ordering right), steep rolling banks go long under every preset, and the `han2005_default` 45° sliding point sits just below its TP B.27 window — per-target values in Section 8. All of these residuals are bit-identical across the settled-rebound branch-selection variants: they are not artifacts of the branch-selection defect. The two direction reversals are further resolved into model-form versus parameter content by extending the one-at-a-time machinery above to the reversed observables themselves, over the full admissible parameter space — the registered sweep ranges as the complete Cartesian product of restitution × μ_w (the preset's own fitted $\mu(v_{\text{slip}})$ curve alongside the flat pins) × nose height, 75/90/72 unique configurations per preset (post-registration analysis, recorded in the artifact bundle). The verdicts split. The *draw-rebound* reversal is model-form: no admissible parameter cell reproduces the source's "draw shortens" direction under the corrected Han form (0 of 75 cells; draw lengthens the 45°

impact rebound by +0.90 to +13.70° across the space, the joint minimum at the corner $e_0 = 0.95$, $\mu_w = 0.10$, nose 0.60) or the Stronge compliant form (0 of 90), while the Mathavan dual-contact class reproduces it in every admissible cell (72 of 72) — a structural boundary of the single-contact impulse class, not a calibration choice. The *speed-direction* reversal is instead governed by an *interaction*: the strongest reversals occur where a fitted $\mu(v_{\text{slip}})$ curve drives a tangential-compliance pairing. Within the admissible space, every flat-μ_w cell restores the source's shortening direction under every restitution and nose setting (75 of 75 Stronge cells, 60 of 60 Mathavan), while every cell retaining the fitted `sp_calibrated` or `hsv_fitted` curve on its compliant pairing reverses across the entire restitution × nose plane (15 of 15 and 12 of 12). A factorial crossing of the four model forms against all seven μ treatments over the full restitution × nose grids — 105/105/84/84 cells for Han, Stronge, and the Mathavan scheme with and without tangential compliance — fixes the scope of that attribution. Every fitted-curve × compliant-pairing cell reverses (54 of 54, +2.30 to +13.93°) and every flat-μ_w cell of every form is source-correct (270 of 270). The non-compliant forms are source-correct in 44 of their 54 fitted-curve cells: the compliance-free Mathavan scheme in all 24 (−3.15 to −0.15°) and the corrected Han impulse in all 15 `sp_calibrated` cells (−0.31 to −0.23°), but under the `hsv_fitted` curve Han flips in 10 of 15 restitution × nose cells — from +0.50° at the default operating point (the Jewett near-independence scale) to +2.72° at the $e_0 = 0.75$, nose-0.67 corner. The interaction attribution therefore governs the reversal's *magnitude* — the compliant reversals reach five times the worst non-compliant flip at matched corners — and not the sign alone, which Han's single-contact

impulse also flips under the `hsv_fitted` curve away from the default point. The same crossing settles the draw ordering beyond the preset spaces: no Han or Stronge cell restores the source direction under any μ treatment (0 of 210), the Mathavan form with tangential compliance restores it everywhere (84 of 84), and its compliance-free variant flips in 6 of 84 cells, all at flat $\mu_w \geq 0.25$ and by at most $+0.52°$. `han2005_default` (flat μ_w by construction) never reverses (-0.23 to $-0.50°$, the near-independence Jewett measured). Table 10's bank-lore row carries this attribution. That defect (the branch-selection defect), discovered by this experiment, produced four discontinuous `stronge_realistic` bank-arrival cells with 0.49–0.85-diamond jumps before the fix; under first-crossing branch selection they collapse to zero, and Section 6 presents the case study.

Corner-5 three-rail system (C2). Experiment C2 drives running-english three-rail kicks over 24 (D, F) cells with $D \in \{2..5\}$, speeds $\{1.5, 2.5, 3.5\}$ m/s, and the commanded english scaled by $\{0.5, 1.0, 1.5\}$. At the reference condition (`han2005_default`, 2.5 m/s, nominal english) the primary corner-5 system $T = D - F$ (Alciatore, n.d.c; Cohen, 2002) holds as a system: the third-rail arrival error has RMSE 0.123 diamond over the 24 cells with $\max|E| = 0.28$ (Fig. 9a), and regressing the measured arrival on the predicted T gives slope 1.077 with intercept -0.189 diamond and RMSE 0.102 (the fit is not through-origin: the 7.7% slope excess and the negative intercept cancel near $T \approx 2.4$, so the system reads on-line mid-range and the deviation grows toward the swept extremes). The validity map is strongly anisotropic: the mean absolute sensitivity of E to a $0.5\times$ english step is 0.22 diamond (up to 0.43 at short $D - F$ cells), versus 0.07 diamond per m/s (at

most 0.13) for speed — english deviation dominates the arrival while speed barely matters (Fig. 9b), consistent with the source's own teaching emphasis (Alciatore, n.d.a). The fourth-rail lore splits: the published benchmark that a 5-3-2 route comes up almost one diamond short plays on-system in the simulator ($n_4 = -0.02$, where positive would be short), while the secondary speed and english direction claims measure sign-reversed — slower arrives long (n_4 -0.201 at 1.5 m/s versus -0.015 at 3.5 m/s) and less english arrives longest (-0.643 at $0.5\times$ versus -0.137 at $1.5\times$). Those claims derive from unspecified "typical table" conditions the battery does not model; both readings are recorded. As a cross-model characterization, `han2005_default` and `stronge_realistic` produce identical arrivals at 2.5 and 3.5 m/s (rolling contacts resolve through the shared stick branch) and separate only at 1.5 m/s, by at most 0.49 diamond; the reference-condition verdict is model-invariant.

Sidespin kicks (C3). Experiment C3 plays perpendicular cross-table kicks with 1/3, 2/3, and full running english at 2 and 3 m/s under all four presets and reads the return offset in diamonds against the 1/2/3-diamond rule (Alciatore, n.d.a,n). One provenance qualification governs this experiment: the default rail-friction intercept $\mu_w = 0.20$ was calibrated against this very anchor family during kernel development (Section 4.3), so the verdict here is calibration-consistent rather than out-of-sample. At 2 m/s every preset lands within the ± 0.3-diamond rule window (`han2005_default`: 0.78, 1.91, 2.80 diamonds); at 3 m/s all remain inside except `grip_fitted`, whose full-english kick misses by -0.64 diamond (marked in Fig. 9c) — the third independent frame in which that preset leaves its validity domain. Two protocol

characterizations attach. First, the 1/3-english cells at 3 m/s reach the rail still slightly sliding (roll ratio 0.90 against the rule's natural-roll assumption); they are flagged per the registration, not excluded. Second, mirror symmetry — exact to about 5×10^{-15} diamond for the closed-form Han and Stronge presets — degrades to about 1.1×10^{-2} diamond for the fitted tangential-compliance presets through last-ulp branch divergence; this is a verification-side numerical characterization, not a physics deviation. These cue-only kicks are bit-identical across the slip-law kernel cascade.

Rebound coupling (C4). Experiment C4 promotes the cushion anchors of Mathavan et al. (2010) to the full simulation surface, and its headline count must be classified before it is quoted (Section 4). The pinned result — 239 of 240 digitized points from the source's Figs. 7–10 inside the per-point composite acceptance tolerance $u_y + 0.05\,|D|$ (digitization uncertainty u_y plus a 5%-of-$|D|$ acceptance allowance inherited from the CI anchor suite; Fig. 10) — was produced by the kernel's port of the source's own scheme at the source's own fitted constants (`SimOptions.mathavan_2010()`: $ee = 0.98$, $\mu_{\text{wall}} = 0.14$, the values the source fitted to its Fig. 7a camera data; *not* the `mathavan_fitted` catalog preset, which appears only in this experiment's later parts). Figs. 8–10 are the source's numerical output of that same scheme, so their 227/228 points are *verification of the port*, not independent validation. The validation content proper is Fig. 7a's 12 in-domain camera-measured points: 12/12 inside tolerance with $\max |E| = 0.067$ m/s — the one comparison in this experiment where E meets a measured datum. Its validation uncertainty combines both recorded digitization components: the fixture carries $u_x = u_y = 0.04$ m/s per point, and because the simulation is evaluated

at the measured incident speed, u_x propagates through the local slope of the rebound-speed response ($\mathrm{d}S/\mathrm{d}V = 0.91$, constant to within 0.003 over the in-domain segment) to 0.036 m/s, giving $u_{\mathrm{val}} = (u_y^2 + (0.91\,u_x)^2)^{1/2} \approx 0.054$ m/s. Stated plainly, the worst point *exceeds* u_{val} by about 20% — under the V&V 20 reading, an estimated model-form error of $|E| - u_{\mathrm{val}} \approx 0.013$ m/s at that point (against the vertical component alone the same point would read $1.7\,u_y$); the other eleven points resolve within u_{val}. Out of sample the picture is disclosed rather than flattered: re-scoring the same 240 points (a post-registration analysis, recorded in the artifact bundle) puts the no-Mathavan-inputs default at 52/240 inside tolerance (`han2005_default`, flat $\mu_w = f_c = 0.20$; worst errors 16.8° on angle observables and 0.55 m/s on speeds; 1 of 12 on the Fig. 7a camera points), and a model-form-swap arm — the corrected Han impulse form driven at the source's own fitted wall friction, $\mu_{\mathrm{wall}} = 0.14$ — at 57/240 (0 of 12 on Fig. 7a). Both Han arms extrapolate to the snooker $7R/5$ nose geometry the Han model was not formulated at, and neither was tuned; the numbers delimit what the Han family can and cannot do on snooker rebound coupling, and they are the honest out-of-sample companions to the in-sample 239/240. The single miss in the pinned run is a tolerance-edge flip — Fig. 9a, $k = 2$ overspin at $\alpha = 10°$: $E = +0.0577$ against a tolerance of 0.056, with a zero-drift extrapolation of the 1 ms launch protocol landing inside the band. Seven further points are internally inconsistent between the source's equations and its printed figures; the kernel reproduces the equations to within 0.005 m/s and 0.07° and genuinely disagrees with the figures — an upstream artifact, excluded from the validation statistics with that attribution. Fig. 7a points beyond

the source's own declared 2.5 m/s rigid-cushion domain are measured but excluded, with the overlay overshoot growing monotonically from +0.018 m/s at 2.53 m/s to +0.359 m/s at 3.40 m/s — the source's domain boundary made explicit. The english-asymmetry extension finds the running-minus-reverse rebound-speed difference peaking at 40° from the rail perpendicular for both `han2005_default` and `mathavan_fitted` at both 1.5 and 3.0 m/s (peak differences 0.32 and 0.64 m/s for Han) — on the steep edge of, but inside, the FAQ's 20–40° window (Alciatore, n.d.a). Shallow approaches show the claimed littleness (peak-normalized asymmetry 0.06 at 80°), and the rebound angle remains monotone in incidence for all four presets, both arrival spins, and both speeds.

Taken together, the diamond systems hold *as systems* near their teaching conditions, while the published speed and spin correction lore attached to them is where validity fails first: it is the speed-direction and fourth-rail refinements, not the systems themselves, that the measurements reverse (Table 10).

5.4. Group D — curved paths (Coriolis)

Group D tests the curved-path results of Coriolis (1835), in the translation and modern notation of Nadler (2005) and the treatment of Alciatore (2008).

Table 9 summarizes the three experiments; the per-experiment prose is Supplementary Material Section S4. Three findings carry the group. First, the tangent-frame waiver: experiment D1's registered acceptance criterion — quadratic-fit $R^2 > 1 - 10^{-6}$ in the frame aligned with the measured post-impact velocity — fails on all six runs ($1 - R^2$ between 2.2×10^{-6} and 3.7×10^{-6}), because the real-ball impact model leaves the friction acceleration

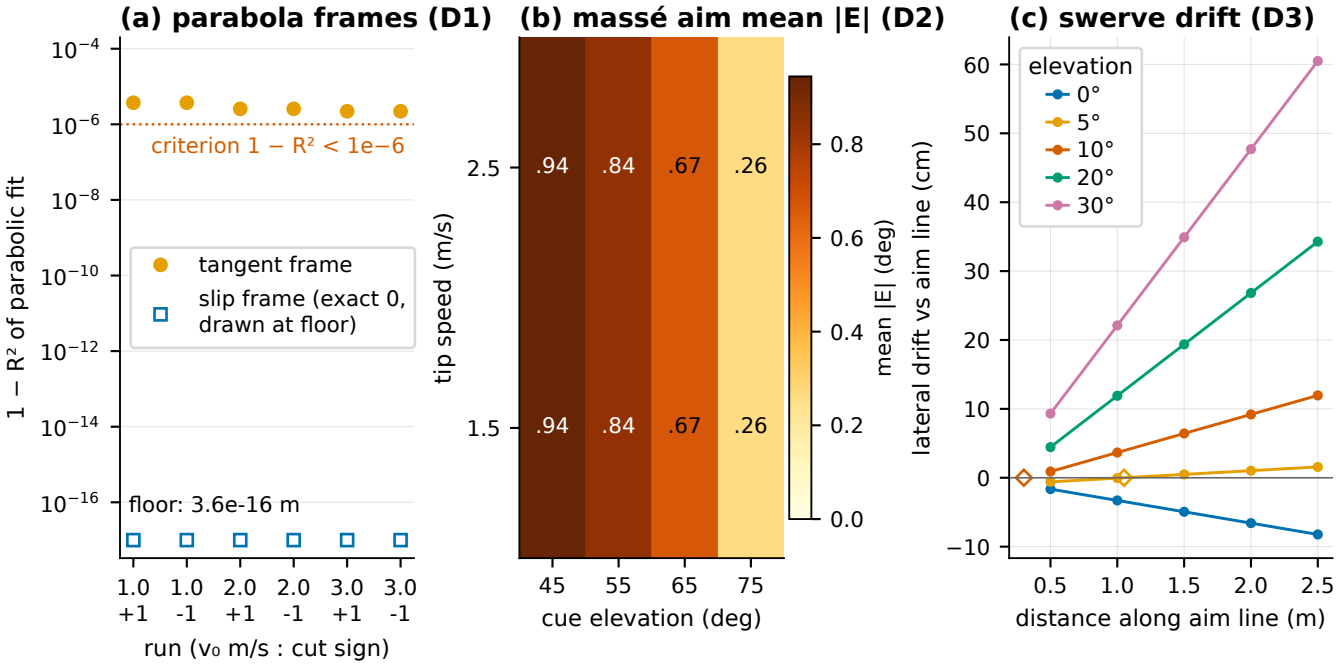


Figure 11: **Verification.** Curved-path (Coriolis) results (experiments D1–D3). (a) Post-impact sliding-path parabola: the slip-aligned-frame quadratic fit is exact (worst time-domain residual 3.6×10^{-16} m; slip points drawn at the 10^{-17} floor) while the registered tangent-frame criterion misses on every run, shown per run for the frame-artifact record. (b) Massé BAR-aiming (B = ball contact point, A = cloth aim point, R = CB resting point) mean $|E|$ over cue elevation × tip speed (azimuth-averaged); the two tip-speed rows are identical — Coriolis' impact-friction invariance; 88% of cells come up slightly short, the source's own caveat (TP A.19; values in the text). (c) Swerve drift versus distance along the aim line at full english and 2 m/s tip speed, one curve per elevation; the level-cue curvature is exactly zero (the elevation-0 line's residual straight drift is squirt, not swerve) and the squirt–swerve cancellation crossings are marked (◇); values in the text.

2.9–4.1° off perpendicular to that axis, injecting a cubic term a quadratic fit cannot absorb; in the slip-aligned frame — the parabola's own axis frame — the fit is exact to machine precision ($1 - R^2 = 0.0$ on every run, worst time-domain residual 3.6×10^{-16} m). The miss is kept as this paper's worked example of a frame-sensitive verification criterion. Second, the massé BAR aiming rule — "B for ball contact point, A for cloth aim point, R for CB resting point," the source's own naming (Alciatore, n.d.a) — is speed-invariant in exactly Coriolis' sense: friction acting along the fixed slip direction cannot rotate the final direction (Coriolis, 1835), and the per-cell heading errors at the two tip speeds are identical to the full recorded precision of the output in all 32 (elevation, azimuth) pairs — with mean heading error 0.68° (maximum 1.60°) and 88% of cells coming up "a little short," the source's own caveat reproduced (TP A.19, Alciatore, n.d.c). The 45–55° rows run in the airborne jump-launch regime, where the flight leg is inert to the comparison by a conservation identity: under any impulsive strike the natural-roll heading — the azimuth of the planar combination $(5\mathbf{v}_{0,xy} + 2R\,\boldsymbol{\omega}_0 \times \hat{\mathbf{z}})/7$ — is exactly invariant to hops, to the ball–slate restitution, and to the friction law (Coriolis' own robustness remark, quoted in Bouleau, 2006), so those rows exercise the strike's impulse–momentum mapping and the solver's conservation structure but cannot discriminate strike realism beyond the impulsive idealization — the scope note, with the falsifiable in-sim check, is Supplementary Material Section S4. Third, the delayed-onset swerve rule is made quantitative through the squirt–swerve cancellation distance ("squerve"), the distance at which swerve re-crosses the aim line that the 1.9° full-english squirt displaced: measured cancellation distances span 0.08–2.65 m on the grid, moving out-

ward at 5° elevation and full english from 0.26 m at 1 m/s to 1.05 m at 2 m/s to 2.36 m at 3 m/s (the marked crossings in Fig. 11c) — a derived, teachable quantity linking the stick–ball and curved-path channels.

Table 9: Group D (curved paths) summary: headline result, measured domain boundary, and figure panel per experiment. Per-experiment prose in Supplementary Material Section S4.

Experiment	Headline result	Domain boundary	Fig.
D1 parabolic slide path	exact in the slip-aligned frame ($1 - R^2 = 0.0$; TP A.4 slide duration to 2.8×10^{-17} s); path extent scales $v^{2.09}$–$v^{2.14}$ vs the registered $\sim v^2$	registered tangent-frame criterion misses every run — a frame artifact, not mechanics	11a
D2 massé BAR aiming	mean $\lvert E\rvert = 0.68°$, max 1.60° over 64 cells (two ill-conditioned cells excluded); per-speed errors bit-identical (Coriolis invariance)	degrades at low elevation/high speed (the source's own caveat); executed elevation subset declared	11b
D3 swerve maps	all three registered structural rules, zero violations; level-cue drift exactly zero; squerve span 0.08–2.65 m	onset delayed at speed (as taught); sliding legs closed-form to 2.6×10^{-13} m — the D1 verification floor	11c

The 1835 closed forms are exact in their own frames and under their own assumptions; every apparent deviation in this group traces to a frame or protocol choice, not to mechanics (Table 10). Group D is the battery's cleanest verification showcase, and its frame-artifact row is the methodological caution it contributes.

5.5. Group E — frozen contacts and the break

Group E exercises the regime that motivated the compliant frozen-cluster resolver of Section 3: shots into topologies in which balls touch before the cue ball arrives. Every experiment in the group runs both resolvers — the default compliant chain (Hertz contact with Lankarani–Nikravesh damping,

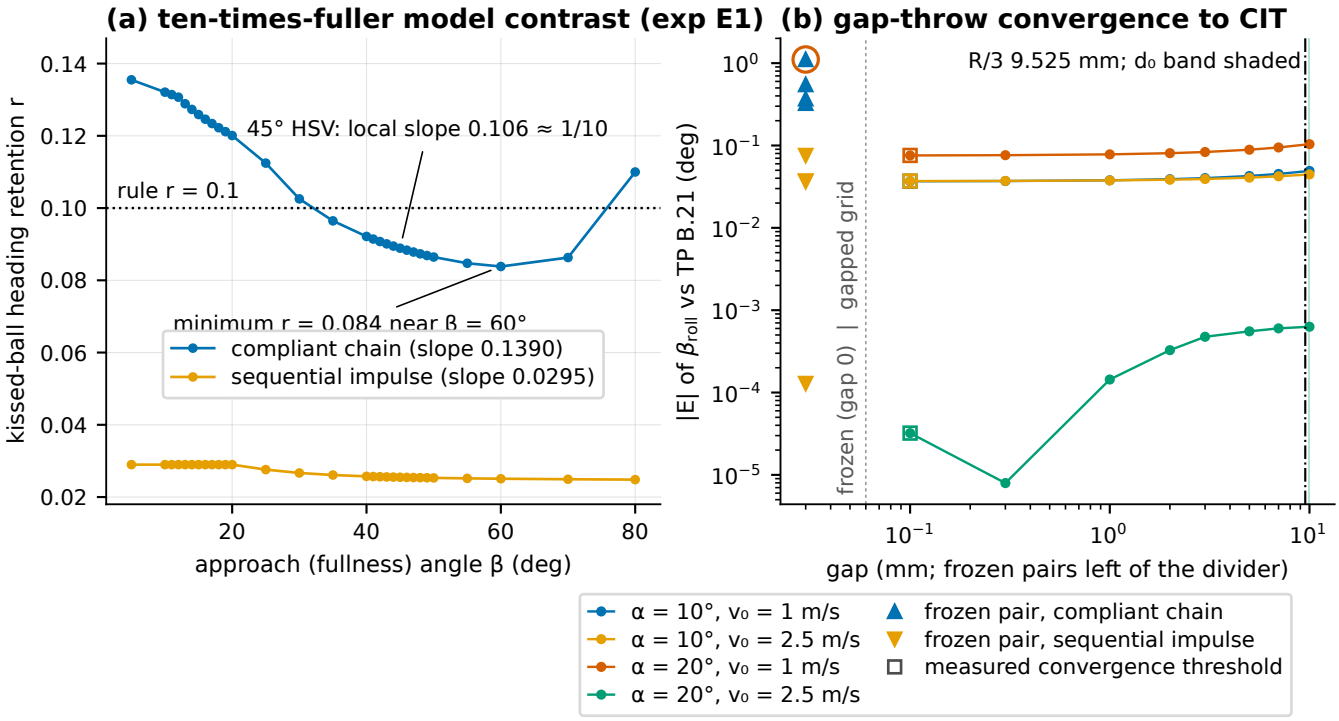


Figure 12: **Validation / comparison.** Frozen-contact rules (experiments E1, E2): panel (a) benchmarks the lore-anchored ten-times-fuller rule (E1; instructional constant, no measured retention datum), panel (b) carries E2's measured-datum content. (a) Kissed-ball heading retention r versus the approach (fullness) angle β for a frozen pair, against the ten-times-fuller rule form $\tan\theta_T = r\cot\beta$ with $r = 0.10$ (Jewett), compliant versus sequential resolver; the 45° HSV reference cell (E vs rule +0.0056) and the measured retention minimum (with the rise toward thin grazing hits) are annotated. (b) Combination throw versus gap: every comparison family converges to the two-body cut-induced-throw law from the smallest tested gap; the source's $R/3$ optimum (dash-dot line, 9.525 mm) is drawn with the measured straightest-departure gap band $d_0 = 9.85$–9.99 mm (shaded, near-straight cells), and the open frozen-limit throw excess is ringed; frozen-pair markers aggregate over the $\alpha \times v_0$ conditions; values in the text.

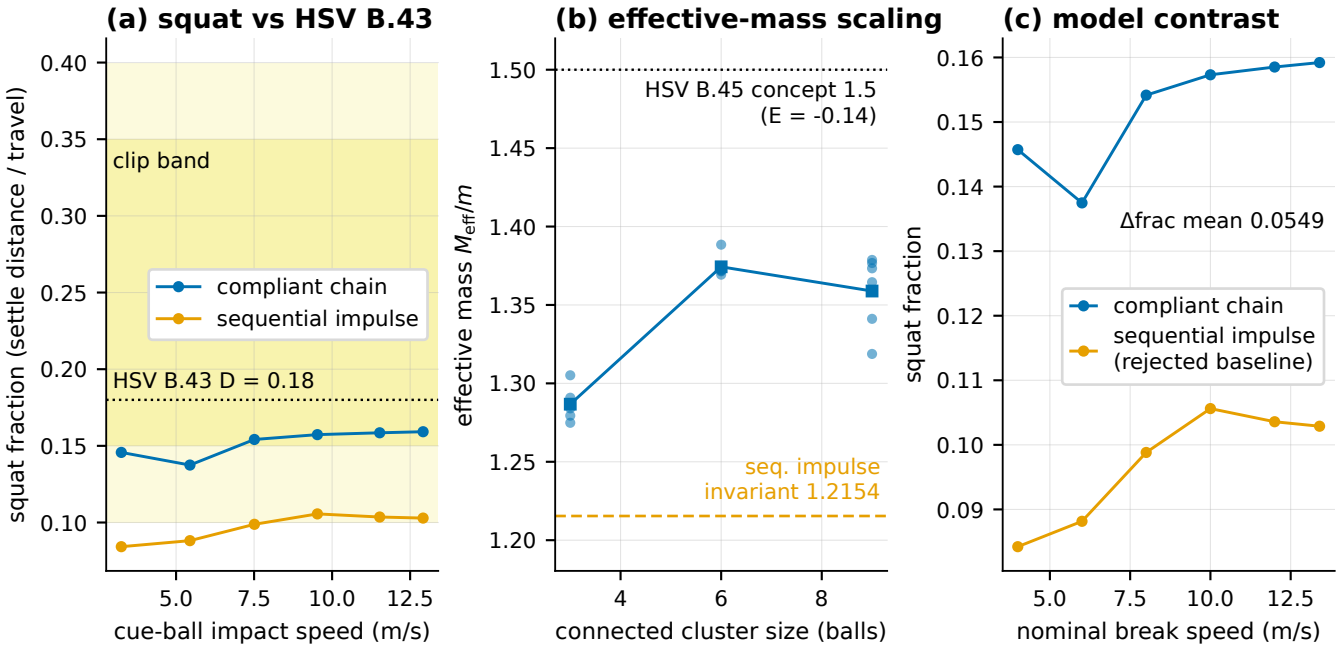


Figure 13: **Validation.** Break mechanics (experiment E3). (a) Square-hit squat fraction (settle distance over travel, the observable defined in the text) versus break speed against the frame-tracked HSV B.43 evidence bands — the inner "clip" band (0.15–0.35) is the tighter frame-tracked target, the outer band (0.1–0.4) its conservative envelope, both read from the pinned output — compliant versus sequential resolver. (b) Effective-mass scaling over connected cluster size: faint dots are per-topology cluster values from the compliant seeds, squares their means, against the HSV B.45 concept value 1.5 and the cluster-size-invariant sequential value. (c) Cluster-model contrast: squat fraction across break speed for both resolvers ($\Delta_{\text{frac}} = 0.0615$ at 4 m/s, mean 0.0549) — the contrast survives the slip-law re-baseline; values in the text.

advanced concurrently across the cluster (Lankarani and Nikravesh, 1994))
and the legacy sequential-impulse pairwise arm — so each rule doubles as a model contrast (Fig. 12).

Ten-times-fuller (E1). The ten-times-fuller system for frozen pairs — published by Jewett (*Billiards Digest*, December 2024) and documented on high-speed video (Alciatore, n.d.b) — asserts that the kissed ball departs one tenth of the way from the tangent line toward the shot line per unit of fullness: $\tan\theta_T = r\cot\beta$ with retention constant $r = 0.10$. The anchor class is lore under the taxonomy of Section 4: the numeric datum is the instructional constant 0.10, and the video documents the scenario rather than a measured retention, so this row is benchmarking, not validation. Under the compliant resolver the measured retention drifts from 0.136 at fullness $\beta = 5°$ to 0.084 at 60°, crossing the rule's 0.10 at $\beta \approx 32°$ (interpolated); the local slope at the 45° video reference is 0.106 ($E_r = +0.006$), so the rule holds at its own teaching point, and the heading error stays within about 1.5° for $\beta \geq 25°$. Past its minimum near 60° the retention rises again toward the thinnest, grazing hits (0.086 at 70°, 0.110 at 80°; $\beta = 0$ is dead full, so large β is a thin hit — the up-hook annotated in Fig. 12a), and the mechanism is normal-channel geometry, not friction: the frictionless baseline $r_{\text{free}}(\beta)$ introduced below carries the same up-hook (0.078 at 60° rising to 0.106 at 80°) while the friction-attributable difference $r - r_{\text{free}}$ *shrinks* monotonically over the same range (0.006 to 0.004) — as the hit thins and $\tan\beta$ grows, the retention read $r = \tan\theta_T \tan\beta$ increasingly magnifies the residual normal-channel deflection that the frozen chain's damped contacts leave on the kissed ball, even as the deflection θ_T itself vanishes. The flat-retention idealization is the

domain boundary: E grows to $+4.7°$ at $15°$ and $+8.3°$ at $5°$ fullness. Because the frozen-pair tangential excess (an open investigation; experiment E2, below) rides this same compliant channel, its possible contamination of the low-fullness conclusion is bounded rather than assumed away: deflating the friction-attributable part of the retention — $r - r_{\text{free}}(\beta)$, measured against a per-fullness frictionless-geometry baseline r_{free} (the same ladder re-run with $\mu \equiv 0$: 0.096 at $\beta = 5°$, 0.078 at $60°$) — by the E2 hot factors normalized to the $\sim 1.04\times$ measured anchor, $h = 1.132/1.04$ to $1.554/1.04 = 1.09$–1.49 (the excess is measured *relative* to the anchor, so the raw 1.13–1.55× ratios are not the divisor), gives $r_{\text{corr}}(5°) = 0.1225$–$0.1323$ against 0.136 uncorrected, moves the interpolated rule crossing from $32.1°$ to 28.3–$31.1°$, and leaves the low-fullness excess over the rule's 0.10 standing in every arm, with E at $5°$ fullness still $+5.7$ to $+7.7°$ against $+8.3°$ uncorrected (post-registration analysis, recorded in the artifact bundle). The curve is also step-converged: halving or doubling the compliant resolver's 10^{-6}-s internal step (episode cap held at 3 ms) moves no retention point by more than 6.3×10^{-4} and the $5°$-fullness heading error by at most $0.12°$. The sequential arm contradicts both the rule and the three-ball chain analysis of TP B.29 (Alciatore, n.d.c): its fitted slope is 0.029 — in the output's characterization a "forty-times-fuller" system, the pairwise $(1 - e_b)/2 = 0.025$ residual — and on the square frozen hit it pushes the cue ball *forward* (+0.024 of impact speed) where the compliant chain rebounds it (-0.056, sign-matching TP B.29's elastic-chain value of -0.071). A 1 mm gap erases the contrast (kissed-ball retention collapses $0.103 \to 0.024$ and the two resolvers coincide bit-exactly): gapped topologies bypass the cluster path by construction. The compliant slope is

bit-identical across the slip-law revision (Section 6) because its force-based Coulomb friction consumes no impulse cap, so the model contrast survives re-baselining.

Gap throw and the $R/3$ optimum (E2). The combination-throw experiment asks where a gapped two-ball combination stops being a cluster problem (TP B.21, TP A.15 (Alciatore, n.d.c); FAQ pages (Alciatore, n.d.a)); folk statements place the boundary near a ball width. The measured boundary is two orders of magnitude lower: every comparison family — the source stun form, the measured TP A-14 overlay (Alciatore, n.d.d), and the characterized kernel law — agrees with the plain two-body cut-induced-throw (CIT) prediction to within the 0.1° tolerance from the smallest tested gap, 0.1 mm, onward, over the whole gapped grid. This is by construction: any gap above the kernel's 10^{-5} m cluster tolerance resolves as two ordinary pairwise impacts. Against the kernel's own law the gapped residual is 0.0000–0.0121°; against the sources' $e = 1$ forms it is 0.000–0.104°, the declared restitution family (a declared-model-content ledger row). TP B.21's teaching optimum — a gap of $R/3$ (9.525 mm) sends the second ball straightest, independent of friction — is reproduced to first order: the straightest-departure gap d_0 measures 9.85–9.99 mm on the three near-straight cells, with the kernel-law small-angle optimum (9.811 mm) accounting for most of the offset, while the fourth cell ($\alpha = 20°$ at 2.5 m/s) measures 5.23 mm, outside the source's near-straight claim domain. The frozen limit itself is the group's open item: the compliant resolver throws frozen pairs 1.13–1.55× more than 1 mm-gapped ones, against a measured anchor of ≈1.04× (5.16° versus 4.95°, at a ∼30° cut — beyond the registered ≤ 20° grid, a u_{val} caveat), with $E = -1.11$ to

$-0.33°$ against the TP B.21 $d \to 0$ form and a speed dependence the anchor does not show; the sequential arm shows no excess at all ($E = -0.076$ to $-0.000°$, the pairwise limit). The sign is correct — no gap throws the most — but the magnitude is an open investigation (Section 8), with over-accumulated tangential impulse in the compliant friction integration as the working hypothesis (cf. Cross, 2023).

Break squat and effective mass (E3). The break experiment measures square-hit "squat" — the cue ball's residual motion after a full-rack collision — against frame-tracked evidence bands from high-speed footage (HSV B.43 (Alciatore, n.d.b)) over nominal break speeds 4–13.4 m/s (measured impact speeds 3.2–12.9 m/s; the grid top matches the reported professional break-speed range (Alciatore, n.d.a)). Two registration deviations are declared rather than absorbed: the elevation grid $\{0, 3\}°$ is a subset of the registered 0–5° range, and the registered "momentum-theft baseline" is realized as the in-kernel sequential arm, pooltool's 10% momentum-removal hack (Kiefl, 2024) being a documented rejected alternative. Under the compliant resolver the cue ball rebounds along the reversed incoming line (within 2×10^{-6} degrees) at 0.137–0.159 of impact speed: every row is inside the outer evidence band $[0.10, 0.40]$, every row at ≥ 8 m/s is inside the clip target $[0.15, 0.35]$, no dead-stop row (< 0.05) appears anywhere, and E against the 0.180 anchor value is -0.021 to -0.043 — a comparison the anchor resolves only at band level: the footage's capture-chain u_D (Table 4) supports the frame-tracked evidence bands, not a point datum, so $|E| \leq 0.043$ is reported against the band rather than against a u_{val} tighter than the band itself. The sequential arm saturates at 0.084–0.106 and falls below the outer band for speeds ≤ 8 m/s, while

the kinetic energy delivered to the rack is nearly model-invariant (0.77–0.81 for both): the resolvers differ in where the cue-ball residual goes (Fig. 13c plots that residual contrast), not in the gross energy budget. The effective-mass ladder sharpens the same point. HSV B.45 assigns the struck rack an effective mass $M_{\text{eff}}/m \approx 1.5$ (via TP B.20 (Alciatore, n.d.c)); the compliant resolver reproduces the direction of the scaling — 1.287 for a three-ball wedge, 1.374 for a six-ball cluster, 1.359 for the default nine-ball rack — but caps below the concept value ($E = -0.14$). The cap is a topology statement: the default rack's seeded back-row micro-gaps truncate its connected frozen cluster at six balls, so the nine-ball rack measures at the six-cluster value. The sequential arm is cluster-size-invariant at 1.2154 ($E = -0.28$): a pairwise resolver never feeds back-row inertia into the cue ball. The compliant model's topology sensitivity — and its complete erasure under the sequential resolver — is the break-scale evidence for contribution C3. Like E1's retention curve, these break-row observables are step-converged in their own right: a post-hoc bracket re-running the square/elevation-0 ladder and the effective-mass probes at compliant steps $\{5 \times 10^{-7}, 10^{-6}, 2 \times 10^{-6}\}$ s (episode cap held at 3 ms) moves no squat fraction by more than 2.8×10^{-4} and no M_{eff}/m value by more than 7.7×10^{-4}, with every evidence-band verdict invariant across the bracket (analysis recorded in the artifact bundle). The declared e_b bracket is likewise applied to the break rows (the compliant resolver consumes e_b directly through its Lankarani–Nikravesh damping), with the scope stated exactly: each arm holds one *constant* e_b for every contact, so these runs measure constant-e_b parameter sensitivity — they do not bound a speed-dependent $e_b(v)$, which would mix effective values across the break's contact

spectrum and which the kernel does not model (Section 4). Over a 62-point constant-e_b grid spanning $[0.92, 0.98]$ the squat fraction moves by up to 0.0201 and M_{eff}/m by up to 0.0551 — observed grid maxima of a response that is non-monotone and *event-discrete* in e_b, not interval bounds: both maxima are realized across an interior jump between the adjacent grid points 0.95775 and 0.958, where the 8 m/s squat drops discontinuously from 0.156 to 0.136 as the cascade's contact sequence reorders, so no finite arm set bounds the declared interval — the outer-band, dead-stop, and direction verdicts are invariant at every grid point, and the only membership change is at the clip target's 8 m/s edge, inside for $e_b \in [0.935, 0.95775]$ at grid resolution and outside on both sides of that window — so the "every row at ≥ 8 m/s" clip statement above holds at the frozen $e_b = 0.95$ and is e_b-sensitive at exactly that edge (analysis recorded in the artifact bundle). Two characterization remarks close the group. First, the post-break hop apex measures ≤ 0.039 ball diameters everywhere, satisfying the one-sided footage bound ($\leq\sim$0.3 diameters, no registered minimum); the gap to the observed $\sim$0.23 is recorded as a vertical-channel remark, not a band miss. Second, three extreme quarter-off rows (12–13.4 m/s) flip a downstream scratch-outcome bit across kernel revisions: individual outcome bits after long slip cascades are chaotic across model changes even though every output is bit-exact for a fixed binary — a deliberate scoping of what the reproducibility contract of Section 7 does and does not promise.

5.6. Group F — kinematics closed forms and the jump

Closed-form verification (F1). Between events the kernel evolves the constant-friction closed forms of TP 4.1/B.5 (Alciatore, n.d.c), so this chan-

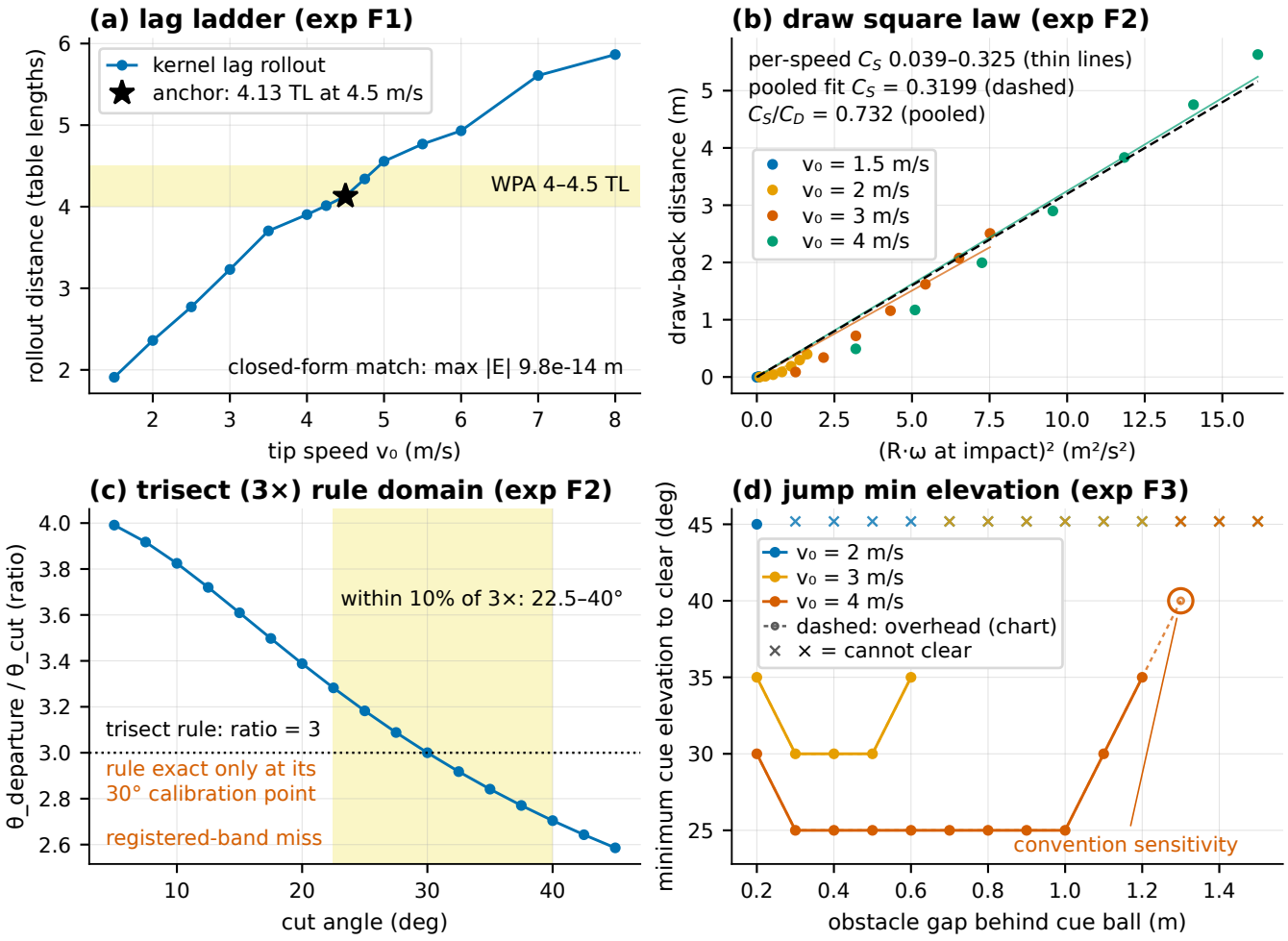


Figure 14: **Comparison.** Kinematics and rule-domain flagship (experiments F1–F3; panels (b) and (c) both belong to experiment F2). (a) Lag table-lengths versus cue speed on the frozen default catalog, with the 4.0–4.5 window and the anchor stroke. (b) Draw square law $S = C_S\,(R\omega)^2$: pooled through-origin fit (dashed, $R^2 = 0.986$) plus per-speed fits (thin lines) — the pooled functional form holds, while the per-speed fits degrade at low spin/speed ($R^2 = 0.42$ at 1.5 m/s, 0.94 at 2 m/s, $\geq$0.98 above) with a strongly speed-dependent coefficient against the perfect-ball program (TP B.8 at kernel constants). (c) Trisect ratio θ/cut versus cut, exact at the 30° calibration point, with the within-10%-of-3 window shaded. (d) Minimum jump elevation versus obstacle gap under the sphere criterion (primary solid curves), with the chart-style overhead criterion as a dashed comparison overlay — the two separate only at the ringed far-edge cell; × marks uncleared cells and the maximum airborne over-cut is 2.779° (slip-branch); values in the text.

nel is verification in the V&V 20 sense (Section 4): E is a numerical floor, not a physics comparison. Over the 25-cell (v, ω) launch grid, slide-to-roll event times and distances match the closed forms to at most 5.4×10^{-14} s and 9.8×10^{-14} m (the roll-onset time is exact), and the v^2 stop law holds to 2.1×10^{-14} m. The one systematic offset is a convention: the kernel's $g = 9.80665$ versus the source's 9.81 (0.03%), isolated in the u_{val} note.

One parameter set, two datasets (F1). The battery's anti-overfit centerpiece is a one-parameter-set, two-datasets test (Fig. 14a). The default parameter catalog — sliding friction 0.2, rolling friction 0.01, Han-model cushion restitution $e_0 = 0.85$ (Han, 2005), WPA cushion-nose height at 63.5% of ball diameter — was frozen before any battery sweep ran. This single set produces a lag curve monotone in cue speed whose anchor stroke travels 4.130 table lengths, inside the 4.0–4.5 lag window; and it is the same set that is scored against group C's bank dataset (Section 5.3), placing five of six through-diamond aims inside the TP B.28 five-table measured envelope (Alciatore, n.d.h) at 3 m/s with an RMSE of 0.116 diamond against the closest measured table — a registered miss of the frozen all-six criterion, stated as such below, not a pass. Lag and banks are independent datasets, and neither was used to tune the other, so each result is out-of-sample with respect to the other — in the scoped sense of Section 4.3: both datasets pre-existed as continuous-integration anchors during kernel development (lag from 2026-07-01, banks from 2026-07-03), so the claim is no-joint-re-tuning under battery-scored defaults unchanged for nineteen days before registration (chronology scoped per literal in Section 4.3), not blindness. Both caveats belong next to the showcase. The sixth bank aim is the registered miss: aim 1.0 measures 2.612

against an envelope minimum of 2.72, i.e. 0.108 diamonds long. And the pairing is catalog-specific: the opt-in `mathavan_fitted` preset — in-domain for snooker rebound coupling — lags 3.45 table lengths, and 4.83 at the literature restitution value, long the other way. The TP B.6 table-lengths overlay (rolling rollouts with cushions in play) is, by contrast, a characterized model-content difference: $E > 0$ on every rail-contacted pinned row, $+0.033$ to $+0.681$ table lengths, because the kernel's Han-model rolling rebound retains more speed than the source's stun-at-restitution-0.7 recursion assumes — a simplification the source itself flags; the rail-free "touch" row's $E = -0.0007$ is purely the gravity convention. Because each pinned row pairs a different speed with a different rail count, no per-rail-count trend is claimed; a per-rail attribution would require a matched-speed design, which the frozen registration does not include.

Draw square law (F2). TP B.8/B.9 predict draw-back distance proportional to $(R\omega)^2$ at impact (Alciatore, n.d.c). The functional form holds as a pooled law: a least-squares fit through the origin over the 26 backspin arrivals gives $R^2 = 0.986$ with $C_S = 0.320$ (Fig. 14b). The pooled qualifier matters — the per-speed fits degrade toward low spin/speed, with $R^2 = 0.42$ at 1.5 m/s and 0.94 at 2 m/s against $\geq$0.98 at 3–4 m/s. The coefficient, in turn, sits 27% below the perfect-ball value $C_D = 0.437$ obtained from TP B.8's own formula at the kernel's cloth constants ($C_S/C_D = 0.732$), and the gap is fully attributed rather than fitted away: re-running TP B.8's branch program from the measured post-impact state reproduces the simulated draw distance exactly (to the output's 10^{-8} rounding) — between events the kernel *is* the TP B.8 kinematics — so the entire rule-level error is the perfect-ball impact

idealization. Concretely, the declared restitution $e_b = 0.95$ leaves the cue ball moving forward at $(1 - e_b)/2 = 2.5\%$ of its pre-impact speed, which enters the natural-roll velocity $v''' = (5v'' + 2u'')/7$ with five-fold weight, and impact friction bleeds 3–6% of the arriving backspin on strong-draw cells. The single-coefficient law also degrades at low spin — per-speed C_S runs 0.039 at 1.5 m/s to 0.325 at 4 m/s — and follow-branch arrivals are excluded exactly as the source's own program excludes them.

The trisect rule (F2). The trisect rule (Alciatore, n.d.a) states that, with "good action" draw, the cue ball's total departure is about three times the cut angle for cuts below roughly 40°. Good action was calibrated once, per the source's definition — bisection to perpendicular cue-ball departure at a half-ball hit, giving commanded offset $b = -0.377$ and spin-to-speed ratio $\rho = -0.687$ at impact, versus the perfect-ball geometric -0.625 — then frozen across the cut sweep. The measured departure-to-cut ratio is not a plateau at 3: it declines monotonically from 3.99 at a 5° cut to 2.59 at 45°, passing through exactly 3.0 at the 30° half-ball calibration point (Fig. 14c). The ratio is within 10% of 3 only for cuts of 22.5–40°, and the absolute error $E = \theta - 3 \times$ cut peaks at +9.1° (15° cut) and −11.8° (40°) *inside* the rule's claimed domain. The window is immobile under the one-at-a-time brackets of Section 4: the grid edges (22.5°, 40.0°) do not move in any arm; the interpolated edges (22.08°, 40.18°) shift by at most 0.01° per endpoint over $e_b \in [0.92, 0.98]$ and by $+0.09°/-0.08°$ under the $\mu(v)$-revision spread; grid quantization bounds the true edges in $(20.0°, 22.5°]$ and $[40.0°, 42.5°)$; and the small-cut ratio spans only 3.99–4.01 across all arms. The 2.5°-grid-step brackets, not the bare edges, are what Table 10 carries. The small-cut excess

is not a kernel artifact, and its operating-point decomposition separates the geometric, restitution, and friction contributions. In the perfect-ball overlay the departure obeys $\tan\theta = 5\sin c\cos c\,/\,(2k - 5\sin^2 c)$, where $k = |\rho|$ is the magnitude of the spin-to-speed ratio at impact — the rolling-transfer fraction that "good action" fixes through the half-ball anchor $\theta(30°) = 90°$ — so the ideal small-cut limit is $\theta/\text{cut} \to 5/(2k) = 4.00$ at the perfect-ball calibration $k = 0.625$. Evaluating the same ideal formula *counterfactually* at the frozen calibration's $k = 0.687$ gives 3.64 — a number we quote only to size and sign the calibration shift, not as a limit of the nonideal system: the overlay is derived for frictionless perfect balls, so once $e_b < 1$ and impact friction act its collision terms change as well, and substituting a recalibrated k alone is not a valid evaluation off the ideal point. The calibration shift itself is the impact channels' doing: matching $\theta(30°) = 90°$ through the kernel's impact — where $e_b = 0.95$ leaves a forward cue-ball residual that enters the natural-roll velocity with five-fold weight, and impact friction bleeds 3–6% of the arriving backspin — requires $|\rho| = 0.687$ rather than 0.625, with the e_b channel carrying about 0.048 of the 0.062 shift (linearized from the bracket: the recalibrated $|\rho|$ runs 0.716 to 0.659 across $e_b \in [0.92, 0.98]$) and the friction channel the remainder. What the measurements then show is that the calibration anchor pins the small-cut behavior near the ideal value: the 5°-cut ratio (a finite sample, not a limit) measures 3.99 at the frozen catalog and stays at 3.99–4.01 in every bracket arm precisely because each arm recalibrates its own ρ. The breakdown of the 3× rule is therefore geometric in this calibration-anchored sense — ideal geometry at the shared anchor gives 4.0, and the measured small-cut ratio stays within 0.01–0.02 of it at every

admissible operating point — while e_b and friction set only the calibration's operating point, not the small-cut excess. The 3× line is tangent to the physics only at the half-ball point — the battery's sharpest instance of a rule that is exact at its calibration point and only there. (The departure direction at the rolling transition matches the closed-form roll direction with zero residual on every run, so the finding is physical content, not numerics.)

Jump envelope and airborne over-cut (F3). The jump experiment measures the minimum cue elevation that clears a full obstacle ball as a function of gap, mirroring the published jump chart (Alciatore, n.d.a); cells that fail to clear *are* the measured envelope (the deliverable), and all registered qualitative jump rules pass with zero violations. The envelope is a speed-dependent window: at 2 m/s only the 0.2 m gap clears (at 45°); at 3 m/s gaps to 0.6 m clear with minima of 30–35°; at 4 m/s gaps to 1.2 m clear under the sphere criterion, with the minimum falling to 25° at mid-range gaps and rising toward both edges. Within the window, longer gaps need *less* elevation (short jumps need height early), and the far edge is set by first-hop landing reach. First-landing distance peaks at 35° elevation at all three speeds, so the handout's "less elevation gives a longer jump" holds only on the descending branch. At the window's far edge the chart-style overhead clearance criterion diverges from the true sphere criterion by at most one grid step ($v_0 = 4$ m/s at a 1.3 m gap: 40° passes overhead while the descending flank clips the ball); Fig. 14d reports the sphere criterion with the overhead overlay. Airborne ball–ball contacts over-cut (HSV B.37 (Alciatore, n.d.b)): the object ball is driven along the plan projection of the tilted three-dimensional line of centers — and simultaneously downward into the slate — and the measured

effective cut tracks that geometry with a friction/throw residual of at most $|E| = 2.78°$, growing with contact height on both branches (airborne impacts ride the slip branch of the ball–ball law). The strike-channel closed forms verify as in experiment F1: cue-strike speed ratio to 6.3×10^{-6} (exactly the recorded mass-ratio rounding), launch direction $v_z/v = -\sin\theta$ to machine zero, and hop apex to the output's rounding. A design boundary completes the picture: the geometric cue-clearance floor binds only within $\sim$0.1 m of a rail, where it reaches 13.8–39°; elsewhere the ballistic requirement ($\geq 20°$ to clear a full ball on this grid) governs, so open-table jump feasibility is ballistics, not cue geometry.

The group's summary is the paper's central exhibit: the closed-form kinematics underneath the rules sits at machine precision (verification), while the teachable rules built on top of it — the trisect multiplier and the jump chart — carry the narrowest measured validity windows in the battery.

5.7. Domains-of-validity synthesis

Table 10 collects the rule-level outcome of the battery: each empirical law, its anchor class under the taxonomy of Section 4 (measured datum / derived idealized reference / practitioner lore / specification window — rows anchored only to lore carry no estimable u_{val} and are characterizations of the rule as printed), the variable it was scored on, its validity domain as measured, and the first mechanism that breaks it. ASME V&V 20 places extrapolation beyond validation points out of scope (ASME, 2009); the table makes that boundary itself the result — to our knowledge the first systematic quantification of the domains of validity of the empirical laws of pool (contribution C1, Section 1.1). Three patterns recur. The rules hold at or

near their teaching conditions, often exactly at a calibration point (the trisect multiplier, the ten-times-fuller slope, the gearing-english percentages). The first failures are, in order of prevalence: idealized parameters in the reference forms ($e = 1$ restitution, flat retention, frictionless-elastic peaks), speed extrapolation beyond the teaching condition, and measurement or protocol conventions rather than mechanics. Per-row output provenance is given in Supplementary Material Section S1.

Table 10: The empirical laws of pool: measured domains of validity. One row per rule; comparison error $E = S - D$ per anchor; per-row provenance in Supplementary Material Section S1. The anchor-class column applies the taxonomy of Section 4: *meas.* = measured datum (instrumented measurement), *deriv.* = derived idealized reference (a source's own closed form, construction, or numerics; "(port)" where the kernel implements the source's own model), *lore* = rule-of-thumb practitioner claim (u_{val} not estimable), *spec.* = published specification window. Diamond units abbreviated "dia"; table lengths "TL". Lettered marks ([a]–[l]) resolve to the notes at the end of the table.

Rule	Class	Validation variable	Measured validity domain	First failure mechanism
Group A — ball–ball impact and throw				
90° stun line; 85° real-ball rule	deriv. + lore	separation angle	ideal form verified; real balls measure 82.49° at slow-speed half-ball	$\mu(v)$ friction and e_b: the real-ball value drifts with speed and cut
30° rule; 33.7° peak	deriv.	carom deflection	peak location 33.71° and 27–34° plateau at all registered speeds and in every bracket arm; $33.7 \pm 0.3°$ peak height at slow speed only	frictionless-elastic reference form; $E = -0.83°$ at 2 m/s; band membership e_b-fragile[a]
CIT: $\leq 20°$ speed collapse; 4× roll/draw reduction	deriv. (port)	throw vs cut, speed, spin	collapse holds to 17.5° — crossover in $(17.5°, 20.0°]$ at the 2.5° grid step; closed-form cap departure 18.0° [b] — ∼4× reduction at medium/fast speed	$\mu(v)$ crossover; saturation relaxes the reduction to ∼2.3× at slow speed
SIT maximum at stun, slow, ∼50% english	deriv.	SIT peak locus	peak at 47.1% english, inside the registered $50 \pm 5\%$ band	none observed on the registered grid
Gearing english 80/60/40%	deriv.	zero-throw crossings	reproduced at 0/25/50% ball fraction; invariant across the slip-law revision	none observed on the registered grid

Continued on next page

Table 10 *(continued)*

Rule	Class	Validation variable	Measured validity domain	First failure mechanism
Spin transfer CB→OB	deriv.	OB spin pickup	friction-capped transfer curves confirmed	declared e_b: slip ratio $(1+e_b)/2 = 0.975$ vs the $e = 1$ worksheets
Cling (chalk-spot) envelope	deriv.	worst-case throw	bounded at $+5.15°$ ($k = 3$, $50°$ cut, medium speed); friction-independence boundaries $32.5/25.0/18.0°$ [c]	envelope content by design; source revision drift recorded
Group B — stick–ball				
Endmass squirt closed form	deriv.	squirt angle	exact match; near-linear in offset; speed-invariant	protocol conventions only
Natural pivot length (8–14 in band)	meas.	pivot length	closed form reproduced; catalog measures 10.5–64.9 in under the adopted tip-center convention[d]	measurement convention: tip-center vs contact offset
Miscue limit 0.5–$0.55R$	lore	maximum side offset	$0.5R$ bound conservative (implied μ_{tip} 0.39–0.58); $(0.5, 0.55]R$ unreachable[l]	commanded-offset input surface, no tip-friction model
Center of percussion (70% height)	deriv.	slide-phase duration	duration $\to 0$ at $0.7 \cdot 2R$ tip height; overspin above	none mechanical; long-slide rows censored by table runway
Group C — cushion and diamond systems				
Sliding-bank through-diamond tables	meas.	arrival diamond	5/6 aims inside the TP B.28 envelope at 3 m/s; twice-plus-tenths replicates	aim 1.0 banks 0.108 dia long; shallow-aim cushion calibration

Continued on next page

Table 10 *(continued)*

Rule	Class	Validation variable	Measured validity domain	First failure mechanism
Bank speed/spin correction lore	lore	bank-angle shift direction	not reproduced as stated	direction reversals resolved by the admissible-space sweep and cross-form crossing (Section 5.3): draw ordering is a Han/Stronge model-form boundary[e]; sliding-45° speed direction splits by cushion family[f]
Corner-5 three-rail system ($T = D - F$)	lore	arrival diamond	RMSE 0.10–0.12 dia over $D = 2$–5	fourth-rail speed/english lore splits against itself
1/2/3-diamond sidespin kicks	lore	cross-table arrival	confirmed on cue-only kicks at reference speed (calibration-consistent: μ_w was pinned on this anchor family, Section 4.3)	natural-roll protocol assumption at 3 m/s
Cushion rebound/spin coupling	meas. + deriv. (port)	rebound angle and speed	measured Fig. 7a camera points 12/12 within tolerance[g]; port at source constants 239/240; out-of-sample `han2005_default` 52/240[h], below 2.5 m/s	source's own rigid-cushion domain above 2.5 m/s; one tolerance-edge point
Group D — curved paths				
Coriolis parabolic slide path	deriv.	quadratic path fit	exact ($1 - R^2 = 0$) in the slip-aligned frame	frame-sensitive criterion only
Massé BAR aiming	deriv.	final CB direction	parallel to the aim line; 88% of cells slightly short, as the source expects	low-elevation/high-speed degradation (the source's own caveat)
Swerve	lore	lateral drift	zero at level cue; drift $\propto$ elevation $\times$ spin	onset delayed at speed (as taught); none beyond

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Table 10 *(continued)*

Rule	Class	Validation variable	Measured validity domain	First failure mechanism
Group E — frozen contacts and the break				
Ten-times-fuller	lore	kissed-ball heading slope	0.106 at the 45° reference; within ∼1.5° for fullness ≥ 25°	flat-retention idealization[i]
Frozen ≈ small-gap combination throw; $R/3$ optimum	deriv. + meas.	combination throw vs gap	two-body CIT from a 0.1 mm gap; straightest gap 9.85–9.99 mm vs 9.525 mm	compliant frozen-limit excess 1.13–1.55× (open)
Break squat; effective mass	meas.	CB residual fraction; M_{eff}/m	squat 0.137–0.159 inside the evidence band at 3.2–12.9 m/s; M_{eff}/m 1.287–1.374, scaling with cluster size	connected-cluster topology caps M_{eff} below the 1.5 concept value; scaling erased by the pairwise resolver
Group F — kinematics and the jump				
Slide→roll and stop-law closed forms	deriv.	event times and distances	machine precision ($\leq 10^{-13}$)	verification floor; 0.03% gravity convention
Lag benchmark (4.0–4.5 TL)	spec.	table lengths	4.13 on the frozen default catalog, jointly with the bank envelope (out-of-sample)	catalog-specific: `mathavan_fitted` lags 3.45 TL
Draw square law	deriv.	draw distance vs $(R\omega)^2$	pooled form holds ($R^2 = 0.986$)	degrades at low spin/speed[j]; perfect-ball idealization: $C_S/C_D = 0.732$ (e_b plus impact-friction spin bleed)
Trisect 3× rule	lore	departure/cut ratio	exact at 30°; within 10% of 3 for 22.5–40° cuts[k]	geometry itself: ratio → 4.0 as cut → 0
Jump minimum-elevation chart	lore	minimum elevation vs gap	speed-dependent window measured (to 1.2 m at 4 m/s)	clearance-criterion convention at the far edge

Continued on next page

Table 10 *(continued)*

Rule	Class	Validation variable	Measured validity domain	First failure mechanism
Airborne over-cut	deriv.	effective cut angle	tracks the tilted line-of-centers plan projection; $\max \lvert E \rvert = 2.78°$	three-dimensional contact geometry; grows with contact height

Notes. [a] ± 0.03 on e_b shifts the peak family $\mp 0.9°$. [b] cap departure 17.9–18.2° over the e_b bracket; 20.9° under the $\mu(v)$ revision; source crossover $\approx 18°$. [c] boundaries move $\pm 0.34/0.24/0.15°$ over the e_b bracket; 34.1/27.7/20.9° under the $\mu(v)$ revision; 10°-grid brackets $[30, 40]/[20, 30]/[10, 20]°$. [d] in-sample calibration consistency: the catalog end masses derive from the same Platinum pivot data, Section 4. [e] draw ordering: 0 of 210 cells restore it; compliant Mathavan correct in all 84, compliance-free flips 6 of 84, $\leq +0.52°$. [f] sliding-45° speed direction: all flat-μ_w cells restore it, all fitted × compliant cells reverse, Han × `hsv_fitted` flips 10 of 15 (to $+2.72°$). [g] $\max \lvert E \rvert = 0.067\,\mathrm{m/s} = 1.2\,u_{\mathrm{val}}$ with u_x propagated through the response slope. [h] model-form swap at source μ_{wall}: 57/240. [i] $E = +8.3°$ at 5° fullness; +5.7 to +7.7° under the full frozen-pair-excess deflation bound, rule crossing 28.3–31.1° — the excess survives. [j] per-speed R^2 0.42 at 1.5 m/s, 0.94 at 2 m/s. [k] true edges in $(20.0, 22.5]$ / $[40.0, 42.5)°$ at grid step; interpolated edges immobile over the e_b bracket, $+0.09/-0.08°$ under the $\mu(v)$ revision. [l] registered convention; revised post-registration to bind the contact offset (Section 8); accepted inputs reproduce bit-exactly, the nickel-tip rejection probes would now be accepted (artifact unchanged), $(0.5, 0.55]R$ still unreachable.

6. Cross-simulator agreement

The battery's fourth anchor tier is an independent simulator. We use pooltool (Kiefl, 2024) at a pinned release (version 0.6.0, commit `9a8abfe0`) as a cross-simulator oracle wherever the two model registries coincide, and we state at the outset that every comparison in this section is *verification* in the ASME V&V 20 sense (ASME, 2009): code against code, under matched models and identical inputs. The oracle harness (ORACLE-X) re-executes the registered A3 and A6 designs — 38 matched shots — through both engines and compares event times, event sequences, post-impact observables, and rest states point by point (Fig. 15).

6.1. Agreement under the parity convention

ORACLE-X feeds both engines identical initial states and strike parameters and matches the model registries exactly: the same ball parameters on both sides, the Han cushion family (Han, 2005), the same Alciatore $\mu(v)$ ball–ball friction form (Alciatore, n.d.d), and matched stick–ball models. One pin is deliberate and permanent. pooltool's ball–ball resolver carries the pre-fix slip-impulse convention (the case study below), so the kernel side of every oracle run explicitly selects the retained legacy law rather than the corrected default. We call this the *parity-convention law*: oracle parity is defined against the legacy slip law permanently, which makes the oracle harness a fixed-convention verification instrument rather than a moving target.

Under that convention the two engines agree at the numerical floor of the comparison. Across all 38 points, ball–ball impact times agree to 7.4×10^{-14} s at worst; throw angles to $\max|E| = 1.12 \times 10^{-4}\,^{\circ}$; post-impact object-

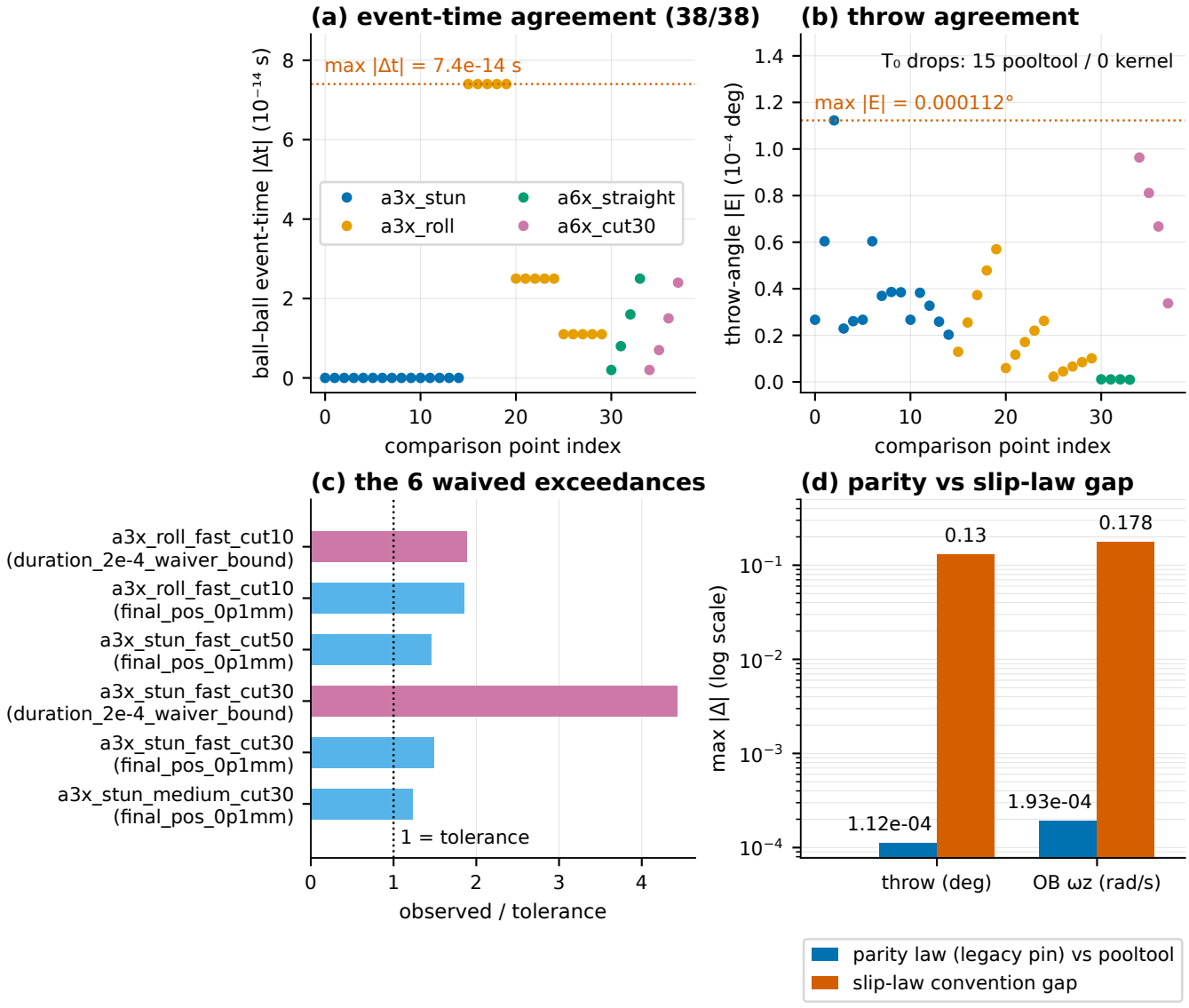


Figure 15: **Verification.** Cross-simulator verification against pooltool (ORACLE-X: 38 matched points on the registered A3/A6 subsets, run under the legacy parity convention). (a) Ball–ball event-time agreement; the panel title carries the 38/38 event-sequence match under the declared zero-length-transition drop rule, whose per-side drop counts are noted in panel (b). (b) Throw-angle comparison error. (c) The six waived exceedances, all attributed to the oracle's `make_kiss` re-seat. (d) The measured slip-law convention gap between the parity law and the corrected default; values in the text.

ball spin ω_z to 1.9×10^{-4} rad/s; and post-strike cue-ball states to 5.2×10^{-11} m/s. Event sequences match 38/38 after one declared bookkeeping rule: transitions within 10^{-9} s of the strike are dropped from both streams, with the drop count recorded per point. The rule exists because the rolling strike sits exactly on the $\omega R/v = 1$ classification boundary, where pooltool emits a zero-length sliding-to-rolling transition at $\sim 6 \times 10^{-17}$ s that the kernel folds into the strike post-state; this occurs in 15 of the 38 runs and is bookkeeping, not dynamics.

Six tolerance exceedances remain, and all six carry one attributed oracle-side mechanism. Four final rest positions exceed the 0.1 mm gate (worst case 0.185 mm) and two rollout durations exceed even the 2×10^{-4} s waiver ceiling (worst case 8.9×10^{-4} s), exclusively on shots that roll out for up to $\sim$7.5 s. pooltool re-seats ball positions by $\sim 1\,\mu$m (its documented `make_kiss` step) before resolving a ball–ball contact; the kernel resolves at the exact quartic contact point and never re-seats, so the micrometer-scale perturbation is amplified through the cushion and roll legs of long rollouts. This is a documented convention difference, not a physics disagreement, and it is waived with exactly that attribution.

Finally, the harness cross-references its shared points against the registered battery outputs, which ride the corrected default. That comparison is scoped as a *measurement* of the slip-law convention gap, never as an identity gate: the maximum throw difference on the 30 shared A3 points is 0.1301°, the maximum ω_z difference on the 8 shared A6 points is 0.1783 rad/s, and stick-branch points coincide. These numbers *are* the slip-law shift, by design (panel (d) of Fig. 15).

6.2. Case study: two defects found by the methodology

A useful test of a verification and validation methodology is whether it finds defects its authors did not know were there. The battery's first full run found two.

The slip-law defect (ball–ball slip impulse). Experiment A1's slip-law characterization recovered a friction ratio of 0.97424 against the Coulomb-cap form of the technical-proof derivations (Alciatore, n.d.c), within 1.2×10^{-4} of the value $2e_b/(1+e_b) = 0.97436$ predicted if the friction impulse were mis-scaled — slip-regime throw ran approximately 2.6% low, an offset subsequently visible across the A3, A4, A6, D1, E2, and F3 channels. The root cause: the kernel scaled the ball–ball slip friction impulse with the post-impact normal velocity difference, $\mu\, e_b\, v_n$ — a convention inherited from the reference-implementation lineage the kernel shares with its oracle. The Coulomb cap instead requires the normal impulse per unit mass, $\mu\,(1+e_b)/2\, v_n$, a +2.6% relative slip-impulse difference at $e_b = 0.95$, worth about 0.12–0.13° of throw at a slow half-ball hit. The corrected law is now the kernel default — slip-regime throw is exactly $\arctan\mu$, the reference form of Alciatore (n.d.d) — with the legacy law retained behind an explicit option.

The branch-selection defect (settled-rebound branch selection). Independently, the C1 bank grid showed through-diamond arrival discontinuities of 0.49–0.85 diamonds at mid aims under the Stronge-compliant cushion preset (Stronge, 2018), with no intermediate arrivals. The root cause: the model's terminal stick-to-slip equation was solved by a whole-bracket root finder, and at the structural double root at stick onset the bracket converged onto

whichever solution family 10^{-16}-scale floating-point noise favored. Sliding and stun cushion contacts near that tangency therefore jumped between rebound families (the tangential ratio stepping from 0.60 to 0.74) as the aim varied smoothly. The fix selects the first crossing by a deterministic scan and is continuous by construction; the legacy selection is likewise retained behind an explicit option.

Neither defect was findable by oracle agreement. The slip-impulse convention is shared with the oracle by inheritance, so ORACLE-X agrees at the 10^{-14} s level on both sides of the fix; and the matched-registry oracle runs exercise only the Han cushion family, never the Stronge branch. The independent published-curve anchors were load-bearing — which is the point of a multi-source battery, and, in our reading, the strongest single piece of evidence for contribution C2.

Predicted-shift verification. Before the post-fix battery re-run, each affected experiment received a written prediction of the direction and mechanism of its shift, and no registered band or catalog parameter was re-tuned. Seven prediction classes were registered, covering every experiment in the battery, and all seven confirmed. Table 11 extracts the two classes that carry the case study — one per defect; the full seven-class table is Supplementary Material Table S3.

Legacy preservation. Both pre-fix laws remain available behind explicit options, with byte-replay proofs: of the 14 archived event streams re-simulated on the corrected binary, 9 are byte-identical, and the 5 registered slip-law-class divergences each reproduce their pre-fix stream byte-identically under the explicit legacy options (Table 12, rows 5 and 10). Model evolution is

Table 11: Predicted-shift verification across the slip-law and branch-selection fix cascade: the two prediction classes carrying the case study, one per defect (extract; the full seven-class table — all seven classes confirmed, no acceptance band or catalog parameter retuned — is Supplementary Material Table S3). "Bit-identical" means byte equality of the affected rows after masking kernel-version stamps.

Experiments	Registered prediction	Measured outcome
C1 (branch selection)	the four `stronge_realistic` discontinuous bank cells collapse; other presets untouched	cells $4 \to 0$; the three-crossing cell (aim 3.0, 1 m/s) monotone; `han2005`, `mathavan`, and `grip` rows bit-identical
A3 (slip law)	slip tan-argument ratio $\to 1$ (throw exactly $\arctan \mu$); stick regime unchanged	ratio $0.9745 \to 1.00014$; stick ratio 1.02564 unchanged; worst slip-channel $\lvert E \rvert$ vs the transcribed master curves $0.0886 \to 0.0012°$

thereby held to the same determinism doctrine as the shipped binary itself: a superseded law is not deleted but pinned, and its historical outputs remain reconstructible to the byte.

6.3. Verification/validation boundary in this section

To restate the boundary: everything in this section is verification — code against code, or code against its own closed forms — in the vocabulary of ASME V&V 20 (ASME, 2009). No validation claim in this paper is derived from oracle agreement; validation against physical measurement and published data lives in Section 5, and the parity-convention law above exists precisely to keep the two from contaminating each other.

7. Reproducibility

This section states the reproduction contract behind every number and figure in the paper, the evidence that the contract holds, and what of it the public source release supports. The contract is deliberately narrow and mechanical: one canonical binary, one manifest of expected hashes, and hash equality as the verification criterion.

7.1. Source availability

The kernel, its WebAssembly and Python bindings, the TypeScript SDK, and the application that consumes them are published as source in the project's public repository under the MIT license; the portions of the kernel derived from pooltool (Kiefl, 2024) remain under the Apache License 2.0, and the repository's notice files identify the upstream project and list the derived files. The public repository receives curated snapshots of the project's

development repository, and its history begins at a snapshot later than the kernel revision this paper reports on. Two limits follow, and we state them plainly. First, the canonical kernel of record, `0.1.0+fb3425ad`, is not part of the public history: a build from the public tree is a later kernel revision, the hashes and digests quoted in this paper cannot be regenerated bit-for-bit from it, and this paper reports no measurement made on it. Second, the canonical `pool_wasm` binary of record (SHA-256 prefix `5e3b6ee4`) with its Python wheel and npm SDK tarball, the experiment scripts, the oracle harness, the 25 pinned outputs, the expected-hash manifest, the waiver ledger, and the validated-scenario video pack described below belong to the artifact bundle, which is not part of the public release and has not been published. The registered results quoted in this paper are those of the `0.1.0+fb3425ad` battery, and checks made on later canonical stamps state their stamp (the lineage notes below; Table 12).

7.2. Tier-1 protocol

The reproduction protocol's first tier is designed as *anyone, minutes — execute the canonical artifact*; until the bundle is published, it can be executed only within the project. Its bundle contains the pinned canonical `pool_wasm` binary, whose version-hash string is identical to the stamp recorded in replay files, together with the scripts, fixtures, expected hashes, and a results map tying every figure and table in this paper to the script that regenerates its data. The bundle also carries the validated-scenario video pack: sixteen slow-motion renders with predicted-overlay annotations, each replaying a solved strike through the then-current shipped canonical binary with its pinned record digest reproduced *before* any frame is captured (per-

entry kernel stamps recorded in the pack's gate reports), accompanied by per-entry machine-readable gate reports and a pack ledger carrying per-entry SHA-256 hashes of the delivered bytes together with the render and verification history, including the cue-model provenance (presentation methods in Supplementary Material Section S6); the videos present verified records, and no claim of this paper rests on them. Verification *is* hash equality: the reproducer's success criterion is a byte-for-byte match, not a visual one. The entire kernel of Fig. 2 ships as this single hash-identified binary.

A second tier — rebuilding the kernel from source — exists through the public repository, with a narrower scope than tier 1. The shipped WebAssembly binary remains the paper's normative physics definition, a posture with precedent in deterministic-WebAssembly runtimes that treat the hash-identified binary as the consensus artifact (WebAssembly Community Group, 2015). Native and PyO3 builds are development conveniences: they are held to the wasm by byte comparison where we run them (Table 12), but the `.wasm` alone defines expected outputs. A rebuild from the public source therefore lets a reader inspect and test the published models, not regenerate this paper's hashes, which are bound to `0.1.0+fb3425ad` (Section 7.1). The costs of this posture are discussed in Section 8.

WebAssembly can carry this obligation because its numeric semantics are IEEE-deterministic up to two specified escape hatches — NaN payload bits and relaxed SIMD (WebAssembly Community Group, 2015) — and the kernel excludes both by construction: simulation state is NaN-free by an invariant enforced at the API boundary, and the binary uses no SIMD and no threading. Every transcendental function is compiled into the binary

rather than imported from the host, so engine-to-engine variance in host math libraries never enters the physics. Replay is version-hashed end to end, and model evolution is held to the same doctrine: an archived corpus of event streams re-simulates byte-identically on the committed binary, with the five streams that predate the corrected slip law reproducing their pre-fix bytes only under the explicit legacy options (Table 12, row 5; Section 6).

The bundle is designed to the FAIR principles (Wilkinson et al., 2016) and their research-software profile (Chue Hong et al., 2022): findable and accessible once published, interoperable (self-describing JSON outputs carrying kernel version, shot parameters, and fixture identifiers), and reusable (per-fixture provenance blocks). The bundle inventory and the step-by-step five-step procedure are the mechanics of Supplementary Material Section S2.

7.3. Baseline identity and the two-sided proof

The paper's baseline of record was recorded under the canonical wasm stamped `0.1.0+fb3425ad`; the battery outputs themselves were produced by a native Python-surface build stamped `0.1.0+a321d64f`. The bridge between the two stamps is a source-identity proof, not an assumption: the version-control diff between the two pins over every compiled kernel source is empty (the only intervening changes are test-only files not compiled into the library), and the rebuilt environment passes all 13 smoke checks. The bridge is no longer load-bearing on its own: every kernel simulation the battery executes has since been replayed end-to-end through the shipped binary itself with digest equality (matrix row 11 below). Four lineage notes apply: after the baseline was recorded, the canonical stamp advanced to `0.1.0+c1c3de9a` through a catalog-only change (two visual ball-set entries,

with fixture re-pins proving the event streams byte-identical), and subsequently to `0.1.0+8efd660e` through the post-registration input-envelope revision (Section 8), proven bit-exact for every previously accepted input at its repin — two supersessions that leave the physics definition of record for every simulated battery input unchanged (the B3 rejection-probe disclosure is the amendment note of Section 8); and thirdly to `0.1.0+b4d6e418` through the pinch-strike revision — the first supersession that changes simulated physics, confined to strikes at cue elevations above 60°. Its battery effect (the 65–75° cells of experiment D2) and the materialized B3 rejection-probe class are measured and re-baselined in the study's baseline manifest. Fourthly, the stamp advanced to `0.1.0+e654eb93` through the ball–ball at-contact window guard — a detection-correctness fix inside the frozen impact semantics (an airborne re-collision immediately after a same-instant compound event went undetected), deliberately shipped without a legacy option because the pre-fix behavior is a defect, not a model choice. It is, to date, the only supersession beyond the three above that has moved any battery output: exactly one trajectory-digest leaf of experiment F3's verification cell 3, with the cell's verdict and every measured column unchanged. The drift is carried as a hash-bound waiver of the continuous-reproduction job and as a numbered exclusion in the study's ledger, with its census recorded in the baseline manifest, and is retired at the submission-freeze battery re-baseline. Later canonical advances are catalog and serialization repins proven battery-empty, chronicled in the same manifest. In every case the registered results quoted in this paper remain those of the `0.1.0+fb3425ad` battery.

The baseline manifest is proven two-sided. In one direction, the current

outputs are pinned by their SHA-256 hashes. In the other, the superseded pre-cascade baseline (build `0.1.0+5be7647`) was regenerated from a clean working tree at its own pin and reproduced all 25 pinned pre-cascade hashes byte-identically, including the oracle run. Every drift row in the manifest is therefore a measured old-versus-new comparison on byte-verified endpoints — never an inference from hash inequality.

Raw file hashes are commit-bound by construction: every output embeds the kernel build stamp, which contains the source-control short hash. The battery therefore also fingerprints trajectories in a stamp-independent form: every serialized shot record carries a digest computed with the version stamp replaced by a fixed token — a cross-build trajectory fingerprint that identity checks *compare*, never strip (checker mechanics in Supplementary Material Section S2). At the re-record that established this normalization, a JSON tree comparison excluding digest fields showed zero differences across all 25 outputs; all 3157 digest leaves were re-recorded; and the full battery, run twice, produced 25/25 byte-identical outputs across runs.

7.4. Determinism audit matrix

Table 12 transcribes the determinism evidence matrix: eleven verdict rows, each citing a mechanically checked run — a fixture-corpus byte comparison, a continuous-integration job, a browser boot pair, or the battery reconstruction above — rather than an argument. The comparison methods are: (A) the native command-line kernel versus the canonical-source wasm under Node, byte comparison of seven serialized shot records; (B) the seven native records byte-compared across operating systems and architectures; (C) an archived corpus of 14 event streams re-simulated on the committed wasm

and byte-diffed, with registered divergences re-proven through explicit legacy options; (D) hermetic browser boots with submitted scenario bytes byte-compared; (E) cross-surface byte identity (command line, TypeScript SDK on the committed wasm, and Python SDK); (F) the rules layer's three-leg byte identity against frozen corpus bytes, raw comparison with no normalization; (G) the paper battery's byte-stable reconstruction described above; and (H) the battery's full recorded-input corpus re-executed through a shipped wasm binary under Node — an opt-in tap on the battery's simulation entry points records one row per kernel simulation (the serde-exact inputs plus the stamp-normalized trajectory digest of the native record; tap-on and tap-off battery outputs proven byte-identical), and the replay harness feeds each row verbatim to the shipped binary through the committed loader and compares digests, after a ten-record preflight proving native and wasm record JSON byte-identical modulo the version stamp (the harness, the per-experiment counts, and the corpus SHA-256 are recorded in the bundle's replay report).

Table 12: Determinism audit matrix at the paper's kernel pin: artifacts compared, platform and runtime, method (letters defined in the text), and verdict. Every row cites a mechanically checked run. The matrix's untested cells — non-V8 JavaScript engines, wasmtime, Windows, further OS/architecture pairs, wheel stamp coherence, and third-party attestation — are stated as untested in the text and in Section 8. Lettered marks resolve to the notes at the end of the table.

#	Artifacts compared	Platform / runtime	Method	Verdict
1	native CLI vs canonical-source wasm	macOS arm64, Node 26 (V8)	A	PASS, 7/7 byte-identical
2	native CLI vs canonical-source wasm	Ubuntu x86_64, Node 22 (V8)	A	PASS
3	native CLI vs canonical-source wasm	macOS arm64 (CI), Node 22 (V8)	A	PASS
4	native records across OS and architecture	Ubuntu x86_64 vs macOS arm64	B	PASS, byte-identical
5	committed wasm vs archived pre-fix capture	Ubuntu x86_64, Node 22 (V8)	C	PASS: 9/14 streams byte-identical; 5 divergences reproduce pre-fix bytes[a]
6	committed wasm, in-browser	macOS arm64, Chrome (V8)	D	PASS, bytes identical across hermetic boots

Continued on next page

Table 12 *(continued)*

#	Artifacts compared	Platform / runtime	Method	Verdict
7	CLI vs TypeScript SDK (wasm) vs Python SDK	Ubuntu x86_64	E	PASS
8	rules layer, three legs vs frozen corpus bytes	Ubuntu x86_64 and macOS arm64	F	PASS (raw `cmp`, no normalization)
9	paper battery, 25 outputs (native Python SDK)	macOS arm64, CPython 3.12	G	PASS, 25/25 hashes reproduced
10	committed wasm, break-fixture event bytes across model revisions	Ubuntu x86_64, Node 22 (V8)	C	PASS: divergence registered honestly (195 → 189 events); pre-fix stream retained and reproduced
11	paper battery, all 16,933 recorded simulations vs shipped wasm	macOS arm64, Node 26 (V8)	H	PASS, 16,933/16,933 digest-identical on both binaries[b]

Notes. [a] registered slip-law-class divergences, re-proven under explicit legacy options. [b] on BOTH the paper-canonical (`0.1.0+fb3425ad`) and the superseding committed (`0.1.0+c1c3de9a`) binaries; 10/10 preflight records byte-identical modulo stamp.

One measured platform sensitivity is reported deliberately, because it argues *for* the binary-canonical doctrine rather than against it. A regression probe that sits intentionally on the structural double root of the branch-
1955 selection defect (Section 6) fails on Linux x86_64 native builds while passing on macOS arm64 native builds: under the explicit *legacy* bracket se-

lection, the probe measures a tangent-ratio step of 2.7×10^{-5} on Linux instead of the >0.1 jump recorded on macOS, because code-generation and math-library differences between native targets flip which solution family the legacy path settles on at the 10^{-16}-noise tangency. No shipped-artifact obligation is breached — the corrected default is continuous by construction, and the canonical wasm is one binary that native code-generation variance cannot reach — but the probe demonstrates that native rebuilds are not bit-trustworthy at measure-zero critical points, which is exactly why the expected outputs are defined by the `.wasm`.

The matrix's untested cells are stated as untested, not argued away; they are enumerated, with the structural argument for why the untested engine cells are expected to pass, in Section 8. The cell that mattered most for reading this paper is now closed by measurement rather than argument: row 11 replays all 16,933 recorded simulations of the battery end-to-end through the shipped binary with stamp-normalized trajectory-digest equality (16,933/16,933), and the replayed digest set covers all 3,157 digest-pinned evidence records carried in the pinned outputs, so the battery grids no longer rest on the native surface plus the source-identity bridge. The remaining reader-facing cell is that no non-author has yet executed the tier-1 protocol. The doctrine is to measure, not argue: each remaining cell is an open pre-submission item.

7.5. Continuous reproduction

In the development repository the battery also runs as a continuous-integration target; the job and its inputs are not part of the public release. A scheduled (weekly, plus on-demand) reproduction job — deliberately in-

formational, never a merge gate — rebuilds the Python surface from pinned sources with a pinned numerical stack, runs the 13-check smoke test, executes all 24 experiments plus the oracle harness, and checks every output for physics identity against the committed normalized manifest under the normalization above (mask version stamps only; compare trajectory digests). Where the runner lacks the pinned oracle environment, the oracle harness records an explicit skip reason in its output: a recorded skip is data; a silent skip is a failure. The runner platform is itself part of the baseline identity — one least-squares fit in experiment D1 differs in full precision between BLAS back ends with unchanged trajectory digests — so the job runs on the platform the baseline was recorded on.

Any unwaivered mismatch fails the job. A mismatch can be downgraded only by a waiver entry carrying an identifier, a physical rationale, and a tracking reference — the loader rejects entries missing any of the three — and this hash-bound waiver ledger is empty by design at the baseline of record. The paper-level ledger travels the same way: it ships verbatim in the artifact bundle, and Section 8 is organized by its categories, so the paper and the bundle reference one machine-readable object.

Two items close the loop before submission: the wheel and the battery are rebuilt at the submission-freeze commit, so that raw hash equality holds with no normalization at all; and a third party (a CODECHECK-style attestation) executes tier 1 on all three operating systems and confirms the hashes. Both are recorded above as open cells of the audit matrix.

8. Limitations

Every limitation in this section is either a row of the machine-readable deviation ledger kept in the artifact bundle (38 rows across nine categories; per-category counts in Table 5, Section 4) or a declared scope boundary of the study itself. The resolved rows and the oracle waiver are treated in Section 6; the declared-model-content rows and the frame artifact are reported alongside their experiments in Section 5. The remaining categories — those that bound what this paper's conclusions may claim — are collected here, by category.

8.1. From the waiver ledger, by category

Registered-band misses (seven rows). Seven registered acceptance bands or source claims were missed. Following the anti-overfit doctrine of Section 4 — and ASME V&V 20's position that validation is not pass/fail (ASME, 2009) — each miss is measured and characterized, no band was re-tuned, and each row's full characterization lives in the ledger itself (artifact bundle). (i) The 30°-rule peak height is in-band only at slow speed ($E = -0.83°$ at 2.0 m/s; band membership itself e_b-fragile, Section 5.1). (ii) One of 240 anchored cushion-rebound points from Mathavan et al. (2010) misses at the band edge ($E = +0.0577$ against 0.056), a tolerance-edge flip recorded as a miss all the same. (iii) Under the calibrated Stronge cushion pairing, the 30° sliding-bank speed-independence anchor spreads +2.71° against its $\leq 1.9°$ window, and the 45° sliding bank *lengthens* with speed where the source claims shortening; the admissible-space sweep of Section 5.3 attributes the *magnitude* of the speed-direction reversal to the interaction of the fitted $\mu(v_{\text{slip}})$ curves

with the tangential-compliance pairings — over the full cross-form crossing every flat-μ_w cell restores the source direction (270 of 270) and every fitted-curve × compliant-pairing cell reverses (54 of 54), while the fitted curves leave the non-compliant forms source-correct in 44 of 54 cells (the exceptions, Han under the `hsv_fitted` curve, flip by +0.28 to +2.72° off the default operating point, still well below this row's reversals at matched corners) — so this row's reversal magnitude is content of the calibrated compliant pairing. (iv) Three further banking targets fail as pinned, documented gates — steep rolling banks go long under every preset (`han2005_default` tangent ratio 1.701 against the ≤ 1.45 cap), and draw *lengthens* the Han impact rebound where the source ordering says shorten; the same sweep and crossing find the draw-ordering reversal in *every* Han/Stronge cell — all 210 crossing cells, both fitted curves included — and in no cell of the shipped tangential-compliance Mathavan form (0 of 84; its compliance-free variant flips in 6 of 84 cells at flat $\mu_w \geq 0.25$, by at most +0.52°), i.e. a model-form boundary of the single-contact impulse class rather than a parameter choice. All of these banking residuals are bit-identical across cushion branch-selection variants — none is a remnant of the resolved branch-selection defect. (v) The frozen all-six bank criterion is met at only five of six aims — five are inside the envelope and the criterion fails on one: aim 1.0 banks 0.108 diamonds beyond the five-table TP B.28 envelope (Alciatore, n.d.h). (vi) Break effective mass scales with the connected cluster but caps at $E = -0.14$ below the high-speed-video concept value of 1.5 (Alciatore, n.d.b), a scaling the sequential baseline erases entirely. (vii) The trisect rule holds within 10% of its 3× factor only for 22.5–40° cuts, a miss of the registered "below 40°" phrasing

that we regard as the study's flagship domains-of-validity result rather than a defect (Section 5.6).

Convention sensitivities (four rows). Four results depend on an input or measurement convention; both readings are recorded per row, one adopted with rationale. The squirt catalog's tip-center reading of the robot protocol undershoots all four robot-measured shafts by 0.32–0.42°, with the contact re-reading closing the gap (Section 5.2). The miscue limit is a one-sidedly conservative commanded-offset envelope with no tip-friction model, implied tip friction 0.39–0.58. The jump chart's overhead clearance criterion overstates reach by up to one grid step at the window's far edge, against the sphere criterion the paper adopts. And oracle parity is pinned permanently to the pre-fix slip law, so the oracle channel measures the slip-law convention gap (0.1301° in throw) by design (Section 6).

Post-registration input-envelope amendment. After the registered baseline was recorded, the kernel's input-envelope convention was revised (kernel `0.1.0+8efd660e`): the miscue limit now binds the ball-surface *contact* offset — the variable in which the 0.5–0.55R literature bracket is stated — raising the commanded tip-center cap from $0.5R$ to $0.5\,(1 + r_{\text{tip}}/R)$, i.e. $0.686R$ for the shipped nickel tip (contact offset exactly $0.5R$ at the cap). The revision widens input validation only, and bit-exact reproduction holds for every previously *accepted* input — which includes every simulated trajectory in the battery. One behavioral change class exists and is disclosed rather than absorbed: B3's registered rejection probes are boundary probes *of* the registered envelope itself, and the nine run under the nickel tip (eight sweep-tail

cells at commanded 0.525–0.6R, contact 0.383–0.438R, plus one combined-axis probe) would be accepted and simulate under the revised convention, while the eight contact-mode cells still reject (their contact offsets exceed the 0.5R contact limit under every convention). Those recorded rejections are convention-bound results of the registered kernel, not reproducible behaviors of the revised one; the pinned `B3` artifact stands unchanged as the registered-envelope record referenced by the registered results, while post-registration maintenance re-baselines under the revised envelope are recorded in the study's baseline manifest. No simulated registered result changes, and the registered battery results remain pinned to the `0.1.0+fb3425ad` lineage (Section 7). The contact-offset band $(0.5, 0.55]R$ remains unreachable — the envelope stays on the conservative side of the literature bracket.

Source discrepancies (three rows). Three deviations belong to the source material itself; the battery records both readings. The 2005 printing of one technical proof fits the ball–ball friction decay with $c = 0.77$ where the current revision — and the kernel — use $c = 1.088$; each document transcription-checks against its own form (Alciatore, n.d.c,n). Seven points in Mathavan et al. (2010) are internally inconsistent between the paper's equations and its printed figures; the kernel reproduces the equations, and the points are excluded with that attribution. The corner-five fourth-rail lore splits against itself: the "one diamond short" benchmark plays on-system ($n_4 = -0.02$ diamonds) while the secondary speed and english direction claims measure sign-reversed.

Preset validity domains (three rows). The fitted equipment presets carry documented validity domains rather than blanket endorsements. `grip_fitted`

is out of domain on bank/kick observables in three independent measurement frames (worst: a +9.6° bank speed spread against a $\leq$ 1.9° window) and has been demoted from new selection in the application layer, the model retained and persisted selections grandfathered. `mathavan_fitted` is in-domain for kick/rebound spin coupling and the snooker pairing, out-of-domain for shallow-aim US-pool banks and for the lag (3.45 table lengths against the WPA 4.0–4.5 window; Section 5.6); per the anti-overfit doctrine it stays opt-in and is not re-tuned. And the fitted tangential-compliance presets break kick mirror symmetry at up to $\sim 1.1 \times 10^{-2}$ diamonds through last-ulp branch divergence — verification-side numerical noise, recorded per cell.

Open investigations (three rows). One physics investigation remains open: the compliant resolver's frozen-pair combination throw exceeds the measured $\sim 1.04\times$ frozen excess at 1.13–1.55×, speed-dependent, while the sequential baseline shows no excess at all — the sign is right, the magnitude is not, the row stays open as the honest edge of contribution C3, and its possible contamination of the E1 conclusion is bounded, not assumed away (Section 5.5). The other two rows are bookkeeping on internal worksheet and note strings whose residuals equal the slip-law shift by design and touch no measured column.

8.2. Scope limitations outside the ledger

Determinism audit gaps. The determinism audit matrix of Section 7 has gaps, which we state as untested rather than argued away. No trajectory hashes have been collected on SpiderMonkey, JavaScriptCore, or any non-

browser WebAssembly runtime such as wasmtime; there is no Windows leg and no linux-arm64 or macOS-x86_64 leg. (The battery-through-wasm cell is closed by measurement: matrix row 11, Section 7.) The bundle wheel's version stamp is coherent with the expected-hash manifest only up to a declared normalization until the planned submission-freeze rebuild, and no non-author has yet executed the tier-1 protocol. The structural argument for the untested engines — kernel-internal transcendentals compiled into the binary, a NaN-free state invariant, no relaxed SIMD (WebAssembly Community Group, 2015) — is given in Section 7; the posture remains *measure, don't argue*.

Model scope. The physical model is bounded by design. There is no air drag and no Magnus force: the deliberate constraint that keeps ball motion piecewise polynomial and event detection closed-form. Cloth wear and humidity are not modeled — precisely the unspecified "typical pool table" condition class that the fourth-rail lore rows implicate. Cushion models inherit their sources' validity domains: the rigid-cushion analysis of Mathavan et al. (2010) is declared valid to 2.5 m/s by its authors, and beyond that domain the kernel overlay overshoots monotonically, from +0.018 m/s at 2.53 m/s to +0.359 m/s at 3.40 m/s. Vertical-channel (hop) physics after the break is constrained only by a one-sided bound: the kernel's post-break cue-ball hop apex is ≤ 0.039 ball diameters everywhere, satisfying the footage's one-sided $\lesssim 0.3$-diameter expectation, but the observed clip value of ~ 0.23 diameters shows the vertical channel is under-constrained by the available evidence.

Anchor-material caveats. Three caveats travel with the validation data. The first-party table-session footage carries a leg-rotation orientation drift (an

open item). The high-speed-video footage used for cushion anchors was not recorded on a K-66 rail profile, so cushion-geometry conclusions drawn from it do not transfer unmodified to regulation US-pool rails (Alciatore, n.d.b). And the lag residual on the fitted Mathavan pairing is stated with its preset-domain row above.

Source availability and the kernel revision of record. The kernel source is public under the MIT license, with the portions derived from pooltool under the Apache License 2.0, so readers can inspect the implementation and rebuild it. The public history, however, begins at a snapshot later than this paper's canonical kernel `0.1.0+fb3425ad`, and the canonical binary of record, the experiment scripts, pinned outputs, expected-hash manifest, and ledger belong to an artifact bundle that has not been published (Section 7.1). This paper's figures and hashes therefore cannot be regenerated from the public tree, and until the bundle is published the study's independently checkable surface is the public source, the reported outputs, and the kernel's own anchor fixtures rather than a re-run of the battery. Section 7 describes the mitigations that take effect once the bundle is published — a hash-defined artifact of record, one regeneration script per figure, and planned third-party attestation.

Chaotic outcome bits at break energy. Contribution C4 is a per-binary claim. Bit-exactness holds for a fixed binary across platforms, runtimes, and runs; it does not extend across model revisions, where downstream outcome bits of long slip cascades at break-scale energies can flip. At the post-cascade re-baseline, three extreme break rows (quarter-ball offsets at 12–13.4 m/s) flipped a scratch outcome bit, with no registered band attached. The paper

claims bit-exact reproduction of every figure from the canonical binary; it does not claim insensitivity of chaotic outcome bits to model evolution.

9. Future work

Reinforcement learning on a validated substrate. A validated, bit-exact simulator bounds the trust-the-simulator caveat that runs through learning-era work consuming billiards physics (Fragkiadaki et al., 2016; Xu et al., 2025; Memery et al., 2025), as it ran through the planner lineage before it (Smith, 2007; Archibald et al., 2016). Our reinforcement-learning program applies the same registration discipline as this paper's battery — frozen kernel pins per generation and byte-reproducible evaluation — and its generation-1 milestone is complete: planner-guided policy ranking against the frozen kernel promoted an incumbent policy under registered gates. The kernel pin validated in this paper, `0.1.0+fb3425ad`, is adopted at the generation-2 boundary. Those results belong to a second paper; their relevance here is that contribution C1's validity domains directly scope the state regions in which trained play can be physics-trusted.

Footage and high-speed-video expansion. The validation corpus grows next on its measurement side: additional first-party table-session captures, currently staged under quality assurance; wider english and speed coverage on the kick/rebound coupling anchors; and a capture campaign on a K-66 rail profile that would retire the non-K-66 caveat of Section 8 attached to the existing high-speed-video anchors (Alciatore, n.d.b). Every new anchor enters through the same gate as the existing battery: design registered and frozen first, misses ledgered, nothing re-tuned.

Completing the determinism matrix. Before camera-ready, the audit matrix of Section 7 should be completed rather than argued. With the full battery now replayed through the canonical wasm runtime itself (matrix row 11, method H: 16,933/16,933 digest-identical), the open cells are the remaining engines and platforms: trajectory hashes on SpiderMonkey, JavaScriptCore, and wasmtime, and a Windows leg plus the remaining OS/architecture combinations — the replay harness already resolves its module graph through portable `file://` URLs, so the Windows leg needs only the platform, not new tooling. The same pass rebuilds the wheel at the submission-freeze commit so that raw hash equality holds with no normalization.

Companion papers. Two companion manuscripts are planned. A methods paper develops the compliant frozen-cluster break resolver as a multiple-impact resolution technique for frozen-contact topologies, in the lineage of Gismeros Moreno et al. (2022) and Cross (2023) and of the contact-force model it builds on (Lankarani and Nikravesh, 1994), with the frozen-pair throw excess (Section 8) as its opening problem. A short single-question paper quantifies the 30°/90°-rule and diamond-system validity domains for the physics-teaching audience in which those rules were first studied experimentally (Bayes and Scott, 1963; Wallace and Schroeder, 1988; Mathavan et al., 2009).

Community validation loop. Finally, the artifact is built to be exercised by others: event-level reproductions of the technical-proof curves form the basis of a planned outreach and figure-permission workflow with their author (Alciatore, n.d.c); a third-party, CODECHECK-style tier-1 attestation on all three operating systems is scheduled before submission; and cross-simulator

work with pooltool (Kiefl, 2024) continues in the cooperative framing of Section 6, where shared scrutiny has already repaid both codebases.

Acknowledgments

This work draws throughout on David G. Alciatore's technical-proof, high-speed-video, and FAQ corpus (Alciatore, 2008, n.d.c,n,n), used under the posted reuse policy (verified 2026-07-21): free use with clear citation and links. No technical-proof figure or high-speed-video frame is reproduced in this paper or redistributed with it; the validation data derived from that corpus are measurement tables with recorded provenance. Explicit written permission will be sought before any camera-ready reproduction of source figures, and any permitted figures will be enumerated here.

We thank the pooltool project (Kiefl, 2024), which serves as this paper's cross-simulator oracle (Section 6) and as the reference-implementation lineage against which the kernel's departures are documented.

Supplementary material

Supplementary material associated with this article is distributed as a separate Supplementary Material document, built from the same sources (sections numbered S1–S6).

Section S1 (data provenance) maps every figure and table of this paper to its experiment ID, pinned output hash (SHA-256 prefix), and registered source, and derives the abstract's 16,933-simulation count from the replay corpus.

Section S2 (tier-1 reproduction mechanics) states the artifact bundle inventory, the five-step reproduction procedure, the physics-identity checker semantics, the reproducibility caveats, and the licensing terms summarized in Section 7.

Section S3 (governing equations and frozen constants) reproduces the complete governing equations of every model-catalog row of Table 2, with each source's deviations itemized and every frozen constant consolidated with units and provenance.

Section S4 (extended results, groups B and D) carries the per-experiment result prose for the stick–ball and curved-path groups, whose summary tables and non-obvious findings remain in Sections 5.2 and 5.4.

Section S5 (predicted-shift verification) transcribes the full seven-class predicted-shift table behind the two-row extract of Section 6 (Table 11).

Section S6 (video-artifact presentation methods) documents the deterministic time-remap and shutter synthesis behind the artifact bundle's validated-scenario video pack (Section 7); the videos present already-verified kernel records, and no claim of this paper rests on them.

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